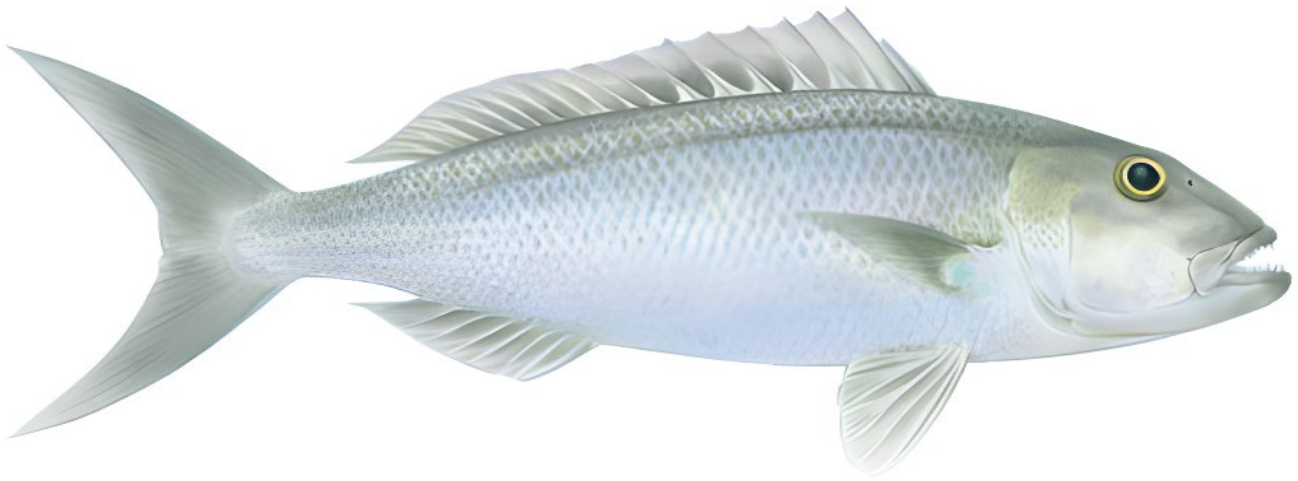




NOAA
FISHERIES



Stock Assessment of Uku (*Aprion virescens*) in Hawai‘i, 2024 Update

Marc O. Nadon

Stock Assessment of Uku (*Aprion virescens*) in Hawai‘i, 2024 Update

Marc O. Nadon¹

¹ Pacific Islands Fisheries Science Center
National Marine Fisheries Service
1845 Wasp Boulevard
Honolulu, HI 96818

NOAA Technical Memorandum NMFS-PIFSC-171

December 2024



U.S. Department of Commerce

Gina Raimondo, Secretary

National Oceanic and Atmospheric Administration
Richard W. Spinrad, Ph.D., NOAA Administrator

National Marine Fisheries Service
Janet Coit, Assistant Administrator for Fisheries

About this report

The Pacific Islands Fisheries Science Center of NOAA's National Marine Fisheries Service uses the NOAA Technical Memorandum NMFS-PIFSC series to disseminate scientific and technical information that has been scientifically reviewed and edited. Documents within this series reflect sound professional work and may be referenced in the formal scientific and technical literature.

Cover drawing: the uku snapper (*Aprion virescens*). Drawing credit: Hawai'i Division of Aquatic Resources. Artist: Les Hata.

Recommended citation

Nadon, M.O., (2024). Stock assessment of Uku (*Aprion virescens*) in Hawai'i, 2024 update. (2024). Department of Commerce. NOAA Technical Memorandum. NMFS-PIFSC-171. doi:10.25923/bsmz-zv87

Copies of this report are available from

Pacific Islands Fisheries Science Center
National Marine Fisheries Service
National Oceanic and Atmospheric Administration
1845 Wasp Boulevard, Building #176
Honolulu, Hawai'i 96818

Or online at

<https://repository.library.noaa.gov/>

Table of Contents

Table of Contents.....	i
List of Tables	ii
List of Figures	iv
Executive Summary	viii
Introduction	1
Biology	1
Distribution and population connectivity.....	2
Fisheries.....	3
Section I: Methods	5
Stock assessment model.....	5
Model inputs	5
Model Diagnostics	21
Reference points	25
Catch projections for 2020-2026.....	25
Section II: Results.....	28
Base-case model.....	28
Projections.....	31
Section III: Discussion.....	32
Acknowledgments.....	35
Literature Cited	36
Section V: Tables.....	41
Section VI: Figures.....	58

List of Tables

Table 1. Key model parameters, including some estimated by Stock Synthesis.....	41
Table 2. Main assumptions of the uku Stock Synthesis model with links to relevant report sections.	42
Table 3. Summary of commercial and recreational catch data for uku.....	43
Table 4. List of predictor variables considered for the logistic and lognormal regression models used for CPUE standardization. Dashes represent variables not available for a model and “Errors” indicate variables resulting in model convergence errors.	45
Table 5. Akaike information criterion (AIC), delta AIC, and percent AIC change from the previous model, used for CPUE standardization model selection (deep-sea handline). The final selected model is shown in bold.....	46
Table 6. Akaike information criterion (AIC), delta AIC, and percent AIC change from the previous model, used for CPUE standardization model selection (inshore handline). The final selected model is shown in bold.....	48
Table 7. Akaike information criterion (AIC), delta AIC, and percent AIC change from the previous model, used for CPUE standardization model selection (trolling). The final selected model is shown in bold.	49
Table 8. Mean and CV of CPUE relative abundance indices used in the base-case model for MHI uku. CVs are estimated from the standardization process (not variance adjusted).	50
Table 9. Negative log-likelihoods (NLL) of data components in the base-case model over a range of fixed levels of virgin recruitment in thousands of recruits in log-scale, $\log(R_0)$. Likelihoods are relative to the minimum NLL (best-fit) for each respective data component. Colors indicate relative likelihood (green: low NLL and better-fit; red: high NLL and poorer-fit). The maximum likelihood estimate of $\log(R_0)$ was 4.50.	52
Table 10. Negative log-likelihoods (NLL) of abundance index data components in the base-case model over a range of fixed levels of virgin recruitment in thousands of recruits in log-scale, $\log(R_0)$. Likelihoods are relative to the minimum NLL (best-fit) for each respective data component. Colors indicate relative likelihood (green: low NLL and better-fit; red: high NLL and poorer-fit). The maximum likelihood estimate of $\log(R_0)$ was 4.50.....	53
Table 11. Mean input log standard error (SE), additional variance added within the model with resulting total log(SE), and root-mean-square-errors (RMSE) for the relative abundance indices used in the base-case model.	53
Table 12. Time series of total biomass (age 1 and older), spawning biomass, age-0 recruitment, and instantaneous fishing mortality estimated in the base-case model. CV = coefficient of variation.	54
Table 13. Estimated biological reference points derived from the Stock Synthesis base-case model where F is the instantaneous annual fishing mortality rate, SPR is spawning potential ratio, SSB is spawning stock biomass, $MSST$ indicates minimum stock size threshold, and MSY indicates maximum sustainable yield.	56

Table 14. Probability of overfishing ($F/F_{MSY} > 1$) at different annual catch values (metric tons) by year. Catch values for a given probability of overfishing in a given year were applied in all previous years (i.e., 2020 to the year of interest).	57
---	----

List of Figures

Figure 1. Map of the main Hawaiian Islands with its four sub-regions. The entire archipelago is visible in the inset, including the Northwestern Hawaiian Islands.	58
Figure 2. Map of the eight main Hawaiian Islands with depth zones (0 m–250 m depths). Red contour lines represent the 3 nautical mile state-waters limit. Islands are not to scale and re-arranged to fit this page. Data source: CREP and Hawaii Mapping Research Group.	59
Figure 3. Boundary of the stock area in the main Hawaiian Islands used for the 2020 and 2024 uku assessment.	60
Figure 4. Proportion of total commercial catch caught by fishing gear type.	61
Figure 5. Proportion of total commercial catch caught by sub-region of the MHI, with Penguin Bank (part of the Maui Nui sub-region) listed separately.	61
Figure 6. History of uku catch by sector (recreational and commercial), with the commercial catch further broken down by fishing gear type. Note that the recreational catch pre-2003 is reconstructed and that it is adjusted by the Stock Synthesis model (expected catch).	62
Figure 7. Summary of all data sets used in the Stock Synthesis model. DSH—deep-sea handline, ISH—inshore handline, Trol—trolling, Rec—recreational data, Divers—diver surveys.	63
Figure 8. Correction factor (multiplier) applied to the HMRFS annual effort estimates between 2003 and 2017, for both the boat-based and shore-based fishery, to correct for the declining number of phone landlines.	64
Figure 9. Comparison of HMRFS boat-based annual fishing effort between the original and corrected time series.	64
Figure 10. Principal Component Analysis (PCA) plots showing the principal axes included in the Direct Principal Component (DPC) analysis for the 3 gear types used for CPUE indices (note: uku are excluded from the actual DPC analyses but are presented here for reference).	65
Figure 11. Diagnostics of the lognormal CPUE standardization model for the 1948–2002 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.	66
Figure 12. Diagnostics of the logistic CPUE standardization model for the 1948–2002 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.	67
Figure 13. Diagnostics of the lognormal CPUE standardization model for the 2003–2023 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.	68

Figure 14. Diagnostics of the logistic CPUE standardization model for the 2003–2023 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.	69
Figure 15. Diagnostics of the lognormal CPUE standardization model for the 2003–2023 period (inshore handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.	70
Figure 16. Diagnostics of the logistic CPUE standardization model for the 2003–2023 period (inshore handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.	71
Figure 17. Diagnostics of the lognormal CPUE standardization model for the 2003–2023 period (trolling gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.	72
Figure 18. Diagnostics of the logistic CPUE standardization model for the 2003–2023 period (trolling gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.	73
Figure 19. Overall nominal vs. standardized CPUE indices for deep-sea handline, generated by combining results from the lognormal and logistic standardization models.	74
Figure 20. Overall nominal vs. standardized CPUE indices for inshore handline (top) and trolling (bottom) for the 2003–2023 period, generated by combining results from the lognormal and logistic standardization models.	75
Figure 21. Distribution of survey sites (red dots) around the MHI for the NOAA diver surveys in each year.	76
Figure 22. Maturity at length (FL) curve used in the stock assessment model, obtained from Everson and Williams (1989).	77
Figure 23. Age and length data collected in the MHI (orange points) and NWHI (blue points) with a two-stage von Bertalanffy curve, including the 95% confidence interval associated with the CV parameters (0.12 for both old and young individuals). Data provided by J. O'Malley, PIFSC.	77
Figure 24. Size structures of uku from the FRS dataset and deep-sea handline gear summarized by decade.	78
Figure 25. LBSPR model fit to size structure data for inshore handline, trolling, and “other” gear types from the FRS data set (2008–2023), as well as for recreational landings reported in the HMRFS dataset (2003–2023).	78

Figure 26. Length-based selectivity used in the Stock Synthesis model for all commercial fishing gear types and the recreational sector. Only the deep-sea handline selectivity was estimated in SS.	79
Figure 27. Fishery management control rules used for bottomfish in Hawai'i.	80
Figure 28. Jitter results for the base-case model, showing spawning biomass (top left), F (top right), recruitment (bottom left), and total likelihood (bottom right) for 150 model runs with different starting parameter values.	81
Figure 29. Profiles of the negative log-likelihoods relative to the minimum value of each component for the different likelihood components affecting the unfished recruitment parameter R_0 in log-scale for the base-case model. Recruitment represents the likelihood component based on the deviations from the stock-recruitment curve, catch is the joint likelihood component based upon the estimated catch for each fleet, survey is the joint likelihood component based upon fitting to the CPUE and OPUE indices, and SizeFreq represents the likelihood component for the deep-sea handline fleet based on the fish size composition data.	82
Figure 30. Profiles of the relative negative log-likelihoods by fleet-specific index likelihood components for the virgin recruitment in log-scale ($\log(R_0)$) ranged from 3.6 to 5.0 of the base-case scenario. DSH—deep-sea handline, ISH—inshore handline, and Trol—trolling.....	83
Figure 31. Plots of observed (open dots) and expected (blue lines) CPUE for the uku base-case model for all abundance indices used in the model. Bars around each observed data point represent the sum of the input SE and additional SE added within the model.....	84
Figure 32. Model fit (lines) to mean weight (kg) of the composition data (points, showing the observed mean age and 95% credible limits around mean age (vertical lines).	85
Figure 33. Pearson residual plots of model fits to the size composition data for deep-sea handline fishery targeting uku.....	86
Figure 34. Comparison of observed (gray shaded area and black dots) and model predicted (green solid line) size composition data for the deep-sea handline fishery used in the stock assessment for uku.	87
Figure 35. Observed (dashed line) vs. estimated (full line) catch in the Stock Synthesis model.....	88
Figure 36. Observed catch by fishing gear.....	89
Figure 37. Annual fishing mortality estimated from the base-case model by fishing gear.	90
Figure 38. Time series of total biomass (age 1 and older, metric tons) for uku estimated in the base-case model. The first year indicates virgin biomass levels.....	91
Figure 39. Time series of spawning biomass (metric tons) for uku estimated in the base-case model. The solid line with circles represents the maximum likelihood estimates and the error bars represent the uncertainty of the estimates (95% confidence intervals). The dashed horizontal line shows the spawning biomass to produce the MSST reference point (SSB_{MSST}).....	91

Figure 40. Estimated log recruitment deviations for the early (1938–1947, blue) and main (1948–2017, black) recruitment periods with associated 95% asymptotic confidence intervals.....	92
Figure 41. Expected recruitment from the stock-recruitment relationship (black line) and estimated annual recruitment (dots) from Stock Synthesis. Estimated virgin SSB and recruitment is indicated with a red diamond.....	92
Figure 42. Bias adjustment applied to the stock-recruitment relationship (red stippled line) and the estimated alternative (blue line) obtained from the r4ss output.....	93
Figure 43. Time series of recruitment (thousands of age-0 fish) for uku estimated in the base-case model. The solid line with circles represents the maximum likelihood estimates and the error bars represent the uncertainty of the estimates (95% confidence intervals).....	93
Figure 44. Time series of instantaneous fishing mortality (average for ages 5–30) for the uku estimated in the base-case model. The solid line with circles represents the maximum likelihood estimates, and the error bars represent the uncertainty of the estimates (95% confidence interval). The dashed horizontal line shows the fishing mortality to produce the MSY reference point (F_{MSY}).	94
Figure 45. Kobe plot of the trends in estimates of relative fishing mortality (average of age 5–30) and spawning stock biomass of uku during 1948–2023. The white dot indicates 1948, the orange dot indicates 2023, and the dotted lines indicate the 95% confidence intervals around the final year values.....	95
Figure 46. Retrospective analysis of spawning biomass (as SSB/SSB_{MSST} , left) and fishing mortality (as F/F_{MSY} , right) consisting of 5 reruns of the base-case model each fitted with one additional year of data removed from the base-case model (black line, 1948-2023).	96
Figure 47. Plot of estimated Age-0 recruits (in 1000s of fish) for each run in the 5-year retrospective analysis. The dark blue line indicates the base-case model (1948–2023).	96
Figure 48. Probability of overfishing (i.e., $F/F_{MSY} > 1$) uku in the main Hawaiian Islands in fishing years 2025 through 2031, for a range of fixed catch scenarios.....	97
Figure 49. Probability of the stock being overfished (i.e., $SSB/SSB_{MSST} < 1$) for uku in the main Hawaiian Islands in fishing years 2025 through 2031 for a range of fixed catch scenarios.....	98
Figure 50. Mean fishing mortality for uku in the main Hawaiian Islands in fishing years 2025 through 2031 for a range of fixed catch scenarios.....	99
Figure 51. Mean spawning stock biomass for uku in the main Hawaiian Islands in fishing years 2025 through 2031 for a range of fixed catch scenarios.....	100
Figure 52. Fishing mortality with F_{MSY} reference line (a), Spawning Stock Biomass with SSB_{MSST} reference line (b), probability of overfishing (c), and probability of being overfished (d) for various fixed catch scenarios from 2024 to 2031.....	101
Figure 53. Comparisons of spawning biomass, fishing mortality, recruitment, and final year stock status (Kobe plot) for (a) the base model used in the current assessment update (blue), (b) the same model without the landline phone correction to the HMRFS catch (yellow), and (c) the 2020 benchmark model..	102

Executive Summary

The uku snapper (*Aprion virescens*, family Lutjanidae) inhabits the coastal waters of the main Hawaiian Islands (MHI) at depths ranging from 20 to 200 meters. The MHI uku population was first assessed with other snappers using a catch-only method applied at the family level (Sabater & Kleiber, 2013). That assessment determined that snappers were not overfished. In 2017, the population was assessed at the species level using a length-based mortality model and a relatively simple numerical population model to obtain fishing mortality rates and spawning potential ratio (SPR). Using this approach, it was determined that the stock was not experiencing overfishing (Nadon, 2017). In 2020, a significant improvement was made by switching to the Stock Synthesis modeling framework (SS3; Methot & Wetzel, 2013). This approach used catch, catch-per-unit-effort (CPUE), diver surveys, and size composition time series in an integrated statistical catch-at-age model to generate maximum likelihood estimates of population parameters, derived outputs, and their associated variability. These outputs were then used to determine stock status and to develop stock projections under different management scenarios. The current assessment is an update to the 2020 benchmark, adding 5 years (2019–2023) of catch, CPUE, and size data, and an extra year of fishery-independent diver surveys (2019). The only change to the modeling approach was to implement correction factors for the recreational catches related to the decline of phone landlines between 2003 and 2016, as done in the recent Deep-7 assessment (Syslo et al., 2024).

All available fishery data from recreational and commercial fisheries in the MHI were used for this stock assessment. Total recreational catches for the 2003–2023 period were obtained from the Hawaii Marine Recreational Fishing Survey (HMRFS) and reconstructed for the 1948–2003 period by relating historical catch to human population trends in the MHI. Total commercial catch was obtained from the Division of Aquatic Resources (DAR) fisher reporting system (FRS). Commercial catches for uku were dominated by the deep-sea handline fishing gear although trolling and inshore-handline catches have increased in recent decades. CPUE data were obtained from all three main fishing gears in the FRS while size composition data were obtained for the deep-sea handline gear only, due to limited data availability. Deep-sea handline data were the only CPUE time-series available for 1948–2002, as trolling and inshore handline data were sparse for this period. Additionally, observations from NOAA diver surveys were incorporated as a fishery-independent abundance index between 2005 and 2019.

Uku catches increased from 1948 to the late 1980s and have been declining slowly since then. Model estimates of population biomass show a gradual decline from 1948 to the late 1980s, followed by a brief period of stability and a consistent increase in biomass starting in the early 2000s and leveling off by 2017. Fishing mortality on the stock (average F on ages 5–30) is currently 0.05 with an F/F_{MSY} value of 0.36 (Figure S1). Fishing mortality has only been above F_{MSY} (0.14) twice, in 1988 and 1989 when F reached 0.17 and 0.15, respectively. The 2023 spawning stock biomass (SSB) of 995 metric tons (mt) is 280% above the SSB_{MSST} (355 mt). Therefore, relative to the reference points defined by the Fisheries Ecosystem Plan, overfishing is not occurring and the MHI uku stock is not overfished (Figure S1).

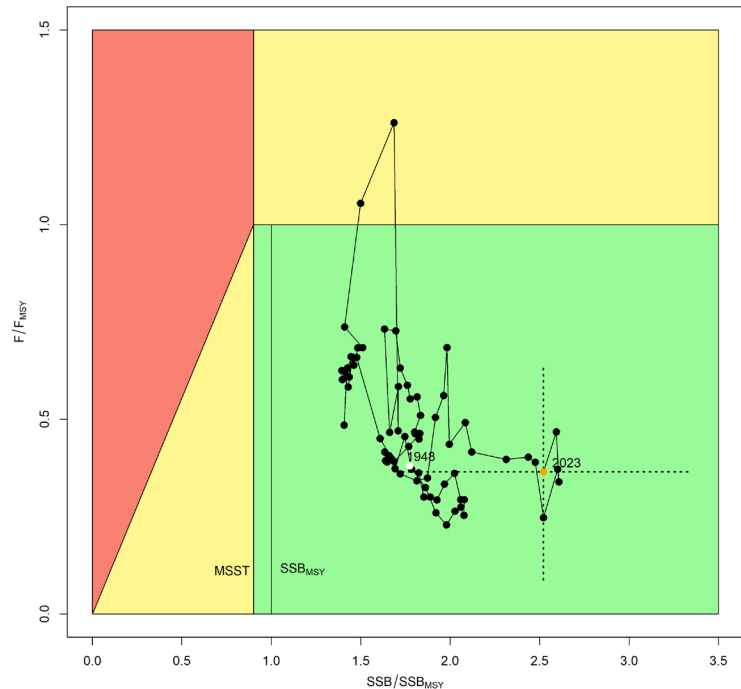


Figure S 1. Kobe plot of the trends in estimates of relative fishing mortality (average of age 5–30) and spawning stock biomass of uku from 1948 to 2023.

Stock projections for uku were conducted using the age-structured projection model software AGEPRO (Brodziak et al., 1998). Stochastic projections were conducted using results from the base-case model to evaluate the probable impacts of constant catch quotas on future spawning stock biomass and yield for uku in the MHI. Results show the projected female spawning stock biomasses and fishing mortality rates under each of the constant-catch scenarios. For example, a constant catch limit of 190 mt each year from 2025 to 2029 would result in a 50% chance of overfishing occurring in 2029.

Introduction

This report presents the updated stock assessment of the ukupalu snapper, commonly known as “uku,” in the main Hawaiian Islands (MHI), covering the period from 1948 to 2023. Uku is a wide-ranging reef-associated snapper (family Lutjanidae) that is found between 35° N and 31° S from East Africa to Hawai‘i. The scientific name for this species is *Aprion virescens* and it goes by several common English names in other regions, typically “green jobfish” or “gray snapper.”

The Hawai‘i uku population was first assessed with other snappers using a catch-only method applied at the family level (Sabater & Kleiber, 2013). That assessment determined that snappers were not overfished in the MHI. In 2017, uku were assessed at the species-level using a length-based mortality model and a relatively simple numerical population model to obtain fishing mortality rates and spawning potential ratios (SPR). Using this approach, it was determined that this stock was not experiencing overfishing (Nadon, 2017), as its spawning potential ratio of 0.33 was above the 0.3 limit defining overfishing. A major overhaul of the assessment approach occurred in 2020, switching to the Stock Synthesis 3.30 (SS3) modeling framework (Methot & Wetzel, 2013) and incorporating time series of catch, catch-per-unit-effort (CPUE), diver surveys, and size composition. This assessment confirmed that the Hawai‘i uku population is not overfished and not experiencing overfishing.

For the current update, we added 5 years (2019–2023) of catch, CPUE, and size structure data to the SS3 model used in the last benchmark assessment, as well as one extra year of fishery-independent diver surveys (2019). We did elicit a significant improvement to the recreational catch data, implementing correction factors for the decline in phone landlines between 2003 and 2016, as was done in the latest Deep-7 assessment (Syslo et al., 2024). No other modifications to the modeling approach were implemented.

Biology

Uku is a coastal semi-pelagic species found in the water column above both hard and soft bottom habitats at depths ranging from around 20 m to 200 m (Asher et al., 2017; Pyle et al., 2016; [Figure 2](#)). They typically grow to about 75 cm fork length (FL; about 6 kg) although larger specimens up to 101 cm (14 kg) have been reported in Hawai‘i (Sundberg & Underkoffler, 2011). They reach maturity around 45 cm (approximately 3 years) and spawn primarily during the summer months (Everson et al., 1989); Penguin Bank appears to be a major spawning ground for this species. They can live up to 32 years in Hawai‘i and elsewhere in the Indo-Pacific region ([Table 1](#); O’Malley et al., 2021).

Distribution and population connectivity

Uku are found throughout the Hawaiian Archipelago, which extends 2,600 km along a SE–NW axis from 19°N, 155°W to 28°N, 178°W ([Figure 1](#)). The archipelago is composed of 18 islands and atolls that are typically divided into two broad regions: the inhabited main Hawaiian Islands (MHI; current population approximately 1.4 million individuals; dbedt.hawaii.gov/census) and the uninhabited Northwestern Hawaiian Islands (NWHI). The MHI consists of eight geologically young, high (4,205 m maximum elevation) volcanic islands while the NWHI are low lying atolls and reefs (275 m max elevation; [Figure 2](#)). It is generally well-accepted that the MHI and NWHI coastal fishes form different stocks, and that minimal larval or adult exchange exists between these two regions given the dominant current direction and the large distances involved (Toonen et al., 2011; Wren et al., 2016).

Focusing on the MHI, the level of connectivity of the uku sub-populations in this region and the significance of larval exchanges or adult movements between the different islands are still not entirely clear. As shown in [Figure 1](#), the longest distance between islands in the MHI is 116 km (O'ahu to Kaua'i), with other islands separated by much shorter distances. Inter-island adult movements have not been detected for tagged uku in the NWHI by Meyer et al. (2007). This study did find significant diel and seasonal movement patterns within-atoll (e.g., up to 24 km in a single tidal cycle) and could not account for a third of their tagged individuals during certain months (suggesting potential long-distance trips). However, the authors concluded that inter-atoll movement must be fairly rare for ukus, given that a similar acoustic array easily detected reef shark inter-atoll movements. In contrast, significant larval dispersal cannot be ruled out. Genetic parentage analyses of groupers in Australia have found parent-offspring pairs at distances up to 250 km, with a median dispersal distance of 110 km (Williamson et al., 2016). Another parentage study conducted on Hawai'i Island found yellow tang surgeonfish parent-offspring pairs separated by up to 184 km, although they did not attempt to find cross-channel pairs (Christie et al., 2010). A recent parentage study of the convict tang on O'ahu found a median dispersal of 25 km, with pairs found up to 80 km away from one another (Coleman et al., 2023). Furthermore, a study of passive pelagic particle connectivity in the MHI, based on a pelagic larval duration of 45 days, found a median distance for successful settlements around 100 km (Wren et al., 2016) and that cross-channel dispersal can be common. Recently, another passive particle simulation study was conducted specifically for uku, using surface currents and Penguin Bank as a release site (Suca et al., under review). The early finding of this study showed that most simulated larvae released during the summer months settled in the O'ahu and Maui Nui area, with a smaller proportion reaching Kaua'i, Ni'ihau, and Hawai'i Islands in each simulated year. Finally, a recent study conducted at the Pacific Islands Fisheries Science Center (PIFSC) assembled an inventory of all uku larval

records from 1967 to 2012 collected around Hawai'i (Schmidt et al., 2023). They also investigated uku larval observations from the literature. They found a majority of uku larvae inside 27 km from shore around Hawai'i, although the literature reports observations up to 88 km offshore. Of note, this study reported that uku larvae mainly occur above 40 m depth and that juveniles from 3 to 9 cm have never been reported anywhere, leaving a blind spot in our understanding of juvenile habitat utilization.

Uku population connectivity within the MHI will require further investigation. In this report, the uku stock was analyzed at the MHI scale ([Figure 1](#), [Figure 3](#)) due to data limitations and current management stock definitions. Based on current research, it appears that this scale is appropriate. Further population connectivity studies may suggest that future stock assessments be conducted at different spatial scales for this species.

Fisheries

Coastal fishes in Hawai'i have been exploited since around AD 1250, when the MHI were first settled (Kittinger et al., 2011). The NWHI were never permanently inhabited; however, they were a focus of commercial fishing, especially in the 19th century (Kittinger et al., 2011). They are now part of the Papahānaumokuākea Marine National Monument, and fishing has been prohibited in that entire region since 2006.

Most fishers report that uku is an important fish, but is rarely their primary target, which is more typically tunas and the snappers of the Deep-7 complex (Ayers, 2022). The uku fishery around the MHI is mostly a small-boat fishery using deep-sea handline gear, combined with a small shore-based component using gears such as spearfishing and hook-and-line. The nearshore component represents only about 10% of the total catch, based on creel surveys and the commercial reported catch between 2013 and 2023, and is entirely recreational/subsistence fishing. The boat-based uku fishery can be split into a commercial and a recreational/subsistence fishing sector, which represent 39% and 61% of the total boat-based catch, respectively (2013 to 2023 average). There are about 9,500 small boats active in coastal fisheries in the MHI, with about 2,000 (21%) of them associated with a Commercial Marine License (source: DOBOR vessel registration list). Of those 2,000 boats with a CML, slightly more than 200 reported catching uku in recent years.

As previously mentioned, the boat-based fishery catches uku mainly with deep-sea handlines (63% of total commercial catch), but also uses inshore handlines (15%) and trolling (10%; [Figure 4](#)). In most years, the majority of the commercial catch comes from the Maui Nui sub-region (54%; [Figure 1](#), [Figure 5](#)) with Penguin Bank, off the western tip of Moloka'i, representing about 36% of the total commercial catch alone ([Figure 2](#), [Figure 5](#)). In recent years, the recreational sector has been catching about 70 mt of uku

annually, while the commercial sector has been catching about 40 mt annually ([Figure 6](#)). Historically, the commercial sector captured around 30 mt of uku annually in the 1950s. Commercial landings increased until they reached a sharp peak in 1988 (156 mt) before declining to about 35 mt in the 1990s ([Figure 6](#)). Commercial landings increased since then, reaching around 50 mt in the mid-2010s, before declining significantly during and after the COVID pandemic (to around 23 mt). Dealer-reported data collected by PIFSC indicate that the price per pound for uku has been steadily increasing from 2003 to 2018, from 3.92 \$/lb to 5.31 \$/lb (adjusted for inflation). Uku are caught year-round, but their catch peaks during the summer months coinciding with their spawning season. The State of Hawai'i has regulations enforcing a minimum size of 1 pound for uku targeted by spear or for sale. Following the 2020 assessment, an annual catch limit of 134 mt (295,419 lb) was established for uku, with commercial and recreational fishers catching about 104 mt (229,000 lb) per year, on average, since 2019.

The 1988–1989 peak in catch is remembered by fishers and scientists as a notable appearance of a large number of ukus in the MHI that could have been associated with a strong recruitment event or even migration from the NWHI correlated with unusually cold temperatures in that region (D. Kobayashi, J. Polovina, R. Morioka, and L. Yamada, pers. comm.). The recent drop in catch for commercial deep-sea handline may be related to a strong increase in shark depredation in 2018, especially on Penguin Bank (R. Morioka, pers. comm.).

Section I: Methods

Stock assessment model

Stock Synthesis (SS) is an integrated statistical catch-at-age model that is widely used for stock assessments in the United States and throughout the world (Methot & Wetzel, 2013). SS takes relatively unprocessed input data in the form of observed catch, size/age composition, and relative abundance indices such as catch-per-unit-effort (CPUE), and incorporates the main population processes (e.g., mortality, selectivity, growth) to recreate population biomass trajectory and derived indicators of stock status. Because many of these inputs are correlated, the theory behind SS is that these should be modeled together, which helps to ensure that uncertainty in the input data is propagated through the assessment. SS is comprised of three subcomponents: (1) a population subcomponent that recreates the numbers- and biomass-at-age using estimates of natural mortality, growth, fecundity, etc.; (2) an observational subcomponent that consists of measured quantities such as CPUE or proportion at length, weight, and/or age; and (3) a statistical subcomponent that uses likelihoods to quantify the fit of the observations to the re-created population. Basic equations and technical specifications underlying SS can be found in Methot (2000). We used SS version 3.30.19.01 with AD Model Builder (ADMB version 12.3, released 04/15/2022).

In the current assessment, our SS model was informed by four types of data: commercial and recreational catches, catch-per-unit-effort time series, body size (weight) frequencies time series, and fishery-independent diver surveys. The sections below describe how each data source was processed to generate the inputs necessary for the SS model while also going into greater detail on the functioning of SS itself. Table 2 presents the main assumptions built into our SS model and includes links to the relevant sections in this report.

Model inputs

We used the following data sets in the current assessment:

- **Fisher Reporting System (FRS)** from the State of Hawaii Division of Aquatic Resources (DAR; Pacific Islands Fisheries Science Center, 2020). This data set is composed of self-reported catch in numbers and weights per record at the species level, with records corresponding roughly to individual trips (see [CPUE standardization section](#) for details). Notably, only commercial fishers are required to report to this system and this data set does not capture the catch from fishers who do not have a Commercial Marine License (CML). Data collection in this system started on January 1, 1948, and continues to this day. This data set was used to obtain **CPUE indices**, **size frequencies**, and annual **commercial catch** from 1948 to 2023, which are the years included in the current assessment.

- **Hawaii Marine Recreational Fishing Survey (HMRFS)** is conducted by the Hawaii Division of Aquatic Resources (DAR) and NOAA Fisheries Marine Recreational Information Program (MRIP; Ma & Ogawa, 2016). This data set provides annual **recreational catch** by using phone (2003–2017) or mail (post-2017) estimating total fishing effort and onsite fisher interviews for CPUE for boat-based and shore-based activities. Data collection in this system started in 2003. Total recreational catch per year was used in the current assessment from 2003 to 2023 (NMFS Office of Science and Technology, 2020). Pre-2002 recreational catches had to be reconstructed as described further below.
- **Pacific Reef Assessment and Monitoring Program (RAMP)** are SCUBA diver survey cruises conducted by PIFSC which provided a fishery-independent **abundance index** every 2 to 3 years starting in 2005, with the latest cruise completed in 2019 (Ayotte et al., 2015; Pacific Islands Fisheries Science Center, 2020b, 2020c).

The data used in our model are summarized in [Figure 7](#) while the sources of key input parameters are presented in [Table 1](#).

Catch reconstruction

The total harvested biomass of uku (referred to in the rest of the report as “catch”) is the sum of commercial and recreational landings by weight in a given year from 1948 to 2023 ([Table 2](#)).

Commercial catch from 1948 to 2023 was obtained from 106,972 catch records in the FRS data set that reported landing uku (DAR species code “20”). We first assigned the uku records to the main Hawaiian Islands (MHI) and Northwestern Hawaiian Islands (NWHI) zones based on the reported DAR fishing areas ([Figure 1](#), [Figure 3](#)). We assigned 102,167 records to the MHI, 4,636 records to the NWHI, and only 169 records had invalid reported areas. The catch from these 169 records was assigned to the MHI zone according to the ratio of MHI/NWHI catch in the corresponding year. Total commercial catch by year was obtained by simply summing the total weight of all uku MHI records. We assumed that the commercial catch was known with relatively low uncertainty and assigned a low coefficient of variation (CV) of 0.05 to the deep sea handline fleet and a CV of 0.1 to the remaining commercial fleets’ data.

Recreational catch from 2003 to 2023 was obtained from the HMRFS program by querying the MRIP website (<https://www.fisheries.noaa.gov/topic/recreational-fishing-data>) for the “green jobfish” (i.e., uku) annual catch time series. This data set provided the number of uku caught by fishing mode (shore or private/rental boat) by year. We did not use the HMRFS estimate of catch in weight given that the mean weight of uku used in that calculation is typically derived from only a few individual fish measurements in each reporting “wave” (there are 6 waves in a year), and the mean weight estimates are sometimes not available in a given wave. Instead, we obtained catch in weight by

multiplying the number of uku caught in a year by the overall mean weight of uku in the HMRFS data set between 2003 and 2023. Unlike the 2020 assessment, we removed two weight outliers that were recognize as data entry errors (25 kg in 2007 and 18 kg in 2012). This resulted in a mean uku weight in the recreational catch of 3.04 kg from 205 individuals (versus 3.26 kg in the 2020 assessment). Due to a recognized error in the household count for Maui County, the 2003–2010 estimates of catch had to be divided by 1.22 (Ma, 2013). Since the HMRFS program also captures some commercial fishing effort from non-fulltime fishers, it was necessary to remove the HMRFS uku catch that was claimed to be sold to avoid this catch being accounted for in both the commercial and recreational tallies. To do so, we obtained the proportion of uku catch sold by year from HMRFS fisher interviews and multiplied the total HMRFS uku catch by one minus the year-specific proportion (Hongguang Ma, pers. comm.). To reconstruct recreational catch before 2003, we first calculated the mean recreational catch between 2003 and 2007, as well as the mean Hawai'i human population size for those years. We multiplied the mean 2003–2007 recreational catch by the ratio of the mean 2003–2007 human population size to the human population size in any given year from 1948 to 2002:

$$Rec.catch_{year} = Rec.catch_{2003-2007} \cdot \frac{Population_{year}}{Population_{2003-2007}} \quad \text{Eq. 1}$$

We used the 2003–2007 mean recreational catch instead of simply using the 2003 catch to avoid attributing too much weight to that single year given the high uncertainty associated with the HMRFS recreational catch estimates. The assumption behind this approach is that pre-2003 recreational catch is proportional to the Hawai'i population size, and participation in the recreational fishery would be expected to increase as the overall population in Hawai'i increases. An alternative approach would have been to apply the ratio of recreational to commercial catch in the 2003–2023 data sets to the 1948–2002 commercial catch to re-create the recreational catch (this scenario seems less plausible but was tested as a sensitivity run in the 2020 assessment). As previously mentioned, the recreational catch estimates are highly uncertain and a mean CV of 0.4 was used in our model for all years. This CV was obtained directly from the queried HMRFS data set.

As noted previously, in 2018, the HMRFS program switched from a phone to a mail survey to estimate recreational fishing effort. Starting in 2017, the program conducted both surveys to calibrate the estimates and found that effort estimates derived from mail surveys were 2.89 and 2.33 times the same estimates from phone surveys for shore- and boat-based activities, respectively (data from the NMFS Office of Science and Technology). The differences were likely related to a progressive reduction in the number of households with phone landlines. In the 2020 assessment, we elected not to

implement correction factors for declining phone landlines between 2003 and 2016, but rather wait for official correction factors from MRIP. However, as of 2024, we have not received an official product from MRIP; therefore, in our 2023 Hawai'i Deep-7 assessment, we decided to implement a linear factor between 1999 to 2017, starting with a value of “1” in 1999 and ending with a final value of “2.33” in 2017 for the boat-based catch and “2.89” for the shore-based catch (Ma, 2023). For the current uku assessment update, we decided to implement this change given that we know that there was some form of progressing negative bias in the HMRFS fishing effort estimate between 2003 and 2017 that needed to be addressed. This was an important change from the 2020 assessment that resulted in a large positive scaling of the uku population in Hawai'i (see [Discussion](#) section). The sensitivity of the current assessment update to this change is explored in the Results and Discussion sections. [Figure 8](#) shows the values of the effort correction multiplier by year for the boat-based and shore-based fisheries. [Figure 9](#) shows the impact of those correction multipliers on the boat-based fishing effort.

CPUE abundance indices

Overview: We used the FRS data set to produce fishery-dependent indices of uku abundance based upon methods used in the Deep-7 bottomfish assessment (Langseth et al., 2018). Starting in January 2003, the FRS data reporting system was improved by requiring fishers to report the number of hours fished per record, thus allowing fishing effort to be standardized to an hourly rate. Before this, fishing effort was provided as individual fishing days. This difference in fishing effort reporting resulted in different CPUE scales, leading us to split the CPUE time series into two time periods, 1948–2002 and 2003–2023, each with an estimated catchability coefficient (q). This being said, it is important to note that the 2020 WPSAR review explored many different variations on merging and splitting the CPUE time series and no important impact on CPUE trends were found (Nadon et al., 2020). Furthermore, we selected the three dominant fishing gears used to catch uku in the MHI to generate CPUE indices: deep-sea handline (DAR gear code “3”), inshore handline (gear code “4”), and trolling (gear codes “6,” “61,” “62,” “63,” and “70”). Before the early 2000s, inshore handline and trolling fishing gears were used only sporadically to catch uku ([Figure 4](#)). We therefore only generated CPUE indices for the recent (2003–2023) period for those gears due to small sample sizes. In total, our assessment model included four CPUE indices:

- Deep-sea handline 1948–2002
- Deep-sea handline 2003–2023
- Inshore handline 2003–2023
- Trolling 2003–2023

The data processing and filtering steps used to clean the data for CPUE standardization were built on the many improvements brought forward from various bottomfish data workshops (Yau, 2018) and the latest Deep-7 assessment (Langseth et al., 2018). These include the ability to track individual fishers through the entire time series with names assigned to practically all records between 1977 and 2023, and most records for the rest of the time series. Notable exceptions were the 1954–1958 period where less than 50% of records had unique fisher names and 1976 where no fisher name information could be located (Langseth et al., 2018). Furthermore, the Deep-7 assessment and its associated data workshops spent significant time improving the following steps: (1) selecting targeted records, (2) accounting for multi-day trips in the 1948–2002 period, (3) selecting records representative of the fishery, and (4) identifying factors affecting CPUE trends. These steps are described below.

Selecting uku-targeted records: Depending on the species targeted, the three gears selected for CPUE indices (deep-sea handline, inshore handline, and trolling) can be deployed in different configurations and different habitats. Therefore, targeting may have a non-negligible impact on CPUE. Unfortunately, the FRS data set does not include species targeting information. We controlled for species targeting in two separate steps: (1) we applied certain basic filters to fishing records (see below) and (2) we used a principal component analysis (PCA) on species composition in the catch to generate principal components that were then used in the CPUE standardization model (Winker et al., 2014; see [CPUE standardization section](#) below).

We filtered for non-uku fishing events mainly for records before 2003, when the FRS reporting form was less detailed and individual trips were defined at a much broader daily time scale (note: these steps only apply to the deep sea handline data given that this was the only CPUE index for the 1948–2002 period). The first step was to filter records with inappropriate or missing data fields: (1) a filter removed records missing a Commercial Marine License (CML) number since this would prevent us from defining individual trips, (2) another filter removed records with no fisher name which would prevent us from using this important variable in the standardization model. These two steps removed 44,864 records out of the original 784,580 deep sea handline records.

It is unlikely that fishers targeting uku failed to catch any while successfully capturing certain pelagic or Deep-7 species. Therefore, trip records were removed if no uku was caught in an entire day of fishing but at least one pelagic management unit species (PMUS; WPRFMC, 2009), Deep-7 species, or unknown species (species code “0”) was reported. This step reduced the number of records from 726,369 to 343,234. Furthermore, for management grids off the southern tip of Hawai’i Island (grid 100–102, 108, 120–122, and 128 in [Figure 3](#)), catch records before 1985 with less than 50 pounds of uku containing PMUS were considered non-uku targeting and filtered out (removing another 7,603 records). The steep bathymetry in this area can lead fishers

targeting PMUS to catch uku and other semi-pelagic species. Before 1981, gear types targeting tunas did not have unique codes. Some of these fishing events were recorded as deep-sea handline.

Two final data filtering steps related to targeting were implemented based on feedback from fishers during various workshop: (1) we removed all records from fishers that never reported catching uku, and (2) we removed records from fishery-independent scientific fishing surveys. These steps removed 5,763 out of 335,631 records. It is highly unlikely that certain fishers targeted uku under an active commercial license but failed to catch a single one throughout their period of commercial activity.

Accounting for multi-day trips: As previously identified, the pre-2003 fishing records do not provide a clear estimate of fishing effort. The only way to re-create effort for the CPUE calculation was to combine all reported catch in individual fishing days by combining CML numbers and dates to generate a unique trip identification. However, this raised the issue that catch reported under a single CML–date combination could be aggregated over multiple fishing days (Yau, 2018). To deal with this issue, we used the approach presented in Yau (2018) and Langseth et al. (2018), which consisted of identifying potential multi-day trips by calculating the distance traveled between fishing areas and different ports. To reduce the number of distances needing manual measurements, a series of centrally located “common” ports were used (J. Ault and S. Smith, U. of Miami, pers. comm.). Some records could not be assigned distances due to missing port information and were filtered out (2,549 out of 180,599 pre-2003 records). Only 56 out of 71,069 trips had multiple ports and these were filtered out as well. For the 772 trips that reported more than one fishing area, we used the furthest to assign distance traveled. Once distances were available for all trips, we analyzed the frequency distribution of distances in 10-year time blocks and compared our results to those from the Deep-7 assessment (Yau, 2018). We found similar patterns and decided to use the 30 nm cutoff to define a single day of fishing (distances between 0 and 30 nm were assigned a single fishing day, distances between 30.1 and 60 nm were assigned two days of fishing, etc.). Overall, 34,387 out of 177,750 pre-2003 records were assigned effort values greater than one fishing day; the majority (24,115) were for two fishing days. Another concern was that certain fishers before 2003 were reporting the entirety of their monthly catch on either the first or the last day of the month (Yau, 2018). To control for this issue, we removed all 1,033 records from fisher-year combinations that only reported on those two days. After reassembling the pre-2003 data with the 2003–2023 data, the remaining number of catch records was 327,019.

Explanatory variables for CPUE standardization: In addition to *year*, *area*, *fisher name*, and *month*, we added the following variables to the CPUE data set: (1) *fisher experience*, measured as the cumulative number of fishing events associated with an individual fisher, (2) *wind speed*, and (3) up to 2 principal components (*PC1* and *PC2*)

related to species catch composition (see details below). The original *area* variable taken directly from the DAR reporting grid had 279 distinct values in the MHI (shown in [Figure 3](#)). We simplified this variable slightly by combining the corresponding offshore/inshore areas together (e.g., areas 300 and 320). This reduced the total number of areas to 230. These offshore/inshore paired areas are not drastically different given that they are typically split at the 2 nm line extending from land. At this distance from shore, some areas of the ocean are already very deep while other areas are still relatively shallow, exhibiting no clear pattern, thus making this distinction unhelpful for CPUE standardization purposes.

Wind data starting in July 1987 were available at a spatial scale similar to the reported DAR fishing area, with some gap in coverage. Average wind speed and direction values at a 0.25-degree scale were downloaded from the [Coast Watch website](#) (accessed 4/01/2024). These data were spatially merged with the CPUE data set based on fishing date and the GPS coordinates of the center points of each DAR fishing area. Given the restricted availability, we could only use this variable for the 2003–2023 CPUE models.

As previously mentioned, we also explored a new approach to infer the degree of targeting of uku in individual fishing events. The Direct Principal Component (DPC) approach consists of including principal components scores (PCs) derived from the species composition in the catch as predictors in the CPUE model (Winker et al., 2014). This procedure builds on the common assumption that the species composition of the catch is directly related to the extent of targeted effort (Pelletier & Ferraris, 2000). For example, a deep sea handline fishing record containing semi-pelagic species (e.g., kahala, ulua) and few Deep 7 species would likely indicate greater targeting of the semi-pelagic uku, as reflected by the relationship between the different PCs and uku CPUE. If there were a temporal trend in uku targeting for deep-sea handlines, this would be directly reflected in changes in the catch composition and their associated PC values. The first step in this procedure was to select the top species caught by each of the three gears used for CPUE indices. Following Winker et al. (2014), we kept species representing a minimum of 1% of the total catch by gear, which resulted in keeping 12 species for deep-sea handline, and 7 for inshore handline and trolling. The second step was to calculate the proportion of each of the species for each fishing record and doing a fourth-root transformation on these values to reduce the influence of the more abundant species on the PCA (Winker et al., 2014). The third step was to run the PCA analyses using the “prcomp” base R function and extracting the PC scores for each fishing record (labeled as “PC1,” “PC2,” etc.). A final step was to select the principal components to keep for the CPUE standardization model. Following Winker et al. (2014), we used the “nFactors” R package to obtain the optimal coordinate solution for Cattell’s scree test (Raïche et al., 2013) while also selecting PCs which had eigenvalues higher than 1 (Kaiser-Guttman rule). Using the AIC and BIC values from the CPUE

standardization model to select PCs was not recommended by Winker et al. (2014) as it tends to select for unnecessarily complex models with a high number of PCs. Following these rules, we selected PC1 and PC2 for deep sea handline, inshore handline, and trolling. [Figure 10](#) shows the full PCA catch composition results for all three gear types. This figure showed some non-linear patterns in catch composition between uku and other groups. For example, the trolling PCA showed that uku were more likely to be targeted when PC1 and PC2 are closed to zero which suggested that a u-shaped quadratic relationship might be appropriate for these variables in the CPUE standardization model. For this reason, we also added second-degree polynomial terms to the standardization models (i.e., $PC1^2$ and $PC2^2$).

Final processing of CPUE data: Following the DPC analysis, we discarded the non-uku catch data, which were no longer necessary, and summed the uku catch by the remaining variables (*trip ID*, *year*, *month*, *area*, *effort* (i.e., # of days or hours, depending on data set), *fisher name*, *fisher experience*, *wind speed*, $PC1$, $PC1^2$, $PC2$, and $PC2^2$).

The original unique trip identifier (*trip ID*) was created by combining CML number and fishing date since this was the only way to define an individual fishing activity for the 1948–2002 time series. However, the 2003–2023 time series contains information on fished hours that can differ within a single day for any given CML (i.e., a fisher is reporting different activities conducted over different numbers of hours). These likely represent separate fishing activities. Fishing records from a given CML containing differing information on hours fished were therefore separated into individual fishing activities by creating new trip IDs.

A relatively small number of fishing events contained multiple areas. There was no information to allow us to partition effort by area within those specific events; therefore, we assigned the area with the greatest catch of uku to the rest of the records within these fishing events. If the uku catch was the same between areas, we simply defaulted to the lowest area number (this was the case for only a few fishing events).

The final step was to calculate CPUE for each fishing event which was calculated as the total weight (kg) of uku caught per day (1948–2002 time series) or per hour (2003–2023 time series). This provided us with the following number of fishing events for each CPUE indices:

- 70,870 deep-sea handline events for 1948–2002
- 70,682 deep-sea handline events for 2003–2023
- 69,800 inshore handline events for 2003–2023
- 570,886 trolling events for 2003–2023

CPUE standardization models: All four CPUE indices were standardized using generalized linear models (GLM) and generalized mixed-effects models (GLMM). A

considerable proportion of trips in each CPUE data set was represented by zero catches: deep-sea handline 39% (1948–2002) and 79% (2003–2023); inshore handline 92% (2003–2023); and trolling 99.5% (2003–2023). Given the large proportion of zero catches, a delta-lognormal approach was used to standardize the CPUE indices. In this type of approach, the probability of catching an uku in a unit of fishing effort and the weight of the catch in a unit of effort (when uku are caught) are modeled separately. A binomial distribution was used to model the probability of catching an uku in a given trip (1 = caught, 0 = not caught) with the explanatory variables described previously and listed below using a logit link function. Positive-only CPUE data were modeled with the following explanatory variables using a lognormal response variable implemented by taking the natural logarithm of positive CPUE observations.

The *year* categorical variable was included in all models, given that our goal is to produce annual estimates of uku relative abundance. The following variables were also tested for inclusion in all models:

- Categorical fixed-effect: *month*, *area*, *area*month*, and *area*year* interactions
- Categorical random-effect: *fisher name*
- Continuous: *wind speed*, *fishing experience*, catch-composition *PC1*, *PC1²*, *PC2*, and *PC2²*

These variables were selected using Akaike's Information Criterion: $AIC = 2 \times \text{number of parameters} - 2 \times \log(\text{likelihood})$ (Burnham, 2004). We used a forward-selection process with a 0.1% AIC improvement threshold for variable inclusion (i.e., the model with the extra variable needs to have an AIC value 0.1% lower than the previous, simpler model). We chose this approach given that a large number of data points in CPUE data sets generally leads to large likelihood values and overly complicated models with little improvement to variance explanation (Maunder & Punt, 2004). Maximum likelihood parameter estimation was conducted using the GLM function in R version 3.4 for models excluding the *fisher name* random effect and using the *lmer* function in R package lme4 to run mixed-effect models using this variable.

Overall, models with the *area*year* interaction and the *fisher name* random-effect for the logistic regression models did not converge, similar to the Deep-7 analyses (Langseth et al., 2018). The *area*, *month*, *wind speed*, *PC1*, *PC1²*, *PC2²*, and *PC2* variables were selected in almost all models, while fisher experience was kept in two models and the *area*month* interaction was only kept for the deep-sea handline 1948–2002 model ([Table 4](#); see [Tables 5 to 7](#) for a list of models tested for all four CPUE indices).

Model diagnostics and index calculation: Pearson residual plots (lognormal regression) and quantile residuals plots (logistic regression; Dunn & Smythe, 1996) were used to assess model fit. Quantile-quantile plots were also used to assess normality for the lognormal models. The diagnostic residual plots for all 8 CPUE

standardization models (both logistic and lognormal regression models) showed no major deviation from normality and heteroscedasticity assumptions ([Figure 11 to Figure 18.](#)). Some patterning could be seen on the second-degree polynomial variables for *PC1* and *PC2* in some models, but these were not considered indicative of serious model assumption violations. Furthermore, residual plots for individual explanatory variables showed no obvious patterns.

After selecting the best regression models, we generated indices of relative abundance for the 4 CPUE data sets by (1) exponentiating the *year* effect of the lognormal models and (2) multiplying this value by the back-transformed expected logit value for each year for the logistic regression models (Maunder & Punt, 2004). For the lognormal models, we added half the square of the standard deviation of the *year* effect before exponentiating as a bias-correction factor. For the logistic models, we selected typical factor levels for the categorical variables and the median level for continuous variables to obtain expected logit values for each year (i.e., June for the *month* effect and area 331 for the *area* effect; Maunder & Punt, 2004). The variance of the indices was calculated using a Monte Carlo simulation by drawing random values for the *year* effects (lognormal models) and expected logit value per year (logistic models) derived from the standard deviation of the *year* effect in both models and fitting a lognormal distribution to the resulting data set. [Figure 19](#) and [Figure 20](#) present the resulting standardized CPUE indices of abundance vs. their corresponding nominal values. The final indices used in our SS model are presented in [Table 8](#).

Size frequency data

Size composition data were obtained from the FRS data set where commercial fishers are required to enter total number and weight of fish caught during individual fishing events. Individual fish measurements were generally not available since most records contained multiple ukus. However, it was possible to select fishing records where only a single uku was reported to build annual size frequencies for the dominant fishing gear (deep-sea handline). Other fishing gears, such as inshore handline and trolling, did not have sufficient single-fish reports to generate consistent size frequencies on an annual basis. The reported individual fish weights in pounds were converted to kilograms for consistency within the SS model. The effective sample size in each year was calculated by multiplying the number of weight measurements in each of the four regions of the MHI ([Figure 1](#)) by their respective regional weights. Regional weights were calculated as the proportion of total uku habitat (depth < 200 m) available in each region (Hawai'i: 0.23, Maui Nui: 0.58, O'ahu: 0.11, and Kaua'i-Ni'ihau: 0.08). This down-weighted the effective sample size in years where weight measurements came disproportionately from smaller sub-regions.

It was also possible to convert these individual weight measurements from the FRS data set to their corresponding lengths by using the following relationship:

$$L = \left(\frac{W}{\alpha}\right)^{\left(\frac{1}{\beta}\right)} \quad \text{Eq. 2}$$

where α is the scaling and β is the volumetric parameters of the typical length-weight relationship ($W = \alpha L^\beta$) (see [Table 1](#) for parameter values). These length data were not used in the SS model which used the weight data but were used to estimate selectivity for the remaining gear types (see *Selectivity* in the [Fishery Dynamics](#) section).

Fishery-independent diver survey

Starting in 2005, trained divers from the NOAA PIFSC have been conducting visual surveys around the MHI. The diver surveys were used to obtain a fishery-independent index of abundance in addition to the CPUE indices described above. An in-depth description of this survey program is available in Ayotte et al. (2015). Below is a brief description of the survey protocol.

Diver survey sites were randomly selected within strata defined by depth bins (shallow, 0–6 m; mid, 6–18 m; and deep, 18–30 m). All coastlines from all islands in the MHI were surveyed, except for the small, restricted island of Kaho‘olawe. For practical and safety reasons, surveys were limited to depths above 30 m. During a typical survey day, a NOAA ship deployed 3 to 4 small dive boats that sampled pre-assigned random sites along 10 to 12 miles of coastline. The daily starting location of the ship along different coastlines of the MHI was set systematically to cover as much of the shoreline as possible. At each site, stationary point counts (SPC) were implemented by two paired divers inside contiguous 15-m diameter cylinders that extended from the bottom to the surface. Divers first listed all observed fish species during an initial 5-minute period. The divers then went through this list, one species at the time, recording the number of individuals and estimating the sizes of all fish seen within the cylinder. Fish sizes were recorded as total lengths to the nearest cm. Fishes from species not listed during the initial 5-minute period but observed later in the survey were also recorded but classified in a different data category (i.e., non-instantaneous count). Divers were continuously trained between cruises in size estimation using fish cutouts of various sizes. Diver performance during research cruises was evaluated by comparing size and count estimates between paired divers.

Total numerical density estimates (individuals per 100 m²) were obtained by dividing fish counts in each survey by the survey area (353 m² from two 15-m diameter survey cylinders) and multiplying by 100. An individual survey consisted of the combined fish

counts from the two divers deployed at a single site. To obtain fishery-targeted numerical density, we multiplied total numerical density at a given size by uku-specific selectivity coefficients derived from the L_{S50} and L_{S95} parameters obtained from the Length-Based Spawning Potential Ratio (LBSPR) model for the trolling fishing gear. The overall fishery-targeted numerical density was obtained by (1) averaging site-level density estimates within a MHI sub-region ([Figure 1](#)) and (2) averaging all sub-region-level density estimates together using sub-region weights (the same as those used for the size-frequency effective sample size). Standard deviations were obtained by bootstrapping the diver survey data set by re-sampling survey sites within each sub-region. Fishery-targeted biomass density (kg per 100m²) was calculated directly from fishery-targeted numerical density by converting fish lengths to weight using the length-weight relationship ($W = \alpha L^\beta$, see [Table 1](#) for parameter values). The distribution of survey sites in each year is presented in [Figure 21](#).

It is important to note that the 2005 and 2007 surveys were conducted using a different diver observation method (belt transects). These were calibrated to the SPC method using a comparison data set collected during 2007–2009 cruises when both methods were implemented at all sites. The results of these analyses for uku showed that the positive-only belt transect density needed to be divided by 0.94 and that the logit-transformed observation probability needed to be subtracted by 0.8 to convert these data to SPC equivalent, which was done for the 2005 and 2007 surveys before being used in our SS model.

Population dynamics

This section describes the source of the various biological parameters used in our SS model, as well as some key assumptions related to population demographic structure ([Table 2](#)).

Spatial and temporal span: As discussed in the introduction, the uku stock was analyzed at the scale of the MHI as a single, well-mixed, population ([Figure 1](#), [Figure 3](#)). This assessment covers the 1948 to 2023 period with the first year matching the start of the FRS data collection program, which represents the oldest data available. The end year of 2023 was selected since this was the last complete year of data by the time we ran the current analyses.

Sex structure: There was no evidence of consistent sex-specific patterns related to growth (O'Malley, pers. comm.) or a sex ratio different from 1:1. Therefore, recruitment was split evenly between both sexes. Growth was treated similarly between males and females (see below). The only sex-specific parameter was maturity (see below), which was defined solely for females, given the spawning stock biomass and its relationship to recruitment was strictly related to mature female biomass (Methot & Wetzel, 2013).

Reproduction and recruitment: Maturity was defined as a function of length using a logistic relationship. Everson and Williams (1989) estimated that female uku reach 50% maturity at 44.8 cm and 95% maturity at 45.7 cm ([Table 1](#); [Figure 22](#)). The length at 50% maturity corresponds roughly to age 3 and the first mature female can appear as early as age 2. Fecundity (number of eggs per female) was set to be proportional to female body weight. Total reproductive output for the uku population was therefore set to be directly proportional to total mature female biomass.

Because catch and CPUE data were available annually, a single season per year was used in the base-case model. Therefore, the uku population was assumed to have one spawning and recruitment period at the beginning of each time step, which corresponds to the first month of each year. A standard Beverton-Holt stock-recruitment relationship was used in this assessment. The expected annual recruitment (R) was a function of spawning biomass (SSB) in a given year using the steepness (h) and virgin recruitment (R_0) parameters. Steepness is defined as the fraction of recruitment from a virgin population (R_0) when the spawning stock biomass is at 20% of its unfished level (SSB_0). Deviations around expected recruitment values were assumed to follow a lognormal distribution with standard deviation (σ_R) (Methot, 2000; Methot & Wetzel, 2013). These annual recruitment deviations were estimated based on the information available in the data and the central tendency that penalizes the log (recruitment) deviations in excess of σ_R . The σ_R parameter was fixed to approximate the expected variability of 0.39 based on an independent estimate of σ_R from the FishLife 2.0 R package (Thorson, 2019), derived from other snappers. The log of R_0 (virgin recruitment), annual recruitment deviations, and the offset for the initial recruitment relative to virgin recruitment, R_1 , were estimated in the base-case model for the 1948–2023 period. Early recruitment deviations, beginning 10 years before the main recruitment in 1948, were also included in the model. The estimation of early recruitment deviations allows for recruitment in early periods and the model to develop a non-equilibrium age structure without biasing recruitment estimates in the main period. Recruitment deviations are estimated on the log scale in Stock Synthesis. Consequently, the expected recruitments require a bias adjustment so that the resulting recruitment level on the standard scale is mean unbiased. The years chosen for bias adjustment and the maximum bias adjustment parameter value were obtained from the SS output with the program r4ss (Methot & Taylor, 2011). Recently, Lee et al. (2012) concluded that steepness is estimable for relatively low productivity stocks with good contrast in spawning stock biomass, given a correctly specified model. However, estimating steepness (h) within the assessment model for uku is likely to be imprecise and biased because the contrast in the spawning biomass over the assessment period is relatively poor. Given this, we followed the same approach as for σ_R and used an independent estimate of steepness from the FishLife 2.0 R package (Thorson, 2019) derived from other snappers ($h = 0.81$; [Table 1](#)). Nevertheless, we note here that this steepness estimate is subject to considerable

uncertainty and further work is needed to evaluate steepness for uku. Sensitivity runs on steepness were used to evaluate the model's sensitivity to this parameter in the 2020 assessment.

Age and growth: The maximum age in the model was 32 years based on the maximum observed age for uku in the Hawaiian Islands (O'Malley, pers. comm.). This value was obtained from a large length and age data set ($n = 419$) collected in both the MHI and NWHI and is the oldest recorded value for this species. Note that the 32-year age bin also served as the accumulator for all potential older ages in the SS model.

We modeled the relationship between fork length (FL) and age with a von Bertalanffy growth function using the Schnute (1981) parameterization:

$$L_2 = L_{Amax} + (L_1 - L_{Amax})e^{-K(A_2 - A_1)} \quad \text{Eq. 3}$$

where L_1 and L_2 are the sizes associated with ages, A_1 and A_2 , respectively; L_{Amax} is the length at the maximum age, A ; and K is the growth coefficient. In this assessment, L_1 and A_1 were fixed at 51.6 cm at age 3. The L_{Amax} and K parameters were fixed at 76.5 cm for the maximum age of 32 and 0.136 yr^{-1} ([Table 1, Figure 23](#)). The coefficients of variation (CV) parameters that describe the variability of individual fish length at specific ages were fixed at 0.12 for both younger (ages ≤ 3) and older (ages > 3 years) fish. These parameters were obtained directly from the von Bertalanffy model fitted to the raw age-length data. SS assumes linear growth between ages 0 and A_1 .

Natural mortality: We obtained an estimate of natural mortality (M) equal to 0.10 using the procedure of Alagaraja (1984), similar to Hewitt & Hoenig (2005). This was recently updated by Then et al. (2014), assuming that 4% of a cohort survives to the observed maximum age (a_λ) by which we defined longevity:

$$M = \frac{-\log(0.04)}{a_\lambda} \quad \text{Eq. 4}$$

We used the 4% cohort survivorship value based on the analyses of Nadon et al. (2015) which showed that this is an appropriate survivorship value for coastal fishes in Hawai'i. Without independent estimates of M , we had to rely on this longevity-based approach. Although there are other data-poor methods for estimating natural mortality involving other parameters (e.g., K , L_{inf} , L_{mat} , water temperature), two recent scientific papers on the subject clearly suggest that longevity-only methods are better performing (Kenchington, 2014; Then et al., 2014). It is important to consider the potential difficulty in obtaining a representative longevity value in heavily exploited stocks. However, our estimate comes from a combined data set that included individuals from the remote

NWHI. We also verified the validity of our M estimate by estimating total mortality (Z) in the NWHI, which can be assumed to be close to M given the exceptionally low fishing mortality in this region. For this analysis, we obtained size data from the FRS data set that were available for the NWHI (before it was closed to fishing) and compared the observed mean length with the expected mean length when Z is set to 0.1. We observed a NWHI mean length of 70.1 cm and the model predicted a value of 70.8 cm for an $M = 0.1$, which therefore supports our M estimate.

Weight-at-length: We used a non-sex-specific weight-length relationship to convert between catch-at-weight and weight-at-length data ($W = \alpha L^\beta$) within the SS framework. The α scaling parameter was 0.0118 and the β volumetric parameter was 3.043, obtained from the MHI uku samples collected in Sundberg and Underkoffler (2011) ([Table 1](#)).

Initial conditions: It was necessary to make some assumptions about the structure of the stock before the start of the main population dynamics period (pre-1948). Typically, two approaches are used to achieve this assumption. The first approach starts the model as far back as necessary to satisfy the notion that the period before the estimation of population dynamics was in an unfished or nearly unfished state. In this assessment, the second approach was used in which initial conditions were estimated assuming equilibrium catch. The equilibrium catch is the catch taken from a fish stock when it is in equilibrium with fishery removals and natural mortality balanced by stable recruitment and growth. The initial fishing mortality rates in the assessment model that remove these equilibrium catches were estimated to allow the model to start at an appropriate depletion level.

Fishery dynamics

Selectivity: SS needs gear-specific selectivity parameters to relate catch, size frequency, and CPUE observations to population dynamics. Size structure patterns in the uku FRS data set suggested that selectivity was related to body length and followed a simple logistic pattern that was time-invariant. For example, [Figure 24](#) shows no significant changes in selectivity in different decades for deep-sea handline using size data from the FRS data set. Selectivity curves were gear-specific and assumed to be only a function of size for the four commercial gear types (deep-sea handline, inshore handline, trolling, and “other”) and the recreational sector. The following logistic equation relating selectivity (S) to length was used in the SS model:

$$S_L = \frac{1}{1 + e^{\left[-\ln(19) \frac{L - L_{S50}}{L_{S95} - L_{S50}} \right]}}$$

Eq. 5

where L_{S50} and L_{S95} are the sizes at 50% and 95% selectivity, respectively.

By providing SS with size frequencies from the deep-sea handline fleet, the model was able to estimate size selectivity parameters for this fishing gear. However, SS still required selectivity parameters for the other commercial fishing gears (inshore handline, trolling, and “others”) as well as for the recreational sector. To provide these parameters, we selected one-fish records from the FRS data set for the inshore handline, trolling, and “other” fishing gears and fitted an LBSPR model to these data sets. LBSPR is a relatively simple modeling approach that fits abundance-at-length data by estimating two selectivity parameters (L_{S50} and L_{S95}) and fishing mortality (Hordyk et al., 2016). We also had to group length data by decade instead of by year due to the limited sample size available for these gear types. We did not observe significant differences in length structure between decades and, therefore, selected the most recent data (2008–2023) to fit the LBSPR model and obtain the required selectivity parameters. Lastly, for the recreational sector, we obtained length observations from the HMRFS interviews for 2003–2023 ($n = 244$) and used the LBSPR model to obtain selectivity parameters. All L_{S50} and L_{S95} parameters for these gear types were entered directly in SS as fixed values. The LBSPR model fits are presented in [Figure 25](#) and the associated parameters are in [Table 1](#). The selectivity curves for all fishing gears and sectors are presented in [Figure 26](#).

In the SS model, selectivity curves for CPUE indices were set to mirror their corresponding catch fleets.

Catchability: Catchability (q), the proportion of the stock captured per unit of effort, was estimated (solved analytically) for each CPUE index assuming these were proportional to vulnerable biomass with a scaling factor of q . It was assumed that q was constant over time for each index.

Fishing mortality: Fishing mortality was estimated as a full parameter providing a continuous annual estimate of F for each fleet.

Data observation models: The current assessment model fitted three data components: (1) total catch, (2) CPUE and diver relative abundance indices, and (3) size composition data. SS estimates population and fishing parameters by minimizing the negative log-likelihood of an objective function from the provided input data components or assumptions.

The reported FRS commercial total catches were assumed unbiased and relatively precise and were fitted assuming a lognormal error distribution with standard error (SE) of 0.05 for deep-sea handline and 0.10 for the other three gears (inshore handline, trolling, and “other”). As discussed in previous sections, the recreational fishery catches were uncertain with a CV of 0.4. The high CV on recreational catch and the estimation

of fishing mortality as individual annual parameters (i.e., F method = 2) gave SS a fair amount of freedom to adjust the recreational catch data within the model.

The relative abundance indices were assumed to have lognormally distributed errors with SE in log space, which is approximately equivalent to CV (SE/estimate) in natural space where the $\log(\text{SE}) = \sqrt{\log_e(1 + CV^2)}$. The estimated CVs of each index in this assessment are shown in [Table 8](#). However, the reported CVs for the abundance indices only capture observation errors within the standardization model and do not reflect process errors that are inherent in the link between the unobserved vulnerable population and observed abundance indices and are related to the number of data points in the analysis; larger data sets have smaller CVs. Total observation errors for abundance indices are typically assumed to range from 0.1 to 0.4 (C. Francis, 2011; C. R. Francis et al., 2003), while Francis et al. (2003) suggested that the portion of the observation variance that accounts for the inter-annual variation in catchability can typically range from 0.15 to 0.2. For the uku assessment, all indices had mean CVs below 0.2 and we therefore added 0.10 to the CVs of all indices, including the diver survey (in order to maintain its relative weight in the model) to increase all CVs to over 0.2.

The size composition data were assumed to have a multinomial error distribution with the error variance determined by the effective sample size (see [Size Frequency section](#)). In this stock assessment, size measurements of fish were assumed random samples of fish from the entire population. Size compositions for the deep-sea handline were included (see Size frequency) with the variance determined by the observed sample size and frequencies.

Model diagnostics

Model diagnostics were used to assess issues associated with convergence, model structure, parameter misspecification, and data conflicts in the base-case model. The following diagnostic tools were employed in this assessment: (1) model convergence tests, (2) R_0 likelihood profiles, (3) residual analyses, and (4) a retrospective analysis.

Model convergence

Model convergence was assessed using several criteria. The first diagnostic was whether the Hessian (i.e., the matrix of second derivatives of the likelihood with respect to the parameters) inverts. The second measure is the maximum gradient component, which, ideally, should be low (< 0.0001 is a standard value). The third diagnostic involved altering or jittering the starting values of the parameters to evaluate whether the model converges to a global solution rather than a local minimum (typically by a maximum of 10%).

Residual analysis

Model fit to the CPUE and diver survey abundance indices were assessed by looking at the residuals of the observed versus predicted CPUE indices by gear. We looked for any signs of conflicting information between different time series and autocorrelated patterns in a given period. We also looked at the overall root mean square error (RMSE) for each index, with values below 0.3 suggesting a good model fit to the abundance index data (Winker et al., 2018).

Likelihood profile on virgin recruitment (R_0)

This diagnostic was implemented by sequentially fixing the equilibrium recruitment parameter, R_0 , on the natural log scale, $\log(R_0)$, to a range of values (step size of 0.1). The maximum likelihood estimates of all other model parameters were obtained from the SS output.

The relative change in negative log-likelihood units over the range of fixed values for $\log(R_0)$ (the R_0 profile) was compared among the SS model likelihood components for CPUE, size composition, and recruitment deviations using two diagnostic tests. First, a relatively large change in negative log-likelihood units along the R_0 profile was an indication of a relatively informative data source for that particular model. Second, a difference in the location of the minimum negative log-likelihood along the R_0 profile among data sources was an indication of either conflicts in the data or model misspecification, or both (Carvalho et al., 2017). Ideally, catch and abundance indices should be the primary sources of information on the population scale in a model (Lee et al., 2014).

Retrospective analysis

A retrospective analysis is a way to detect bias and model misspecification (Hurtado-Ferro et al., 2015). A retrospective analysis as described in Carvalho et al. (2017) was applied to the base-case model. The diagnostic was implemented here by sequentially eliminating the five most recent years of data from the full stock assessment model (a 5-year “peel”) and then re-estimating all stock assessment model parameters from each peel and from the base-case model. Although measuring retrospective pattern has proved challenging, the most commonly used metric is the rho (“ ρ ”) statistic proposed by Mohn (1999) which measures the relative difference between an estimated quantity from an assessment with a shorter time series and the same quantity estimated from the full time series. Interpreting the Mohn’s rho statistic is subjective. However, a rule of thumb was proposed by Hurtado-Ferro et al. (2015), which states that values of Mohn’s rho falling outside the range of -0.15 to 0.20 indicate the presence of a noticeable retrospective pattern for long-lived species.

Assessment strategy

During the model development phase, we first focused on improving the fits to the observed data (CPUE and size-frequency) and obtaining a stable likelihood profile with an informative minimum. A key part of the early exploration involved assessing which parameters could be estimated from the input data and which needed to be fixed. The base-case was chosen as the version of the model with the best fit to the data while also making reasonable assumptions about model parameterization. From this base-case, a set of one-off sensitivities was chosen to test the impact on model results of choices made regarding key fishery and biological parameter assumptions. For the current assessment update, we did not revisit these sensitivities analyses, which are found in Nadon et al. (2020). These sensitivities were as follows:

Life history parameters: We tested alternative specifications for L_{Amax} , L_{mat50} , and M . Our estimates of L_{Amax} and L_{mat50} came from robust local studies and while there is a high degree of confidence in these parameters, these parameters are known to disproportionality impact model outcome. Natural mortality is a parameter that is hard to estimate and is influential in most assessments. We tested values representing plus and minus 10% adjustments for each of these parameters (L_{Amax} : 68.9 cm and 84.2 cm; L_{mat50} : 40.3 cm and 49.3 cm; M : 0.09 and 0.11). We also tested the use of age-specific M values under the Lorenzen natural mortality option in SS.

Recruitment: Steepness is a challenging parameter to estimate from observations or within a stock assessment model, but it tends to be very influential on assessment predictions given that it mediates the relationship between spawners and recruits. We examined the impact of the different values of steepness of 0.73 and 0.89 in comparison to the 0.81 used in the base-case. We also tested values representing positive and negative 10% adjustments for the recruitment deviation parameter σ_R (0.35 and 0.43). Finally, we tested an alternative model where σ_R was estimated within SS.

Catch scenarios: Current and historical recreational catch estimates are, in general, highly uncertain particularly for coastal stocks in Hawai'i. While our base-case model did consider the uncertainty in recreational catch (CV = 0.40), we also wanted to test our model's sensitivity to a few alternative catch histories. In the first alternative scenario, we estimated recreational catch as a fixed ratio of commercial catch, which is known with greater certainty. This ratio was calculated by dividing the yearly HMRFS recreational catch estimates by their corresponding commercial catch between 2003 and 2023 (the years for which recreational catch is available) and averaging these yearly ratios to obtain an overall ratio. For the second and third alternative scenarios, we adjusted the pre-2003 recreational catch by a positive and negative factor of 30%.

CPUE standard error: The standard errors estimated from the CPUE standardization model were small especially for deep-sea handline (mean CV 0.065). This forces the assessment model to be more constrained by the CPUE index. We allowed the model to obtain a more relaxed fit to the CPUE in the base-case by adding a constant of 0.10 to all CPUE indices CVs (see [Population Dynamics section](#)). However, we also tested the sensitivity of predictions using the standard error estimated from the CPUE standardization model directly. Finally, we tested our model outcome when the CPUE information was limited to deep-sea handline for the 1948–2002 period combined with either deep-sea handline, inshore handline, or trolling CPUE indices individually for the 2003–2018 period.

Size frequency data: Size frequency data provide a highly informative source of information on trends in fishing mortality that can often overshadow the CPUE indices. We tested the robustness of our model results to a lower importance of this source of data by reducing its weight in the likelihood calculation to 10% (i.e., $\lambda = 0.1$).

Alternate CPUE timelines: Given the paucity of inshore handline data before 2003, we decided not to include a 1948–2003 CPUE time series for this fishing gear. However, we tested extending the inshore handline CPUE data to 1992 (the oldest year with sufficient data) in two scenarios: (1) including a 1992–2018 time series using daily effort units and (2) including a 1992–2002 time series using both daily (pre-2003) and hourly (post-2003) effort units. Similarly, we treated the deep-sea handline CPUE data as a single 1948–2018 time series using daily effort units. Finally, we tested the sensitivity of our results to an alternate model specification that included both the 1992–2018 and 1948–2018 time series for inshore and deep-sea handline gears, as well as using the Lorenzen M estimate.

Alternate model start year: The starting year of a model can sometimes have a significant impact on underlying population dynamics. This is especially true if the early years contain data of lesser quality that may skew certain population parameter estimates (e.g., R_0 , recruitment deviations). We tested the impact of starting our SS model in 1970 as opposed to 1948 to ensure that it was not sensitive to the first few decades of data.

Alternate model specifications: We tested a few alternate specifications to our SS model to ensure that they did not have drastic impacts on our results. We ran alternative models with (1) time-varying catchability (q) as opposed to constant catchability, (2) effective sample size for the size-frequency data as suggested by SS, and (3) no diver survey data.

Reference points

The Fishery Ecosystem Plan for Hawai'i (WPRFMC, 2009), following guidelines from Restrepo et al. (1998), uses a Maximum Fishing Mortality Threshold (MFMT) defining overfishing as the fishing rate that leads to an equilibrium biomass that is below the minimum stock size threshold (MSST) of $(1-M) \times SSB_{msy}$. In our model, SSB_{msy} is defined as the equilibrium spawner biomass at F_{msy} , where F_{msy} is the F that produces maximum equilibrium catch (i.e., MSY), which is found through an iterative search algorithm in SS (Methot & Wetzel, 2013). Note that when SSB falls below the MSST, management of the fishery would shift to using a rebuilding control rule, as represented in [Figure 27](#).

Catch projections for 2020–2026

Stock projections for uku were conducted using the age-structured projection model software AGEPRO (Brodziak et al., 1998). Stochastic projections were conducted using results from the base-case model to evaluate the probable impacts of constant catch quotas on future spawning stock biomass and yield. For the projections, a series of constant catch quotas were run from 130 mt to 260 mt in five metric ton increments to provide the probability of overfishing in each year. We fitted a polynomial model to these data (i.e., probability of overfishing as a function of fixed catch) to obtain a higher resolution overfishing risk table.

Initial conditions for the stochastic projections were based on the estimated initial population size-at-age in the year 2023 from the base-case model. A total of 100 bootstrap replicates of the 2023 uku population size-at-age were calculated in SS to characterize the uncertainty in the initial population size. In each projection, 1000 total simulations were run for each bootstrap replicate to characterize the effects of process errors in future recruitment, life history, and fishery parameters. This gave 100,000 total simulated trajectories to evaluate the central tendency and variability of population and fishery quantities of interest (SSB and probability $F > F_{msy}$) in each projection.

Recruitment for the stochastic projections was based on three hypotheses about future recruitment processes. The first hypothesis was that future recruitment would be similar to recent short-term recruitment. This hypothesis was based on the observation that recruitment estimates have remained relatively high since 2000 and the expectation that this trend may continue in the near future (Thorson et al., 2014). In particular, the short-term recruitment scenario was based on resampling the empirical recruitment distribution of recruitment observed during 2005–2022. The AGEPRO recruitment sub-model 3 resampled the estimated recruitment from those 18 years with equal probability. Under the short-term recruitment scenario, the average recruitment was 108,588 age-1 fish with a CV of 26%. For each projection, 20% of the simulations were

run using the short-term recruitment scenario. The second hypothesis was that future recruitment would be similar to the long-term recruitment pattern. The long-term recruitment scenario was based on resampling the empirical recruitment distribution of recruitment observed during 1948–2004. The AGEPRO recruitment sub-model 3 resampled the estimated recruitment from those 57 years with equal probability. Under the long-term recruitment scenario, the average recruitment was 79,439 age-1 fish with a CV of 28%. For each projection, 20% of the simulations were run using the long-term recruitment scenario. The third hypothesis was that recruitment would follow a Beverton-Holt stock-recruitment curve with lognormal deviations, which is the same assumption made in our SS model. AGEPRO parameterizes the Beverton-Holt stock-recruitment relationship as:

$$R = \frac{\alpha \cdot SSB}{\beta + SSB} \cdot e^w \quad \text{Eq. 6}$$

where,

$$\alpha = \frac{4hR_0}{5h - 1} \quad \text{and} \quad \beta = \frac{B_0(1 - h)}{5h - 1} \quad \text{Eq. 7 and 8}$$

as defined in Brodziak et al. (1998). Our alpha ($\alpha = 96.1$) and beta ($\beta = 93.2$) parameters were derived from the steepness of $h = 0.81$, virgin recruitment of $R_0 = 90.5$ thousand recruits, and virgin SSB (i.e., SSB_0) of 1,495 mt, all derived from our SS model. Since AGEPRO is a pooled-sex model while SS uses female-only SSB , we multiplied our β parameter by 2 to convert our female-only S-R curve to a male+female equivalent curve in AGEPRO ($\beta_{\text{AGEPRO}} = 186.4$). We also doubled our SSB_{msst} threshold in AGEPRO from 355 mt to 710 mt to control for this mismatch in model structure. The recruitment and SSB scaling parameters in AGEPRO were both set to 1000 given that our S-R relationship is in units of 1000s of fish and metric tons, while AGEPRO operates in units of individual fish and kilograms. The variability around the stock-recruitment curve was set to be equal to $\sigma_R = 0.39$ (inputted as a variance of 0.152) also from the SS base model. For each projection, 60% of the simulations used the stock-recruitment relationship to estimate future recruitment.

Fishery selectivity for the stochastic projections was based on the estimated selectivity in the base-case model. All of the gear-specific fishery selectivities by size were logistic in the base-case model; of these, only the commercial deep-sea handline fleet was estimated in the base-case model. The catch data in AGEPRO were entered by individual fishing gear with their respective selectivity-at-age values provided from SS. For stochastic projections, fishery selectivity-at-age was sampled with a multiplicative lognormal process error with a mean of unity and a CV of 10% to represent uncertainty about future selectivity.

Life history parameters for the projections were based on the same values as were used in the base-case model. This included age-invariant natural mortality, maturity at age, and mean spawning weights at age. Mean fishery catch weights-at-age were based upon the catch from each fleet as for selectivity-at-age inputs. For stochastic projections, life history parameters at age were sampled with a multiplicative lognormal process error with a mean of unity and a CV of 10% to represent uncertainty about future values except natural mortality and maturity at age, which were sampled with a CV of 1%.

Catch-based projections ran from 2024 to 2031; total catch in 2024 was set to be the status quo catch (average in 2021–2023, 111 mt) based upon the assumption that 2024 catches would likely be similar to recent catch amounts. Annual catches during 2025–2031 were set to the projected catch. The maximum catch was set to give a 50% probability of overfishing in the final year of the projections.

Section II: Results

Base-case model

Results from the base-case model provided estimates of biological reference points for u_k and annual estimates of total stock biomass, spawning stock biomass, recruitment, and fishing mortality, along with a Kobe plot indicating stock status over time. In this section, we first discuss model fit and diagnostics before presenting the results of the base-case model and projections.

Model diagnostics

Model convergence: All the key estimated parameters in the base-case model were within the set bounds. The final gradient of the model was less than 0.0001, and the Hessian matrix for the parameter estimates was positive definite, which indicated that the model had converged to a local or global minimum. Two minor parameters had gradients slightly above the 0.0001 limit, namely the F estimates in two years (1950 and 1956) for the trolling fleet, at -0.0004 and 0.00011, respectively. This was judged to be of minimal concern given the small amount of catch associated with this fleet and those years specifically. Results from 150 model runs with different random initial starting values for estimated parameters using the internal “jitter” routine in SS supported the result that a global minimum was obtained (i.e., there was no evidence of a lack of convergence to a global minimum, [Figure 28](#)).

Likelihood profiling: [Figure 29](#) presents the likelihood profile of the logarithm of the unfished recruitment parameter ($\log[R_0]$) for each data component. Detailed information on changes in negative log-likelihoods among the various fishery data sources is shown in [Tables 9, 10, Figure 30](#).

Changes in the likelihood of each data component indicate how informative that component is to the overall estimated model fit. Our results showed that the estimation of the catch was the most informative likelihood component followed by the estimated recruitment deviations, fit to the size-frequency data, and fit to the abundance indices, respectively. There were also slight differences in the minimum negative log-likelihood along the R_0 axis among data likelihood components with abundance indices converging on $R_0 = 4.4$ and the catch and size-frequency components converging on $R_0 = 4.5$.

The change in negative log-likelihood of abundance indices was relatively small, particularly in the lower limit of the population scale. The deep-sea handline 1948–2002 index showed the largest changes in negative log-likelihood values across values of R_0 among abundance indices (max Δ 6.0), followed by the trolling index (max Δ 3.9), and

the inshore handline index (max Δ 1.5). The 2003–2023 deep-sea handline index and diver surveys had lower likelihood changes (max Δ 1.2 and max Δ 0.3, respectively; [Table 10](#), [Figure 30](#)).

Goodness-of-fit of abundance indices: Goodness-of-fit diagnostics are presented in [Table 11](#) and plots of predicted and observed CPUE by gear type for the base-case model are shown in [Figure 31](#). Results showed that both deep-sea handline indices had an RMSE less than 0.2, indicating that the model fit this CPUE index well. The inshore handline and trolling indices had an RSME of 0.18 to 0.34, and the diver survey had an RSME greater than 0.3, suggesting that fits to these data were poorer.

Residuals analysis of size composition data: Comparisons between observed and expected mean values of size composition data (Francis, 2011) are shown in [Figure 32](#). It demonstrated that the expected mean weight values passed within 95% confidence intervals for all years.

There appeared to be little pattern in the residuals of the size composition data ([Figure 33](#)) although there were more small fish than expected in the 1970s to early 1980s, and there were higher positive residuals for the larger size classes. Assuming the standardized residuals were normally distributed, 95% of the measurements would fall within two standard deviations of the mean. Some Pearson residuals did not meet this criterion ([Figure 33](#)); in particular, very small fish in 1984–1986 and fish over 10 kg. Overall, the model fitted the weight modes in size composition data fairly well using the imputed effective sample sizes ([Figure 34](#)).

Stock assessment results

Given the uncertainty associated with the recreational catch (CV = 0.4), SS had substantial freedom in adjusting the observed values of catch in the fitting process. This resulted in some differences between the observed versus expected recreational catch. As shown in [Figure 35](#), SS greatly reduced the high peaks in recreational catch seen in 2005 and 2012, which were likely sampling outliers. Interestingly, SS did not dramatically adjust the pre-2003 recreational catch estimates; only a short period between 1972 and 1985 showed some positive adjustment in the catch estimates ([Figure 35](#)). [Figure 36](#) and [Figure 37](#) present catch and fishing mortality broken down by fishing gear, respectively.

Estimates of population biomass (i.e., age 1 and older fish at the beginning of the year) increased from 1,475 mt in 1948 to a high of 1,688 mt in 1954. They then varied while generally declining to a low of 1157 mt in 1996. The population biomass then increased to its largest value at around 2093 mt in the final 3 years of the current stock

assessment time horizon ([Table 12](#), [Figure 38](#)). Over time, population biomass has generally varied between 1,300 mt and 1,800 mt ([Figure 38](#)).

Female spawning stock biomass (*SSB*) oscillated between 600 and 800 mt until the 1990s. *SSB* reached its lowest level of 550 mt in 1997 then increased to a high of 1028 mt in 2021 ([Table 12](#), [Figure 39](#)). The time-series of *SSB* during the past decade averaged 957 mt or 64% of unfished *SSB* (1495 mt). Overall, *SSB* was relatively stable until the 2000s when it began increasing and reached a maximum around 2018. However, the *SSB* seems to have plateaued and has been slightly declining since 2020.

The estimated recruitment and stock-recruitment relationship were generally consistent with the biology of the stock and assumptions in the base-case model. The recruitment deviations from the expected spawner-recruit curve appeared to be relatively well estimated and consistent with the expected distribution of recruitment deviations ([Figure 40](#)). Expected recruitment from the stock-recruitment relationship showed little evidence of a relationship between *SSB* and recruits ([Figure 41](#)). The change in recruitment bias was consistent with expectations and accounted for in the base-case model ([Figure 42](#)). Recruitment (age-0 fish) estimates indicated a long-term fluctuation around a mean of approximately 80,000 with periodic strong recruitments (over 150,000 age-0 fish; [Table 12](#), [Figure 43](#)). The model estimated high recruitments in 1970–1971, 1982, 1988, 2002–2003, 2010–2011, and 2013–2014. Additionally, recruitment was above average and increased between 2002 and 2018 but has decreased to its long-term average since then (~80,000 recruits). While the overall pattern of recruitment from 1975 to 2017 was variable, there was an apparent increasing trend in recruitment strength over time ([Table 12](#), [Figure 43](#)).

Over the assessment period, estimated fishing mortality (arithmetic average of F for ages 5–30) varied around 0.05 per year until the 1970s when it increased to a high of 0.17 per year in 1988 and then declined to 0.05 in the late 2000s. It has since varied between 0.05 and 0.06 per year ([Table 12](#), [Figure 44](#)).

Biological reference points: Biological reference points were computed from the SS base-case model. Based upon the Fishery Ecosystem Plan for Hawai'i (WPRFMC, 2009), F_{MSY} and B_{MSSST} were used to assess relative stock status ([Table 13](#)). The point estimate of the maximum sustainable yield (MSY) was 111 mt. The point estimate of the female spawning stock biomass to produce MSY (SSB_{MSY}) was 394 mt and SSB at the minimum stock size threshold (SSB_{MSSST}) was 355 mt. The point estimate of F_{MSY} , the fishing mortality rate to produce MSY on age-5 to age-30 fish, was 0.14, and the corresponding equilibrium value of spawning potential ratio at MSY (SPR_{MSY}) was 31%.

Stock status: Compared to the established FEP reference points, the current spawning biomass (2021–2023 average) was 280% of SSB_{MSSST} , and the current fishing mortality

(average for ages 5–30 in 2021–2023) was 36% of F_{MSY} . The Kobe plot indicated that the main Hawaiian Islands uku stock is not currently overfished and is not subject to overfishing relative to the FEP reference points ([Table 13](#), [Figure 45](#)).

Retrospective analysis: A retrospective analysis of the base-case uku stock assessment model was conducted for the last 5 years of the assessment time horizon to evaluate whether there were any strong changes in parameter estimates through time. The results of the retrospective analysis are shown in [Figure 46](#). The trajectories of estimated spawning stock biomass (as SSB/SSB_{MSST}) and fishing mortality (as F/F_{MSY}) showed little pattern when removing the most recent years of data ([Figures 46, 47](#)). The estimated Mohn's rho for SSB was -0.01 , which was between -0.15 and 0.20 , and consequently indicated that the retrospective pattern was relatively small (Hurtado-Ferro et al., 2014; Carvalho et al., 2017). The Mohn's rho for recruitment and fishing mortality were -0.001 and 0.05 , respectively.

Projections

The 7-year constant catch projection scenarios showed the distribution of outcomes in the probability of overfishing, spawning stock biomass, fishing mortality rates, and the probability of $SSB < SSB_{msst}$ that would occur under various catch scenarios in the MHI during 2025–2031. Projections indicate that practically none of the catch scenarios explored would result in a probability of being overfished ($SSB/SSB_{MSST} < 1$; [Table 14](#), [Figure 48–Figure 51](#)). [Figure 52](#) shows the evolution of F and SSB (and their associated overfishing and overfished probabilities) through time at various constant catch levels.

Section III: Discussion

This report presents the 2024 update of the 2020 benchmark stock assessment of uku in Hawai'i. The 2020 benchmark model covered the period from 1948 to 2018, while the current model extends this period to 2023. Using the Stock Synthesis 3.30 software, we incorporated all available data sources related to the main Hawaiian Islands uku population into a single analytical framework. These data sources included both recreational and commercial catch data, catch-per-unit-effort indices from three main fishing gears, size-frequency data, abundance data from diver surveys, and local life-history parameters. The main conclusions from this integrated assessment update are as follows. As of 2023, the last year with data in our current model,

- the uku stock is not currently overfished: $SSB/SSB_{MSS} = 2.8$ (from 2.7 in 2018), and
- overfishing is not currently occurring: $F/F_{MSY} = 0.57$ (from 0.36 in 2018).

Furthermore, our analyses suggest that the uku stock was never in an overfished state during the period covered in this assessment (1948–2023), but that overfishing may have occurred for two years, in 1988 and 1989.

The general trend of population abundance for uku in Hawai'i has been of a steady decline from 1948 to the mid-1990s, followed by a 10-year period of stability, and an important increase in biomass from 2005 to 2017 (Figures 38, 39). The decrease in catch and fishing mortality after 1988 likely explains some of this population increase, but most of it is modeled in Stock Synthesis as a period of strong positive recruitment starting in the early 2000s (Figures 40, 43). Of note, another spike in recruitment in the early 1980s led to a notable peak in catch and CPUE in 1988 which is still remembered by fishers and scientists (D. Kobayashi and others, pers. comm.; Ayers, 2022). It is not clear what could be driving this recent trend in strong recruitment. Besides global climate change, which has led to slowly increasing sea surface temperatures around Hawai'i (<http://hahana.soest.hawaii.edu/hot/>), this recent trend in recruitment did not correlate with the Pacific Decadal Oscillation (PDO) index, the Southern Oscillation Index (SOI), or the Oceanic Niño Index (ONI). An alternative explanation for the positive recruitment trend could be that the increase in CPUE that drives this pattern in our model was not due to an increase in uku abundance but to an increase in uku targeting (and catchability) related to improvements in fishing techniques. For example, fishers can deploy deep-sea handlines in a configuration that targets Deep-7 bottomfish versus uku, even though both gear configurations would be categorized as “deep-sea handline” in the reporting system. An increased interest in uku could thus lead to fishers deploying the uku gear configuration more often and thus lead to an increase in CPUE that is not related to an increase in abundance. However, we did control for species targeting in our CPUE standardization using the DPC approach (Winker et al., 2013). Furthermore, the same increasing CPUE pattern is seen in all three main fishing gears used to catch

uku ([Figures 19, 20](#)), and the diver survey index does not contradict this pattern ([Figure 31](#)). These observations suggest that the increase in population biomass between 2005 and 2017 may be real and not an artifact related to targeting or other confounding variables. It is important to note that this pattern of increasing CPUE (and biomass) reached a plateau as of 2017, and spawning biomass has been stable at around 1,000 mt since then. The main deep-sea handline CPUE index is stable, as is the trolling index. However, the **inshore handline index is showing a decline**; it will be important to keep track of these indices as we prepare to work on the next benchmark assessment.

Changes from the 2020 benchmark assessment

In the 2020 benchmark report, we showed that our base model was robust to uncertainty in the data and input parameters (Nadon et al., 2020). We tested model sensitivity to (1) different recreational catch time series; (2) different CPUE time series implementations (e.g., extending inshore handline to 1992, removing diver surveys, single deep-sea handline index for the entire period); (3) different life history parameters (e.g., recruitment deviations, steepness, Lorenzen M); and (4) starting the model in 1970. **We did not re-run these sensitivity analyses for the current assessment update.** We instead focus here on the impact of the new way of calculating recreational catches on our model results.

As previously explained, we calculated recreational catches following the approach used in the latest Deep-7 assessment (Syslo et al., 2024) to **correct for a negative bias in fishing effort associated with a decline in phone landlines**. This approach applied a multiplier to the HMRFS catches that linearly increased from “1” in 1999 to either “2.33” for boat catches or “2.89” for shore catches in 2017 ([Figure 8](#)). This increased the HMRFS catches from 1948 to 2016 compared to the previous assessment (**post-2017 catches are identical since these are derived from the mail effort surveys**). This also removed a spurious increase in fishing effort post-2017 ([Figure 9](#)). The overall impact of this change was to **scale the uku population biomass and overfishing limit (OFL) higher by about 27%**. Specifically, the OFL increased from 150 mt to 190 mt when using 2029 as the end year for the projections. For both models, the stock status in 2023 are almost identical, as show in [Figure 53](#). This figure also shows that, without this change to recreational catches, the current model would have been very similar to the 2020 benchmark for both population status and scale.

Future directions

The Stock Assessment Program at the PIFSC will continue exploring and improving the use of Stock Synthesis for domestic stocks, including integrating potential new sources of data.

One major source of uncertainty in our model is the catch data, for both the recreational sector and, to a lesser extent, the commercial sector. For the commercial sector, there are concerns with under-reporting when catches are not sold through traditional vendors (e.g., social media) or kept for private consumption (Ayers, 2022). For the recreational sector, there are concerns with the high variability associated with HMRFS catch estimates, but also that these may be overestimated (this sector currently represents 61% of the boat-based catches, on average). To address some of these concerns, PIFSC and the State of Hawai'i, in collaboration with fishers, will be launching an experimental mail survey in late 2024. This monthly survey will seek to estimate total uku boat-based catches by mailing forms to a sample of boat owners with questions regarding their uku catches, number of trips, and fishing gear deployment in the previous month. The sampling universe for this survey is the 9,500 boats that are registered in the current DOBOR vessel registry and identified as used for fishing, combined with information from the CML and bottomfishing registries. The results from this survey will be used to evaluate the catch estimates coming from both the FRS and HMRFS programs. Beyond this local project, it is important to note that MRIP is currently conducting a nationwide study on the impact of question order on their mailed effort survey forms, in addition to increased sampling frequency, on fishing effort estimates. We are expecting preliminary results by the end of 2024, with a possible implementation of a correction to 2025 and prior years' effort estimate, followed by a new sampling design for 2026 (contingent upon a successful review of the updated Fishing Effort Survey (FES) calibration model).

Another area of interest is incorporating data from the annual PIFSC Bottomfish Fishery-Independent Survey in Hawai'i (BFISH) that started in 2016. This survey consists of a combination of research fishing and underwater stereo video cameras and covers the entirety of the main Hawaiian Islands. BFISH primarily targets bottom-dwelling, more sessile species; however, uku are frequently observed (or caught) on these surveys and could provide a new fishery-independent abundance index as well as an extra source of size composition data. This fishery-independent index would complement the ongoing PIFSC diver surveys, which only take place every three years and cover only a fraction of the uku habitat (depths above 30 m).

GitHub repository

The data filtering, CPUE standardization, and other R scripts, as well as the SS base-case model files, are available on GitHub at: https://github.com/PIFSCstockassessments/2024_Hawaii_Uku. Other files, such as the necessary data inputs and other SS models, are permanently stored on the Island Fisheries Stock Assessment Program (ISAP) Google Shared Drive.

Acknowledgments

This technical memo was made possible by the work of a large number of people across multiple NOAA offices in Hawai'i. The Ecosystem Science Program at PIFSC collected the diver survey data under the supervision of F. Parrish and T. Kindinger, with the assistance of the crew and officers of the NOAA Ships *Hi'ialakai* and *Oscar Elton Sette*. The Life History Program at PIFSC generated the data needed for uku population demographics (J. O'Malley). M. Sculley and J. Brodziak helped set up the AGEPRO projections, and H. Ma provided information on the HMRIP survey recreational catch data. The WPSAR review panel provided numerous improvements to the analyses and final report (E. Franklin, J. Helyer, and M. Chaloupka).

Literature Cited

- Alagaraja, K. (1984). Simple methods for estimation of parameters for assessing exploited fish stocks. *Indian Journal of Fisheries*, 31(2), 177–208.
- Asher, J., Williams, I. D., & Harvey, E. S. (2017). An assessment of mobile predator populations along shallow and mesophotic depth gradients in the Hawaiian Archipelago. *Scientific Reports*, 10.1038/s41598-017-03568–1. <https://doi.org/10.1038/s41598-017-03568-1>
- Ayers, A. (2022). *Ecosystem & Socioeconomic Profile of uku (Aprion virescens) in the main Hawaiian Islands*. <https://doi.org/10.25923/9F2M-4E10>
- Ayotte, P., McCoy, K., Heenan, A., Williams, I., & Zamzow, J. (2015). *Coral Reef Ecosystem Program standard operating procedures: Data collection for rapid ecological assessment fish surveys*. Pacific Islands Fisheries Science Center. Administrative Report H-15-07.
- Brodziak, J., Rago, P., & Conser, R. (1998). A general approach for making short-term stochastic projections from an age-structured fisheries assessment model. In F. Funk, T. Quinn II, J. Heifetz, J. Ianelli, J. Powers, J. Schweigert, P. Sullivan, & C. Zhang (Eds.), *Fishery Stock Assessment Models* (pp. 933–954). Alaska Sea Grant, University of Alaska Fairbanks. <https://doi.org/10.4027/fsam.1998.52>
- Burnham, K. P. (2004). Multimodel inference: Understanding AIC and BIC in model selection. *Sociological Methods & Research*, 33(2), 261–304. <https://doi.org/10.1177/0049124104268644>
- Carvalho, F., Punt, A. E., Chang, Y.-J., Maunder, M. N., & Piner, K. R. (2017). Can diagnostic tests help identify model misspecification in integrated stock assessments? *Fisheries Research*, 192, 28–40. <https://doi.org/10.1016/j.fishres.2016.09.018>
- Christie, M. R., Tissot, B. N., Albins, M. A., Beets, J. P., Jia, Y., Ortiz, D. M., Thompson, S. E., & Hixon, M. A. (2010). Larval Connectivity in an Effective Network of Marine Protected Areas. *PLoS ONE*, 5(12), e15715. <https://doi.org/10.1371/journal.pone.0015715>
- Coleman, R. R., Kraft, D. W., Hoban, M. L., Toonen, R. J., & Bowen, B. W. (2023). Genomic assessment of larval odyssey: Self-recruitment and biased settlement in the Hawaiian surgeonfish *ACANTHURUS TRIOSTEGUS SANDVICENSIS*. *Journal of Fish Biology*, 102(3), 581–595. <https://doi.org/10.1111/jfb.15294>
- Dunn, P., & Smythe, G. (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics*, 5, 236–244.
- Everson, A. R., Williams, A., & Ito, B. M. (1989). Maturation and reproduction in two Hawaiian eteline snappers, uku, *Aprion virescens*, and onaga, *Etelis coruscans*. *Fishery Bulletin*, 87(4), 877–888.
- Francis, C. (2011). Data weighting in statistical fisheries stock assessment models. *Canadian Journal of Fisheries and Aquatic Sciences*, 68(6), 1124–1138. <https://doi.org/10.1139/f2011-025>
- Francis, C. R., Hurst, R. J., & Renwick, J. A. (2003). Quantifying annual variation in catchability for commercial and research fishing. *Fishery Bulletin*, 101(2), 293–304.
- Hewitt, D. A., & Hoenig, J. M. (2005). Comparison of two approaches for estimating natural mortality based on longevity. *Fishery Bulletin*, 103(2), 433–437.

- Hordyk, A. R., Ono, K., Prince, J. D., & Walters, C. J. (2016). A simple length-structured model based on life history ratios and incorporating size-dependent selectivity: Application to spawning potential ratios for data-poor stocks. *Canadian Journal of Fisheries and Aquatic Sciences*, 73(12), 1787–1799.
- Hurtado-Ferro, F., Szuwalski, C. S., Valero, J. L., Anderson, S. C., Cunningham, C. J., Johnson, K. F., Licandeo, R., McGilliard, C. R., Monnahan, C. C., Muradian, M. L., Ono, K., Vert-Pre, K. A., Whitten, A. R., & Punt, A. E. (2015). Looking in the rear-view mirror: Bias and retrospective patterns in integrated, age-structured stock assessment models. *ICES Journal of Marine Science*, 72(1), 99–110. <https://doi.org/10.1093/icesjms/fsu198>
- Kenchington, T. J. (2014). Natural mortality estimators for information-limited fisheries. *Fish and Fisheries*, 15(4), 533–562.
- Kittinger, J. N., Pandolfi, J. M., Blodgett, J. H., Hunt, T. L., Jiang, H., Maly, K., McClenachan, L. E., Schultz, J. K., & Wilcox, B. A. (2011). Historical reconstruction reveals recovery in Hawaiian coral reefs. *PLoS ONE*, 6(10), e25460. <https://doi.org/10.1371/journal.pone.0025460>
- Langseth, B., Syslo, J., Yau, A., Kapur, M., & Brodziak, J. (2018). *Stock assessment for the Main Hawaiian Islands Deep 7 bottomfish complex in 2018, with catch projections through 2022* (No. U.S. Dept. of Commerce, NOAA Technical Memorandum NOAA-TM-NMFS-PIFSC-69, 217 p.).
- Lee, H.-H., Maunder, M. N., Piner, K. R., & Methot, R. D. (2012). Can steepness of the stock–recruitment relationship be estimated in fishery stock assessment models? *Fisheries Research*, 125–126, 254–261.
- Lee, H.-H., Piner, K. R., Methot, R. D., & Maunder, M. N. (2014). Use of likelihood profiling over a global scaling parameter to structure the population dynamics model: An example using blue marlin in the Pacific Ocean. *Fisheries Research*, 158, 138–146. <https://doi.org/10.1016/j.fishres.2013.12.017>
- Ma, H. (2013). *Catch estimates for major pelagic species from the Hawaii Marine Recreational Fishing Survey (2003-2011)* (No. PIFSC internal report IR-13-006).
- Ma, H. (2023). *Non-commercial catch estimation for Deep 7 bottomfish in the main Hawaiian Islands*. <https://doi.org/10.25923/2MBE-XH91>
- Ma, H., & Ogawa, T. K. (2016). *Hawaii Marine Recreational Fishing Survey: A summary of current sampling, estimation and data analyses* (No. U.S. Dep. Commer., NOAA Tech. Memo., NOAA-TM-NMFS-PIFSC-55, 43p.).
- Maunder, M. N., & Punt, A. E. (2004). Standardizing catch and effort data: A review of recent approaches. *Fisheries Research*, 70(2), 141–159.
- Methot, R. D. (2000). *Technical description of the Stock Synthesis assessment program*. U.S. Dept. Commer., NOAA Tech. Memo. NMFS-NWFSC-43, 46 p.
- Methot, R. D., & Taylor, I. G. (2011). Adjusting for bias due to variability of estimated recruitments in fishery assessment models. *Canadian Journal of Fisheries and Aquatic Sciences*, 68(10), 1744–1760. <https://doi.org/10.1139/f2011-092>
- Methot, R. D., & Wetzel, C. R. (2013). Stock Synthesis: A biological and statistical framework for fish stock assessment and fishery management. *Fisheries Research*, 142, 86–99. <https://doi.org/10.1016/j.fishres.2012.10.012>

- Meyer, C. G., Papastamatiou, Y. P., & Holland, K. N. (2007). Seasonal, diel, and tidal movements of green jobfish (*Aprion virescens*, Lutjanidae) at remote Hawaiian atolls: Implications for marine protected area design. *Marine Biology*, 151(6), 2133–2143. <https://doi.org/10.1007/s00227-007-0647-7>
- Mohn, R. (1999). The retrospective problem in sequential population analysis: An investigation using cod fishery and simulated data. *ICES Journal of Marine Science*, 56(4), 473–488.
- Nadon, M. O. (2017). *Stock assessment of the coral reef fishes of Hawaii, 2016* (p. 212) [NOAA Tech. Memo.]. NOAA-TM-NMFS-PIFSC-60.
- Nadon, M. O., Ault, J. S., Williams, I. D., Smith, S. G., & DiNardo, G. T. (2015). Length-based assessment of coral reef fish populations in the Main and Northwestern Hawaiian Islands. *PLoS ONE*, 10(8), e0133960.
- Nadon, M. O., Sculley, M., & Carvalho, F. (2020). *Stock assessment of uku (Aprion virescens) in Hawaii, 2020* (p. 120). NOAA TM-NMFS-PIFSC-100. doi:10.25923/57nb-8138
- NMFS Office of Science and Technology. (2020). *Marine Recreational Information Program*, <https://inport.nmfs.noaa.gov/inport/item/10906>.
- O'Malley, J., Wakefield, C. B., Kinney, M. J., & Newman, S. J. (2021). Markedly similar growth and longevity of green jobfish *Aprion virescens* over an expansive geographic range between the Hawaiian Archipelago and the Eastern Indian Ocean. *Marine and Coastal Fisheries*, 13(3), 253–262. <https://doi.org/10.1002/mcf2.10155>
- Pacific Islands Fisheries Science Center. (2020a). *Hawaii DAR Fisherman Reporting System Data*, <https://inport.nmfs.noaa.gov/inport/item/5609>.
- Pacific Islands Fisheries Science Center. (2020b). *Pacific Reef Assessment and Monitoring Program: Rapid Ecological Assessments of Fish Belt Transect Surveys (BLT) at Coral Reef Sites across the Pacific Ocean from 2000 to 2009*, <https://inport.nmfs.noaa.gov/inport/item/5565>.
- Pacific Islands Fisheries Science Center. (2020c). *Pacific Reef Assessment and Monitoring Program: Stratified Random Surveys (StRS) of Reef Fish, including Benthic Estimate Data at Coral Reef Sites across the Pacific Ocean from 2007 to 2012*, <https://inport.nmfs.noaa.gov/inport/item/34515>.
- Pelletier, D., & Ferraris, J. (2000). A multivariate approach for defining fishing tactics from commercial catch and effort data. *Canadian Journal of Fisheries and Aquatic Sciences*, 57(1), 51–65.
- Pyle, R. L., Boland, R., Bolick, H., Bowen, B. W., Bradley, C. J., Kane, C., Kosaki, R. K., Langston, R., Longenecker, K., Montgomery, A., Parrish, F. A., Popp, B. N., Rooney, J., Smith, C. M., Wagner, D., & Spalding, H. L. (2016). A comprehensive investigation of mesophotic coral ecosystems in the Hawaiian Archipelago. *PeerJ*, 4, e2475.
- Raïche, G., Walls, T. A., Magis, D., Riopel, M., & Blais, J.-G. (2013). Non-graphical solutions for Cattell's scree Test. *Methodology*, 9(1), 23–29. <https://doi.org/10.1027/1614-2241/a000051>
- Restrepo, V. R., Thompson, G. G., Mace, P. M., Gabriel, W. L., Low, L. L., MacCall, A. D., Methot, R. D., Powers, J. E., Taylor, B. L., Wade, P. R., & Witzig, J. F. (1998). *Technical guidance on the use of precautionary approaches in implementing national standard 1 of the Magnuson-Stevens Fishery Conservation and Management Act*. NOAA Technical Memorandum NMFSF/ SPO-031, NOAA, USA: 54 pp.

- Sabater, M. G., & Kleiber, P. (2013). *Improving specification of acceptable biological catches of data-poor reef fish stocks using a biomass-augmented catch-MSY approach*. Western Pacific Regional Fishery Management Council, Honolulu, HI.
- Schmidt, A. L., Whitney, J. L., Suca, J., & Tanaka, K. R. (2023). *Larval ecology of Uku (Aprion virescens) in the Main Hawaiian Islands: A review from historical data*. <https://doi.org/10.25923/AEVX-HR06>
- Schnute, J. (1981). A versatile growth model with statistically stable parameters. *Canadian Journal of Fisheries and Aquatic Sciences*, 38(9), 1128–1140.
- Sundberg, M., & Underkoffler, K. (2011). *Size composition and length-weight data for bottomfish and pelagic species in Honolulu, Hawaii* [Admin report 11-04].
- Syslo, J., Oshima, M., Ma, H., Ducharme-Barth, N., Nadon, M., & Carvalho, F. (2024). *Benchmark stock assessment for the main Hawaiian Islands Deep 7 bottomfish complex, with projections through 2029*. <https://doi.org/10.25923/5SSG-8D54>
- Then, A. Y., Hoenig, J. M., Hall, N. G., & Hewitt, D. A. (2014). Evaluating the predictive performance of empirical estimators of natural mortality rate using information on over 200 fish species. *ICES Journal of Marine Science*, 72(1), 82–92.
- Thorson, J. T. (2019). Predicting recruitment density dependence and intrinsic growth rate for all fishes worldwide using a data-integrated life-history model. *Fish and Fisheries*, 10.1111/faf.12427.
- Thorson, J. T., Jensen, O. P., & Zipkin, E. F. (2014). How variable is recruitment for exploited marine fishes? A hierarchical model for testing life history theory. *Canadian Journal of Fisheries and Aquatic Sciences*, 71(7), 973–983. <https://doi.org/10.1139/cjfas-2013-0645>
- Toonen, R. J., Andrews, K. R., Baums, I. B., Bird, C. E., Concepcion, G. T., Daly-Engel, T. S., Eble, J. A., Faucci, A., Gaither, M. R., Iacchei, M., Puritz, J. B., Schultz, J. K., Skillings, D. J., Timmers, M. A., & Bowen, B. W. (2011). Defining boundaries for ecosystem-based management: A multispecies case study of marine connectivity across the Hawaiian Archipelago. *Journal of Marine Biology*, 2011, 1–13. <https://doi.org/10.1155/2011/460173>
- Williamson, D. H., Harrison, H. B., Almany, G. R., Berumen, M. L., Bode, M., Bonin, M. C., Choukroun, S., Doherty, P. J., Frisch, A. J., Saenz-Agudelo, P., & Jones, G. P. (2016). Large-scale, multidirectional larval connectivity among coral reef fish populations in the Great Barrier Reef Marine Park. *Molecular Ecology*, 25(24), 6039–6054.
- Winker, H., Carvalho, F., & Kapur, M. (2018). JABBA: Just Another Bayesian Biomass Assessment. *Fisheries Research*, 204, 275–288. <https://doi.org/10.1016/j.fishres.2018.03.010>
- Winker, H., Kerwath, S. E., & Attwood, C. G. (2013). Comparison of two approaches to standardize catch-per-unit-effort for targeting behaviour in a multispecies hand-line fishery. *Fisheries Research*, 139, 118–131. <https://doi.org/10.1016/j.fishres.2012.10.014>
- Winker, H., Kerwath, S. E., & Attwood, C. G. (2014). Proof of concept for a novel procedure to standardize multispecies catch and effort data. *Fisheries Research*, 155, 149–159. <https://doi.org/10.1016/j.fishres.2014.02.016>
- WPRFMC. (2009). *Western Pacific Fishery Management Council [WPRFMC]—Fishery Ecosystem Plan for the Hawaii Archipelago*.

- Wren, J. L. K., Kobayashi, D. R., Jia, Y., & Toonen, R. J. (2016). Modeled Population Connectivity across the Hawaiian Archipelago. *PLOS ONE*, 11(12), e0167626. <https://doi.org/10.1371/journal.pone.0167626>
- Yau, A. (2018). *Report from Hawaii bottomfish commercial fishery data workshops, 2015-2016* (No. NOAA Tech. Memo. NMFS-PIFSC-68, 105 p.).

Section V: Tables

Table 1. Key model parameters, including some estimated by Stock Synthesis.

Parameter	Value	Comment	Source
Natural mortality (M)	0.10 yr ⁻¹	Fixed. Assumes $S=0.04$	Derived from $A_{\max}$
Reference age ($A_{\min}$)	3 yr	Model structure	O'Malley – pers. comm.
Maximum age ($A_{\max}$)	32 yr	Model structure	O'Malley – pers. comm.
Length at $A_{\min}$ ($L_{A_{\min}}$)	51.6 cm	Fixed	O'Malley – pers. comm.
Length at $A_{\max}$ ($L_{A_{\max}}$)	76.5 cm	Fixed	O'Malley – pers. comm.
Growth rate (K)	0.136 yr ⁻¹	Fixed	O'Malley – pers. comm.
CV of length < $L_{A_{\min}}$	0.12	Fixed	O'Malley – pers. comm.
CV of length > $L_{A_{\min}}$	0.12	Fixed	O'Malley – pers. comm.
Length-weight σ	1.18e-05	Fixed	Sundberg & Underkoffler (2011)
Length-weight β	3.043	Fixed	Sundberg & Underkoffler (2011)
Length-at-50% maturity	44.8 cm	Fixed	Everson & Williams (1989)
Slope of maturity ogive	-3.44	Fixed	Everson & Williams (1989)
Spawner-recruit steepness (h)	0.81	Fixed	FishLife 2 - Thorson (2019)
Recruitment variability (σ_R)	0.39	Fixed	FishLife 2 - Thorson (2019)
Unfished log-scale recruitment ($\log(R_0)$)	4.50	Estimated	Stock Synthesis model
Length 50% selectivity deep-sea handline	53.3 cm	Estimated	Stock Synthesis model
Length 95% selectivity deep-sea handline	73.6 cm	Estimated	Stock Synthesis model
Length 50% selectivity inshore handline	34.9 cm	Fixed	LBSPR model
Length 95% selectivity inshore handline	44.8 cm	Fixed	LBSPR model
Length 50% selectivity trolling	47.1 cm	Fixed	LBSPR model
Length 95% selectivity trolling	60.7 cm	Fixed	LBSPR model
Length 50% selectivity other	35.0 cm	Fixed	LBSPR model
Length 95% selectivity other	48.5 cm	Fixed	LBSPR model
Length 50% selectivity recreational	36.3 cm	Fixed	LBSPR model
Length 95% selectivity recreational	50.7 cm	Fixed	LBSPR model
Length 50% selectivity diver survey	47.1 cm	Fixed (trolling)	LBSPR model
Length 95% selectivity diver survey	60.7 cm	Fixed (trolling)	LBSPR model

Note: all lengths are fork lengths (FL)

Table 2. Main assumptions of the uku Stock Synthesis model with links to relevant report sections.

Model structure and assumptions	Report section
Single stock area corresponding to the entire main Hawaiian Islands (MHI)	Distribution and biology Population dynamics
No sex differences in life history and abundance	Population dynamics
Time-invariant life history parameters	-
Fecundity proportional to female biomass	Population dynamics
Single spawning and recruitment period during the first month of the year	Population dynamics
Beverton-Holt stock-recruitment curve with estimated lognormal annual deviations	Population dynamics , Population dynamics
Steepness (h) fixed, not estimated within the model	Assessment strategy
Natural mortality (M) constant with age and linked to longevity with $S_{\text{max age}} = 0.04$	Population dynamics Assessment strategy
Initial conditions derived from equilibrium catch and estimated in Stock Synthesis	Population dynamics
Time-invariant selectivity	Fishery dynamics
Multinomial error distribution for size-frequency data in Stock Synthesis	Fishery dynamics Assessment strategy
Size observations are random and representative of population size-frequency	Fishery dynamics Assessment strategy
Time-invariant catchability	Fishery dynamics Assessment strategy
Fishing mortality estimated as continuous full parameters	Fishery dynamics
Recreational catch is highly uncertain, commercial catch has greater certainty	Fishery dynamics Assessment strategy
CPUE tied to population abundance	CPUE abundance indices Assessment strategy
Diver survey index tied to population abundance	Fishery-independent diver survey Assessment strategy
CPUE coefficients of variation (CV) artificially low and increased by 10% in model	Fishery dynamics Assessment strategy
Projection recruitment: 60% S-R curve, 20% recent, and 20% long-term	Catch projections for 2020-2026 Projections

Table 3. Summary of commercial and recreational catch data for uku.

Year	Commercial (metric tons)				Recreational (metric tons)
	Deep-sea handline	Inshore handline	Trolling	Other gears	
1948	40.4	4.33	0.04	1.27	29.5
1949	33.0	3.42	0.24	1.02	29.5
1950	22.8	2.83	0.00	0.65	29.5
1951	16.1	3.43	0.02	0.92	30.4
1952	25.0	3.81	0.02	0.62	30.6
1953	22.6	9.22	0.00	1.00	30.2
1954	16.6	10.09	0.00	1.45	29.9
1955	14.3	12.30	0.01	7.93	31.9
1956	19.2	12.15	0.00	0.75	33.1
1957	37.3	5.93	0.00	0.53	34.6
1958	28.8	3.56	0.00	0.54	35.8
1959	18.4	1.90	0.00	0.61	36.8
1960	17.2	2.35	0.00	1.02	38.0
1961	15.8	2.86	0.00	0.48	39.0
1962	23.4	3.43	0.02	2.06	40.5
1963	24.8	3.10	0.02	0.87	40.4
1964	40.0	0.48	0.00	0.31	41.4
1965	22.2	0.24	0.00	0.23	41.7
1966	26.1	0.01	0.00	0.14	42.0
1967	26.0	0.38	0.04	0.13	42.8
1968	22.1	0.03	0.03	0.37	43.4
1969	25.5	0.02	0.01	0.61	44.4
1970	21.3	0.01	0.00	0.25	45.6
1971	21.8	0.01	0.02	0.30	47.4
1972	21.5	0.02	0.12	0.21	49.0
1973	30.0	0.05	0.10	0.18	50.4
1974	34.9	0.09	0.16	0.19	51.4
1975	26.6	0.75	0.52	0.30	52.4
1976	23.6	3.20	0.62	0.73	53.5
1977	25.0	3.58	0.82	1.64	54.3
1978	28.0	8.29	0.16	1.72	55.1
1979	36.1	1.85	0.42	1.18	56.4
1980	31.8	0.55	0.69	0.83	57.3
1981	36.7	0.27	0.30	1.35	57.9
1982	43.9	0.31	0.61	0.96	58.8
1983	56.3	0.61	1.33	1.86	59.9
1984	61.5	0.11	0.51	0.87	60.8
1985	21.7	0.25	0.12	0.33	61.5
1986	36.8	4.20	0.68	5.56	62.2

Year	Commercial (metric tons)				Recreational (metric tons)
	Deep-sea handline	Inshore handline	Trolling	Other gears	
1987	21.4	2.85	0.67	0.86	63.2
1988	145.2	4.81	2.64	3.62	63.9
1989	90.9	1.34	1.59	0.69	64.8
1990	39.3	2.07	2.83	0.99	65.8
1991	32.0	5.11	2.64	1.24	66.9
1992	31.2	6.76	0.91	0.97	68.0
1993	27.8	2.19	0.64	1.12	68.7
1994	26.3	5.18	0.33	0.78	69.5
1995	23.7	2.19	0.73	0.61	69.9
1996	18.6	3.91	0.82	0.84	70.1
1997	21.6	7.97	0.61	0.63	70.4
1998	20.0	6.37	0.48	0.85	70.4
1999	34.0	5.31	0.57	0.90	70.1
2000	30.5	5.87	0.78	0.64	71.8
2001	17.5	6.97	1.47	0.59	72.5
2002	19.2	4.36	0.60	1.38	73.4
2003	14.4	2.93	2.46	1.06	62.6
2004	26.1	3.57	3.71	1.52	84.6
2005	20.3	2.44	4.23	1.89	117.1
2006	17.4	4.33	3.23	2.03	71.1
2007	20.7	5.21	3.81	1.67	45.7
2008	28.6	5.89	5.80	1.66	31.4
2009	30.2	4.84	2.52	2.33	40.9
2010	37.8	7.83	4.22	4.97	78.7
2011	34.8	8.29	2.88	3.64	117.8
2012	34.4	8.98	4.82	4.82	194.6
2013	34.6	8.60	5.46	6.28	54.8
2014	25.8	5.51	5.53	7.11	98.6
2015	29.5	5.71	3.92	7.10	68.4
2016	33.3	5.23	6.56	8.99	57.6
2017	38.8	7.70	6.91	6.44	54.6
2018	15.4	7.74	4.18	6.54	84.4
2019	21.9	7.46	3.55	7.75	30.9
2020	11.9	3.67	2.12	3.99	92.5
2021	17.2	2.30	3.35	4.47	71.7
2022	15.9	3.21	1.96	2.90	108.7
2023	13.5	1.73	1.79	3.38	80.8

Table 4. List of predictor variables considered for the logistic and lognormal regression models used for CPUE standardization. Dashes represent variables not available for a model and “Errors” indicate variables resulting in model convergence errors.

Predictor	1948–2002		2003–2023					
	Deep-sea handline		Deep-sea handline		Inshore handline		Trolling	
	Logistic	Lognormal	Logistic	Lognormal	Logistic	Lognormal	Logistic	Lognormal
Year	Y	Y	Y	Y	Y	Y	Y	Y
Fisher	Errors	Y	Errors	Y	Errors	Y	Errors	Y
Area	Y	Y	Y	Y	Y	Y	Y	Y
Month	Y	Y	Y	Y	Y	Y	Y	Y
Log(cumulative experience)	N	N	Y	N	N	N	Y	N
Wind speed	-	-	N	Y	N	Y	Y	Y
PC1	Y	Y	Y	Y	Y	Y	Y	Y
PC1 ²	Y	Y	Y	Y	Y	Y	Y	Y
PC2	Y	N	Y	N	Y	Y	Y	Y
PC2 ²	Y	N	Y	N	Y	Y	Y	Y
Area:month	N	Y	N	N	N	N	N	N
Year:Area	Errors	Errors	Errors	Errors	Errors	Errors	Errors	Errors

Table 5. Akaike information criterion (AIC), delta AIC, and percent AIC change from the previous model, used for CPUE standardization model selection (deep-sea handline). The final selected model is shown in bold.

Model	AIC	ΔAIC	AIC change (%)
1948–2002 Lognormal			
log(CPUE) ~ 1	145484	19568	
log(CPUE) ~ YEAR	145030	19115	0.3
log(CPUE) ~ YEAR + (1 FISHER)	132459	6544	8.7
log(CPUE) ~ YEAR + (1 FISHER) + AREA	129512	3597	2.2
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH	128031	2116	1.1
log(CPUE) ~ YEAR + (1 FISHER) + AREA*MONTH	127518	1603	0.4
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + log(CUM_EXP)	127945	2030	-0.3
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + log(CUM_EXP) + PC1	126299	384	1.3
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + log(CUM_EXP) + poly(PC1,2)	126120	205	0.1
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + log(CUM_EXP) + poly(PC1,2) + PC2	126119	204	0.0
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	126113	198	0.0
log(CPUE) ~ YEAR + (1 FISHER) + AREA*MONTH + poly(PC1,2)	125915	0	0.2
1948–2002 Logistic			
PRES ~ 1	94495	34038	
PRES ~ YEAR	90214	29757	4.5
PRES ~ YEAR + AREA	81296	20839	9.9
PRES ~ YEAR + AREA + MONTH	81198	20741	0.1
PRES ~ YEAR + AREA + MONTH + log(CUM_EXP)	81197	20740	0.0
PRES ~ YEAR + AREA + MONTH + log(CUM_EXP) + PC1	71216	10759	12.3
PRES ~ YEAR + AREA + MONTH + log(CUM_EXP) + poly(PC1,2)	65836	5379	7.6
PRES ~ YEAR + AREA + MONTH + log(CUM_EXP) + poly(PC1,2) + PC2	62866	2409	4.5
PRES ~ YEAR + AREA + MONTH + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	60457	0	3.8
PRES ~ YEAR + AREA + MONTH + poly(PC1,2) + poly(PC2,2)	60477	20	
2003–2023 Lognormal			
log(CPUE) ~ 1	46219	8362	
log(CPUE) ~ YEAR	46203	8346	
log(CPUE) ~ YEAR + (1 FISHER)	40771	2914	11.8
log(CPUE) ~ YEAR + (1 FISHER) + AREA	40634	2777	0.3
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH	39701	1844	2.3
log(CPUE) ~ YEAR + (1 FISHER) + AREA * MONTH	39805	1948	-0.3
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED	39706	1849	0.2
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP)	39713	1857	0.0
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + PC1	37877	21	4.6
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2)	37861	4	0.0
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + PC2	37857	0	0.0
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	37867	11	

Model	AIC	ΔAIC	AIC change (%)
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + poly(PC1,2)	37869	12	
2003-2023Logistic			
PRES ~ 1	67759	15767	
PRES ~ YEAR	67530	15538	0.3
PRES ~ YEAR + AREA	62029	10037	8.1
PRES ~ YEAR + AREA + MONTH	59790	7797	3.6
PRES ~ YEAR + AREA + MONTH + SPEED	59754	7761	0.1
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP)	59359	7366	0.7
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + PC1	54977	2985	7.4
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2)	54659	2666	0.6
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + PC2	52361	368	4.2
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	51994	1	0.7
PRES ~ YEAR + AREA + MONTH + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	51993	0	

Table 6. Akaike information criterion (AIC), delta AIC, and percent AIC change from the previous model, used for CPUE standardization model selection (inshore handline). The final selected model is shown in bold.

Model	AIC	ΔAIC	AIC change (%)
2003-2023Lognormal			
log(CPUE) ~ 1	16371	3087	0.0
log(CPUE) ~ YEAR	16233	2949	0.8
log(CPUE) ~ YEAR + (1 FISHER)	13687	403	15.7
log(CPUE) ~ YEAR + (1 FISHER) + AREA	13588	304	0.7
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH	13627	343	-0.3
log(CPUE) ~ YEAR + (1 FISHER) + AREA * MONTH	13716	432	-0.7
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED	13632	348	0.6
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP)	13639	355	-0.1
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + PC1	13479	195	1.2
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2)	13488	204	-0.1
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + PC2	13496	212	-0.1
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	13290	6	1.5
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + poly(PC1,2) + poly(PC2,2)	13284	0	
2003-2023Logistic			
PRES ~ 1	35729	11820	
PRES ~ YEAR	35325	11416	1.1
PRES ~ YEAR + AREA	28415	4506	19.6
PRES ~ YEAR + AREA + MONTH	28393	4484	0.1
PRES ~ YEAR + AREA + MONTH + SPEED	28363	4454	0.1
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP)	28334	4425	0.1
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + PC1	27208	3299	4.0
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2)	26191	2282	3.7
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + PC2	24634	725	5.9
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	23909	0	2.9
PRES ~ YEAR + AREA + MONTH + poly(PC1,2) + poly(PC2,2)	24076	167	-0.7

Table 7. Akaike information criterion (AIC), delta AIC, and percent AIC change from the previous model, used for CPUE standardization model selection (trolling). The final selected model is shown in bold.

Model	AIC	ΔAIC	AIC change (%)
2003-2023Lognormal			
log(CPUE) ~ 1	10282	2032	
log(CPUE) ~ YEAR	10301	2051	-0.2
log(CPUE) ~ YEAR + (1 FISHER)	8408	158	18.4
log(CPUE) ~ YEAR + (1 FISHER) + AREA	8364	114	0.5
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH	8397	147	-0.4
log(CPUE) ~ YEAR + (1 FISHER) + AREA * MONTH	8534	284	-1.6
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED	8398	149	1.6
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP)	8406	156	-0.1
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + PC1	8401	151	0.1
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2)	8315	65	1.0
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + PC2	8316	66	0.0
log(CPUE) ~ YEAR + (1 FISHER) + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	8292	43	0.3
log(CPUE) ~ YEAR + (1 FISHER) + AREA + SPEED + poly(PC1,2) + poly(PC2,2)	8250	0	0.5
2003-2023Logistic			
PRES ~ 1	41855	8432	
PRES ~ YEAR	41402	7980	1.1
PRES ~ YEAR + AREA	35512	2089	14.2
PRES ~ YEAR + AREA + MONTH	34953	1530	1.6
PRES ~ YEAR + AREA + MONTH + SPEED	34909	1486	0.1
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP)	34840	1417	0.2
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + PC1	34678	1255	0.5
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2)	33956	533	2.1
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + PC2	33931	509	0.1
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	33423	0	1.5
PRES ~ YEAR + AREA + MONTH + SPEED + log(CUM_EXP) + poly(PC1,2) + poly(PC2,2)	33423	0	

Table 8. Mean and CV of CPUE relative abundance indices used in the base-case model for MHI uku. CVs are estimated from the standardization process (not variance adjusted).

Year	Deep-sea Handline Old		Deep-Sea Handline Recent		Inshore Handline Recent		Trolling Recent		Diver Survey	
	Mean	CV	Mean	CV	Mean	CV	Mean	CV	Mean	CV
1948	9.68	0.07	-	-	-	-	-	-	-	-
1949	9.40	0.07	-	-	-	-	-	-	-	-
1950	8.72	0.09	-	-	-	-	-	-	-	-
1951	8.51	0.09	-	-	-	-	-	-	-	-
1952	9.02	0.09	-	-	-	-	-	-	-	-
1953	11.90	0.10	-	-	-	-	-	-	-	-
1954	11.61	0.10	-	-	-	-	-	-	-	-
1955	11.14	0.11	-	-	-	-	-	-	-	-
1956	9.91	0.12	-	-	-	-	-	-	-	-
1957	11.50	0.11	-	-	-	-	-	-	-	-
1958	13.19	0.10	-	-	-	-	-	-	-	-
1959	8.01	0.10	-	-	-	-	-	-	-	-
1960	7.54	0.10	-	-	-	-	-	-	-	-
1961	7.85	0.11	-	-	-	-	-	-	-	-
1962	8.81	0.10	-	-	-	-	-	-	-	-
1963	9.64	0.09	-	-	-	-	-	-	-	-
1964	9.08	0.09	-	-	-	-	-	-	-	-
1965	8.55	0.09	-	-	-	-	-	-	-	-
1966	8.52	0.08	-	-	-	-	-	-	-	-
1967	7.23	0.09	-	-	-	-	-	-	-	-
1968	8.54	0.08	-	-	-	-	-	-	-	-
1969	7.96	0.09	-	-	-	-	-	-	-	-
1970	7.93	0.08	-	-	-	-	-	-	-	-
1971	8.02	0.09	-	-	-	-	-	-	-	-
1972	9.34	0.08	-	-	-	-	-	-	-	-
1973	9.91	0.08	-	-	-	-	-	-	-	-
1974	10.95	0.08	-	-	-	-	-	-	-	-
1975	10.19	0.09	-	-	-	-	-	-	-	-
1976	9.39	0.61	-	-	-	-	-	-	-	-
1977	7.55	0.08	-	-	-	-	-	-	-	-
1978	11.85	0.08	-	-	-	-	-	-	-	-
1979	10.69	0.08	-	-	-	-	-	-	-	-
1980	9.07	0.08	-	-	-	-	-	-	-	-
1981	8.47	0.08	-	-	-	-	-	-	-	-
1982	7.89	0.07	-	-	-	-	-	-	-	-
1983	8.04	0.07	-	-	-	-	-	-	-	-
1984	7.15	0.07	-	-	-	-	-	-	-	-
1985	6.31	0.08	-	-	-	-	-	-	-	-

Year	Deep-sea Handline Old		Deep-Sea Handline Recent		Inshore Handline Recent		Trolling Recent		Diver Survey	
	Mean	CV	Mean	CV	Mean	CV	Mean	CV	Mean	CV
1986	8.02	0.08	-	-	-	-	-	-	-	-
1987	6.76	0.08	-	-	-	-	-	-	-	-
1988	11.59	0.07	-	-	-	-	-	-	-	-
1989	8.11	0.08	-	-	-	-	-	-	-	-
1990	6.34	0.08	-	-	-	-	-	-	-	-
1991	6.47	0.08	-	-	-	-	-	-	-	-
1992	6.97	0.08	-	-	-	-	-	-	-	-
1993	7.19	0.08	-	-	-	-	-	-	-	-
1994	6.50	0.08	-	-	-	-	-	-	-	-
1995	6.08	0.08	-	-	-	-	-	-	-	-
1996	6.01	0.08	-	-	-	-	-	-	-	-
1997	6.69	0.08	-	-	-	-	-	-	-	-
1998	6.02	0.08	-	-	-	-	-	-	-	-
1999	7.35	0.08	-	-	-	-	-	-	-	-
2000	7.25	0.08	-	-	-	-	-	-	-	-
2001	6.97	0.08	-	-	-	-	-	-	-	-
2002	6.02	0.08	-	-	-	-	-	-	-	-
2003	-	-	0.77	0.07	0.67	0.12	0.04	0.17	-	-
2004	-	-	1.03	0.07	1.04	0.11	0.04	0.16	-	-
2005	-	-	1.02	0.07	1.02	0.12	0.04	0.16	0.01	0.54
2006	-	-	0.98	0.07	1.40	0.12	0.03	0.17	-	-
2007	-	-	1.15	0.06	1.51	0.11	0.04	0.17	-	-
2008	-	-	1.14	0.06	1.18	0.11	0.05	0.16	0.12	0.50
2009	-	-	0.86	0.07	0.90	0.11	0.04	0.17	-	-
2010	-	-	1.26	0.06	1.23	0.11	0.09	0.15	0.08	0.43
2011	-	-	1.16	0.07	1.39	0.11	0.07	0.16	-	-
2012	-	-	1.06	0.07	1.43	0.10	0.07	0.16	0.04	0.21
2013	-	-	1.33	0.06	1.42	0.11	0.12	0.15	-	-
2014	-	-	1.17	0.07	1.60	0.11	0.11	0.15	-	-
2015	-	-	1.54	0.07	1.81	0.11	0.09	0.16	0.04	0.36
2016	-	-	1.45	0.07	1.93	0.12	0.09	0.16	0.05	0.30
2017	-	-	1.60	0.07	2.19	0.12	0.11	0.17	-	-
2018	-	-	1.23	0.08	2.38	0.10	0.10	0.16	-	-
2019	-	-	1.57	0.07	2.08	0.11	0.12	0.15	0.06	0.42
2020	-	-	1.44	0.08	2.04	0.11	0.15	0.17	-	-
2021	-	-	1.79	0.08	1.61	0.13	0.15	0.15	-	-
2022	-	-	1.46	0.08	1.78	0.15	0.11	0.17	-	-
2023	-	-	1.51	0.08	1.38	0.18	0.15	0.17	-	-

Table 9. Negative log-likelihoods (NLL) of data components in the base-case model over a range of fixed levels of virgin recruitment in thousands of recruits in log-scale, $\log(R_0)$. Likelihoods are relative to the minimum NLL (best-fit) for each respective data component. Colors indicate relative likelihood (green: low NLL and better-fit; red: high NLL and poorer-fit). The maximum likelihood estimate of $\log(R_0)$ was 4.50.

Log(R_0)	Total	Catch	Recruitment	SizeFreq	Survey
3.8	109.00	49.98	41.37	14.16	6.02
3.9	76.98	38.89	25.33	10.39	4.54
4	50.38	26.88	14.20	8.12	2.95
4.1	29.90	15.99	7.37	6.33	1.59
4.2	15.48	8.09	3.45	4.30	0.66
4.3	6.35	3.33	1.35	2.23	0.16
4.4	1.51	0.93	0.36	0.65	0.00
4.5	0.00	0.04	0.01	0.00	0.12
4.505	0.00	0.02	0.00	0.00	0.13
4.6	1.03	0.00	0.04	0.45	0.48
4.7	3.96	0.33	0.31	1.99	1.06
4.8	8.23	0.70	0.71	4.55	1.88
4.9	13.39	0.87	1.17	7.98	2.92
5	19.03	0.72	1.62	12.11	4.18

Table 10. Negative log-likelihoods (NLL) of abundance index data components in the base-case model over a range of fixed levels of virgin recruitment in thousands of recruits in log-scale, $\log(R_0)$. Likelihoods are relative to the minimum NLL (best-fit) for each respective data component. Colors indicate relative likelihood (green: low NLL and better-fit; red: high NLL and poorer-fit). The maximum likelihood estimate of $\log(R_0)$ was 4.50.

Log(R_0)	Total	Com_DSH_Old	Com_DSH_Recent	Com_ISH	Com_Trol	Divers
3.8	6.019	2.2725	1.2189	0	3.86552	0.28759
3.9	4.538	1.6282	1.0132	0.2051	3.14831	0.16801
4	2.952	0.9823	0.8279	0.388	2.31357	0.0658
4.1	1.594	0.4547	0.6997	0.5392	1.50832	0.01696
4.2	0.665	0.1234	0.617	0.6809	0.86741	0.00177
4.3	0.158	0	0.5492	0.8162	0.41736	0
4.4	0	0.0643	0.4785	0.9348	0.1428	0.00429
4.5	0.12	0.2886	0.3999	1.033	0.0122	0.01171
4.505	0.132	0.3028	0.3959	1.0373	0.0091	0.01211
4.6	0.479	0.6523	0.3144	1.1164	0	0.02068
4.7	1.064	1.1472	0.2257	1.1948	0.0909	0.03029
4.8	1.879	1.7714	0.1392	1.2781	0.2751	0.03996
4.9	2.923	2.5203	0.0617	1.3739	0.5426	0.0494
5	4.175	3.3752	0	1.4863	0.87974	0.05846

Table 11. Mean input log standard error (SE), additional variance added within the model with resulting total log(SE), and root-mean-square-errors (RMSE) for the relative abundance indices used in the base-case model.

Fleet	N	Input log(SE)	Variance adjustment	Total log(SE) in model	RMSE
CPUE_DSH_old	55	0.09	0.1	0.19	0.13
CPUE_DSH_recent	21	0.07	0.1	0.17	0.12
CPUE_ISH_recent	21	0.12	0.1	0.22	0.18
CPUE_Trol_recent	21	0.16	0.1	0.26	0.34
OPUE_Divers	7	0.41	0.1	0.51	0.59

Table 12. Time series of total biomass (age 1 and older), spawning biomass, age-0 recruitment, and instantaneous fishing mortality estimated in the base-case model. CV = coefficient of variation.

Year	Age 1+ biomass (mt)	Spawning Biomass (mt)		Recruitment (1000s recruits)		Fishing mortality (yr ⁻¹)	
	Mean	Mean	CV	Mean	CV	Mean	CV
1948	1475	699	0.16	82	0.35	0.05	0.23
1949	1535	734	0.15	76	0.36	0.04	0.25
1950	1577	757	0.15	78	0.35	0.04	0.28
1951	1620	781	0.14	75	0.35	0.03	0.30
1952	1657	799	0.14	66	0.34	0.04	0.26
1953	1680	812	0.14	63	0.34	0.04	0.25
1954	1688	819	0.14	57	0.33	0.03	0.27
1955	1687	820	0.14	56	0.33	0.04	0.26
1956	1667	812	0.14	60	0.32	0.04	0.27
1957	1639	798	0.14	57	0.32	0.05	0.25
1958	1596	776	0.14	57	0.33	0.05	0.29
1959	1561	759	0.14	60	0.33	0.04	0.34
1960	1533	745	0.14	61	0.33	0.04	0.34
1961	1505	731	0.14	61	0.32	0.04	0.34
1962	1483	719	0.14	59	0.32	0.05	0.30
1963	1451	703	0.14	65	0.32	0.05	0.31
1964	1422	689	0.14	72	0.32	0.06	0.28
1965	1382	667	0.14	70	0.32	0.05	0.34
1966	1368	659	0.14	66	0.33	0.06	0.33
1967	1359	654	0.14	78	0.33	0.06	0.32
1968	1348	649	0.14	83	0.33	0.05	0.34
1969	1345	645	0.14	84	0.34	0.06	0.33
1970	1352	646	0.14	117	0.32	0.05	0.34
1971	1373	655	0.14	107	0.32	0.05	0.34
1972	1408	665	0.14	74	0.33	0.05	0.34
1973	1469	697	0.13	64	0.33	0.06	0.31
1974	1501	722	0.13	82	0.32	0.06	0.31
1975	1492	720	0.13	104	0.30	0.06	0.34
1976	1482	711	0.13	77	0.31	0.06	0.35
1977	1490	711	0.13	72	0.31	0.06	0.33
1978	1505	724	0.13	86	0.27	0.07	0.31
1979	1487	716	0.13	74	0.28	0.08	0.33
1980	1461	701	0.13	88	0.25	0.08	0.35
1981	1444	694	0.13	73	0.27	0.08	0.33
1982	1417	678	0.13	126	0.21	0.09	0.31
1983	1396	669	0.13	98	0.26	0.10	0.28

Year	Age 1+ biomass (mt)	Spawning Biomass (mt)		Recruitment (1000s recruits)		Fishing mortality (yr ⁻¹)	
	Mean	Mean	CV	Mean	CV	Mean	CV
1984	1368	644	0.13	84	0.28	0.10	0.26
1985	1382	655	0.13	59	0.30	0.06	0.34
1986	1411	675	0.13	61	0.28	0.08	0.26
1987	1396	674	0.13	81	0.27	0.06	0.32
1988	1377	665	0.12	155	0.22	0.17	0.18
1989	1243	591	0.14	64	0.32	0.15	0.23
1990	1203	556	0.14	71	0.31	0.10	0.29
1991	1247	596	0.14	74	0.32	0.09	0.30
1992	1224	585	0.14	69	0.31	0.09	0.30
1993	1206	577	0.14	63	0.32	0.09	0.32
1994	1194	572	0.14	76	0.32	0.09	0.32
1995	1172	562	0.14	100	0.29	0.09	0.33
1996	1157	551	0.14	87	0.31	0.08	0.33
1997	1166	550	0.14	75	0.32	0.09	0.31
1998	1190	564	0.14	76	0.32	0.08	0.31
1999	1203	574	0.14	68	0.33	0.09	0.28
2000	1195	570	0.14	69	0.34	0.09	0.29
2001	1183	566	0.14	73	0.35	0.08	0.33
2002	1171	561	0.14	113	0.37	0.08	0.34
2003	1164	555	0.14	162	0.36	0.07	0.33
2004	1200	561	0.15	120	0.40	0.09	0.29
2005	1268	583	0.15	99	0.37	0.09	0.31
2006	1355	634	0.15	83	0.35	0.06	0.30
2007	1432	679	0.15	82	0.36	0.05	0.26
2008	1495	715	0.14	113	0.36	0.05	0.22
2009	1542	739	0.14	107	0.38	0.05	0.23
2010	1590	756	0.14	125	0.39	0.07	0.23
2011	1626	774	0.14	129	0.40	0.08	0.26
2012	1650	782	0.14	108	0.42	0.09	0.26
2013	1663	787	0.15	193	0.37	0.06	0.22
2014	1731	822	0.15	145	0.40	0.07	0.27
2015	1795	836	0.15	101	0.39	0.06	0.26
2016	1929	912	0.15	115	0.40	0.05	0.24
2017	2005	961	0.14	110	0.39	0.06	0.24
2018	2037	976	0.14	99	0.38	0.05	0.32
2019	2070	995	0.15	85	0.37	0.03	0.23
2020	2126	1026	0.14	83	0.37	0.05	0.37
2021	2122	1028	0.14	82	0.38	0.05	0.35
2022	2108	1023	0.15	86	0.40	0.06	0.39
2023	2050	995	0.15	88	0.40	0.05	0.39

Table 13. Estimated biological reference points derived from the Stock Synthesis base-case model where F is the instantaneous annual fishing mortality rate, SPR is spawning potential ratio, SSB is spawning stock biomass, $MSST$ indicates minimum stock size threshold, and MSY indicates maximum sustainable yield.

Reference Point	Estimate
F_{MSY} (age 5–30)	0.14
F_{2023} (age 5–30)	0.05
F_{2023}/F_{MSY}	0.36
SSB_{MSY}	394 mt
SSB_{MSST}	355 mt
SSB_{2023}	995 mt
SSB_{2023}/SSB_{MSST}	2.8
MSY	111 mt
$Catch_{2021-2023}$	111 mt
SPR_{MSY}	0.31
SPR_{2023}	0.61

Table 14. Probability of overfishing ($F/F_{MSY} > 1$) at different annual catch values (metric tons) by year. Catch values for a given probability of overfishing in a given year were applied in all previous years (i.e., 2020 to the year of interest).

Probability of overfishing	Total catch in metric tons						
	2025	2026	2027	2028	2029	2030	2031
0.50	225.5	212.0	203.4	195.5	189.8	184.2	180.4
0.49	225.1	211.6	202.8	194.9	189.2	183.6	179.7
0.48	224.7	211.3	202.2	194.3	188.5	183.0	179.1
0.47	224.3	210.9	201.6	193.7	187.9	182.5	178.5
0.46	223.8	210.5	201.0	193.2	187.3	182.0	177.9
0.45	223.3	210.0	200.4	192.6	186.7	181.4	177.4
0.44	222.8	209.5	199.9	192.1	186.1	180.9	176.8
0.43	222.2	209.1	199.3	191.6	185.5	180.4	176.3
0.42	221.6	208.5	198.7	191.0	185.0	179.9	175.8
0.41	220.9	208.0	198.2	190.5	184.4	179.4	175.3
0.40	220.3	207.4	197.6	190.0	183.9	179.0	174.8
0.39	219.6	206.8	197.1	189.5	183.4	178.5	174.3
0.38	218.9	206.2	196.5	189.0	182.9	178.0	173.8
0.37	218.1	205.6	195.9	188.5	182.4	177.5	173.4
0.36	217.4	205.0	195.4	188.0	181.9	177.1	172.9
0.35	216.6	204.3	194.8	187.4	181.4	176.6	172.4
0.34	215.7	203.6	194.2	186.9	180.9	176.1	172.0
0.33	214.9	202.9	193.6	186.4	180.4	175.6	171.5
0.32	214.0	202.2	193.0	185.8	179.9	175.1	171.1
0.31	213.2	201.4	192.4	185.3	179.4	174.7	170.6
0.30	212.3	200.7	191.8	184.7	178.9	174.2	170.1
0.29	211.4	199.9	191.2	184.2	178.4	173.6	169.6
0.28	210.4	199.1	190.5	183.6	177.8	173.1	169.2
0.27	209.5	198.3	189.9	183.0	177.3	172.6	168.7
0.26	208.6	197.5	189.2	182.3	176.7	172.1	168.2
0.25	207.6	196.6	188.5	181.7	176.1	171.5	167.7
0.24	206.6	195.8	187.7	181.0	175.5	171.0	167.1
0.23	205.7	194.9	187.0	180.3	174.9	170.4	166.6
0.22	204.7	194.0	186.2	179.6	174.3	169.8	166.0
0.21	203.7	193.1	185.4	178.9	173.6	169.2	165.5
0.20	202.7	192.2	184.5	178.1	172.9	168.5	164.9
0.19	201.7	191.3	183.7	177.4	172.2	167.9	164.3
0.18	200.7	190.4	182.8	176.5	171.4	167.2	163.6
0.17	199.7	189.5	181.9	175.7	170.6	166.5	163.0
0.16	198.7	188.5	180.9	174.8	169.8	165.8	162.3
0.15	197.7	187.6	179.9	173.9	169.0	165.0	161.6
0.14	196.7	186.6	178.9	172.9	168.1	164.3	160.8
0.13	195.7	185.7	177.8	172.0	167.2	163.5	160.1
0.12	194.7	184.7	176.7	170.9	166.2	162.6	159.3
0.11	193.7	183.7	175.5	169.9	165.2	161.8	158.4
0.10	192.7	182.7	174.3	168.8	164.2	160.9	157.6

Section VI: Figures

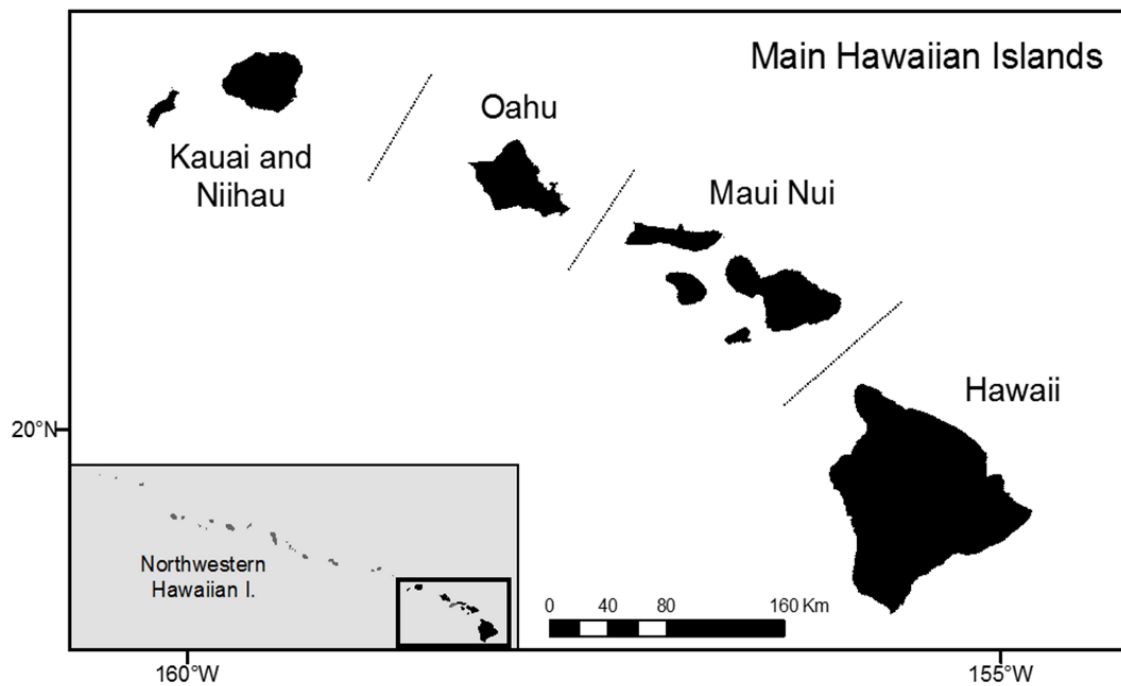


Figure 1. Map of the main Hawaiian Islands with its four sub-regions. The entire archipelago is visible in the inset, including the Northwestern Hawaiian Islands.

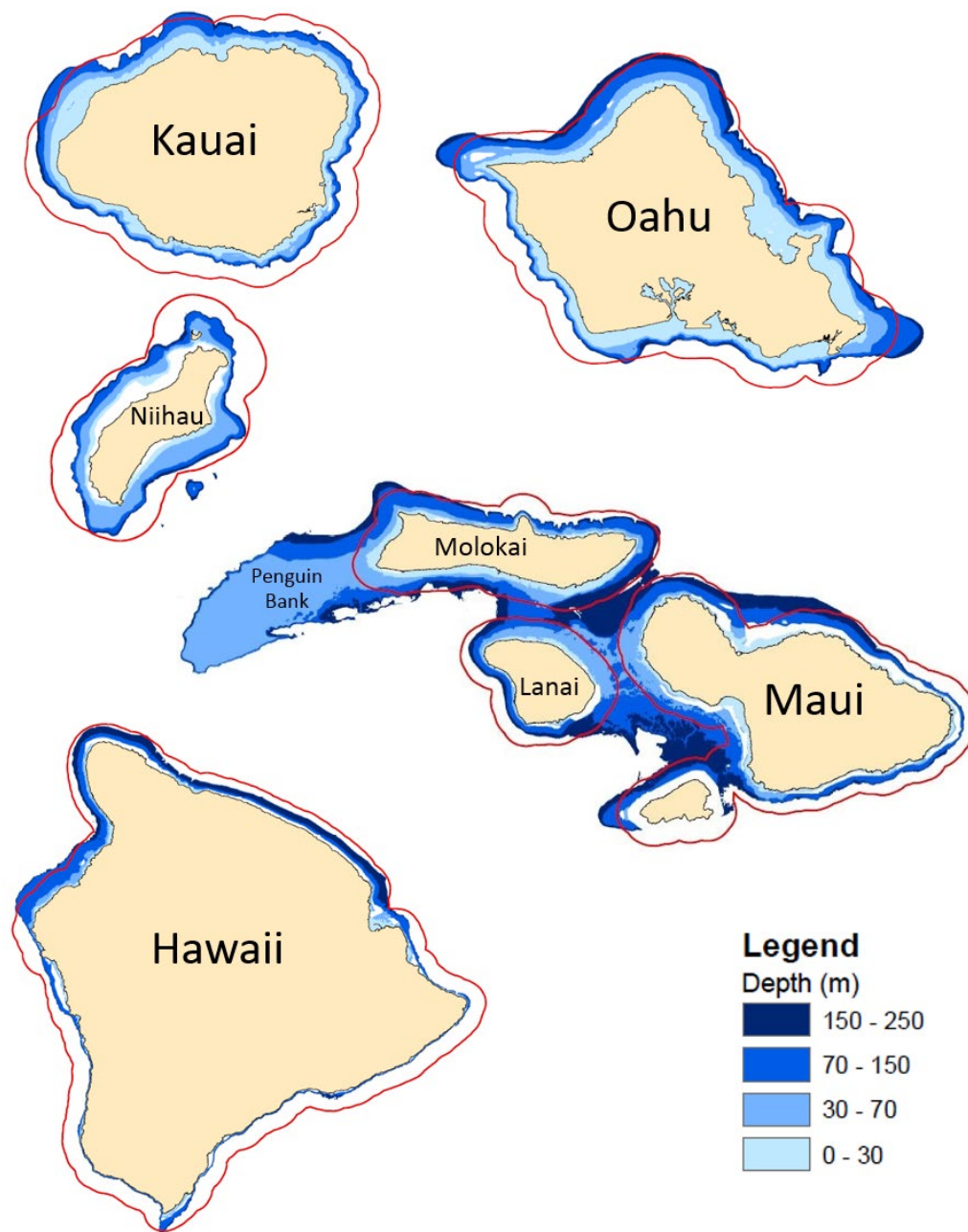


Figure 2. Map of the eight main Hawaiian Islands with depth zones (0 m– 250 m depths). Red contour lines represent the 3 nautical mile state-waters limit. Islands are not to scale and re-arranged to fit this page. Data source: CREP and Hawaii Mapping Research Group.

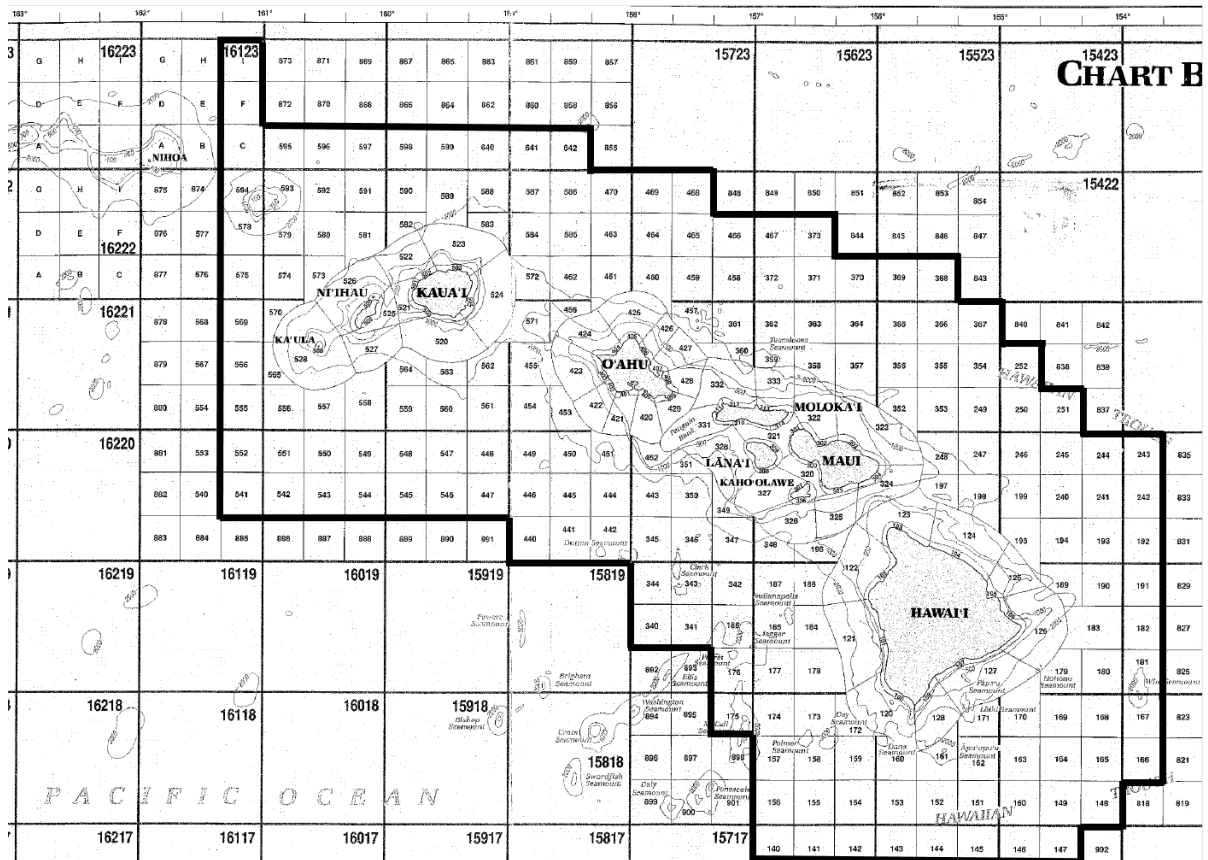


Figure 3. Boundary of the stock area in the main Hawaiian Islands used for the 2020 and 2024 uku assessment.

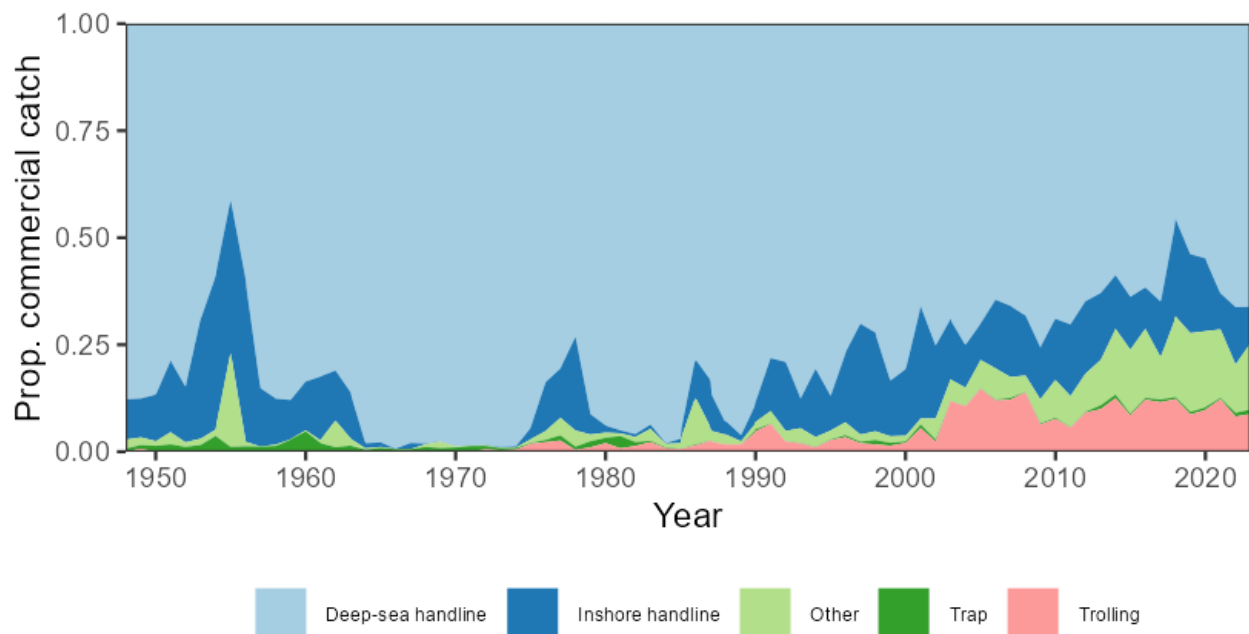


Figure 4. Proportion of total commercial catch caught by fishing gear type.

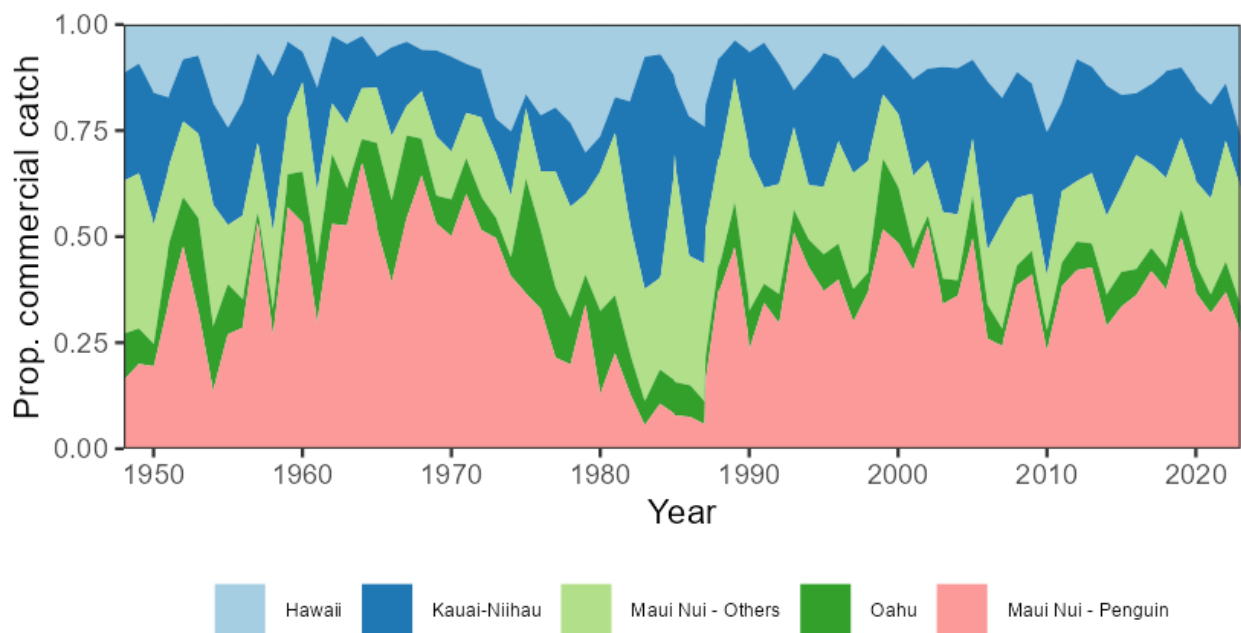


Figure 5. Proportion of total commercial catch caught by sub-region of the MHI, with Penguin Bank (part of the Maui Nui sub-region) listed separately.

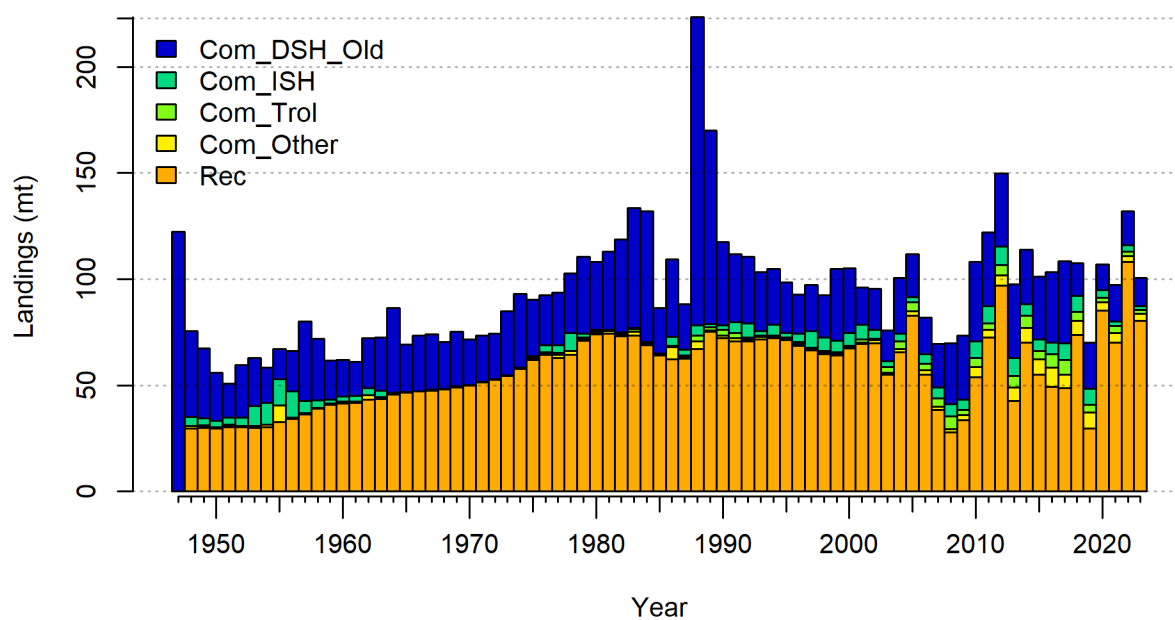


Figure 6. History of uku catch by sector (recreational and commercial), with the commercial catch further broken down by fishing gear type. Note that the recreational catch pre-2003 is reconstructed and that it is adjusted by the Stock Synthesis model (expected catch).

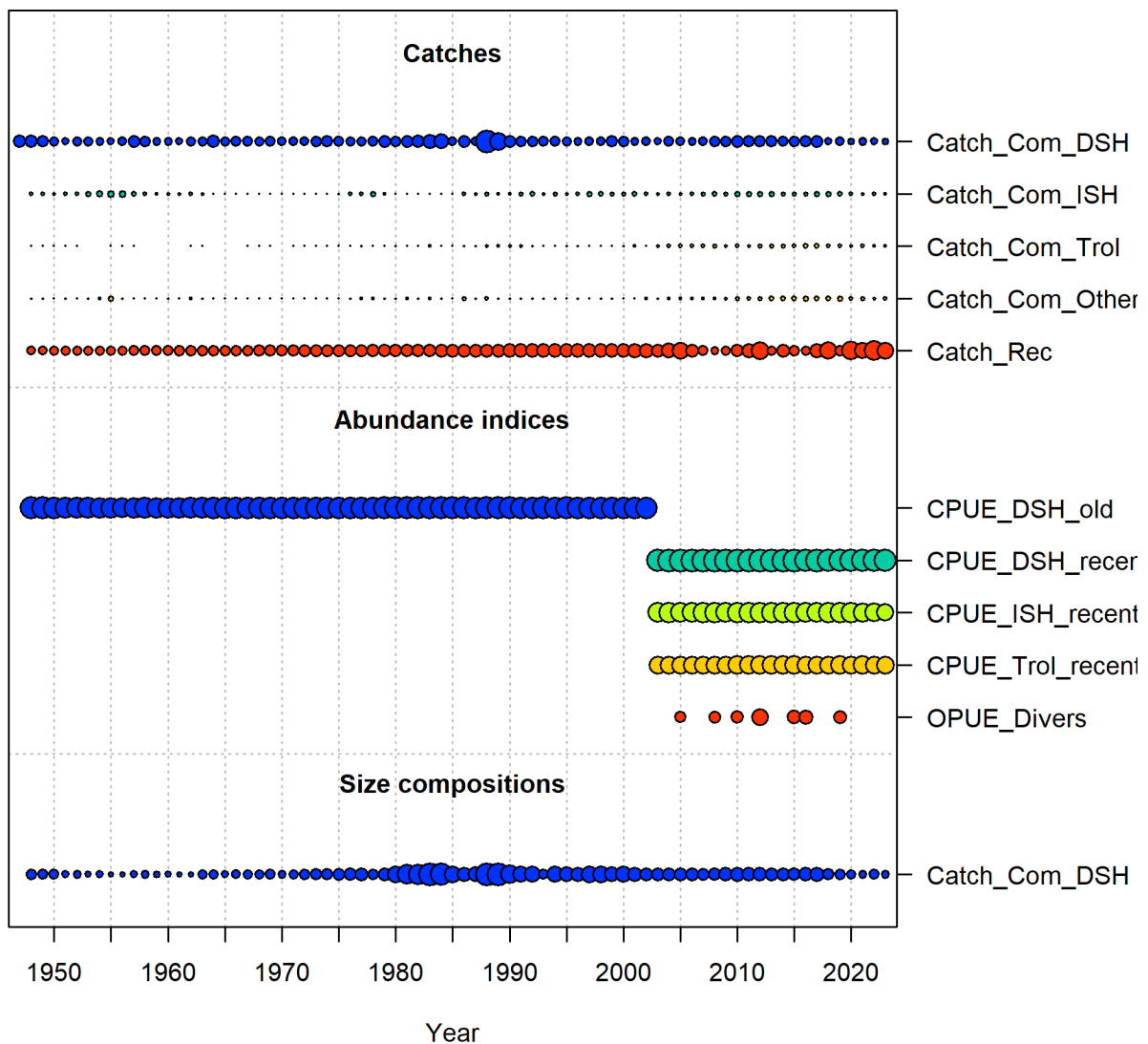


Figure 7. Summary of all data sets used in the Stock Synthesis model. DSH—deep-sea handline, ISH—inshore handline, Trol—trolling, Rec—recreational data, Divers—diver surveys.

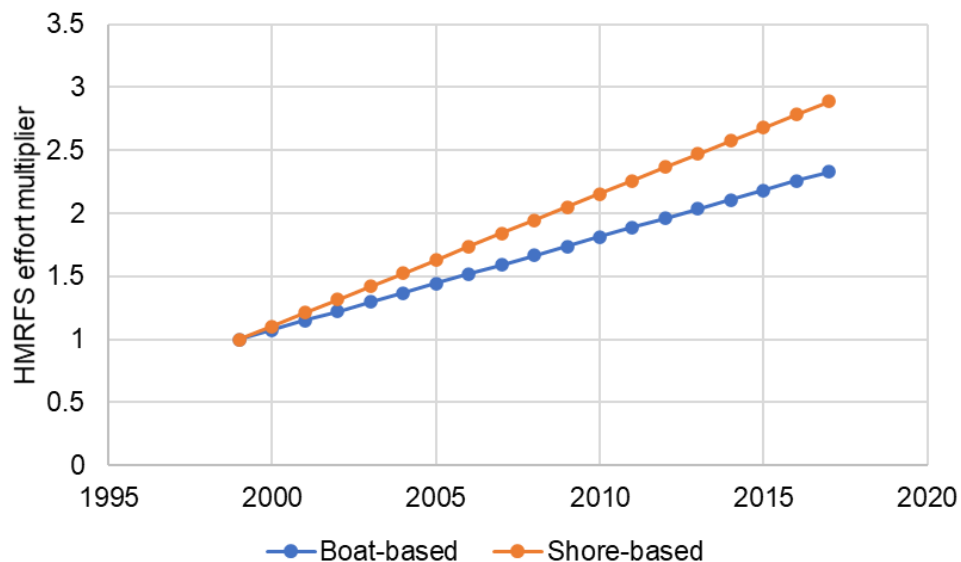


Figure 8. Correction factor (multiplier) applied to the HMRFS annual effort estimates between 2003 and 2017, for both the boat-based and shore-based fishery, to correct for the declining number of phone landlines.

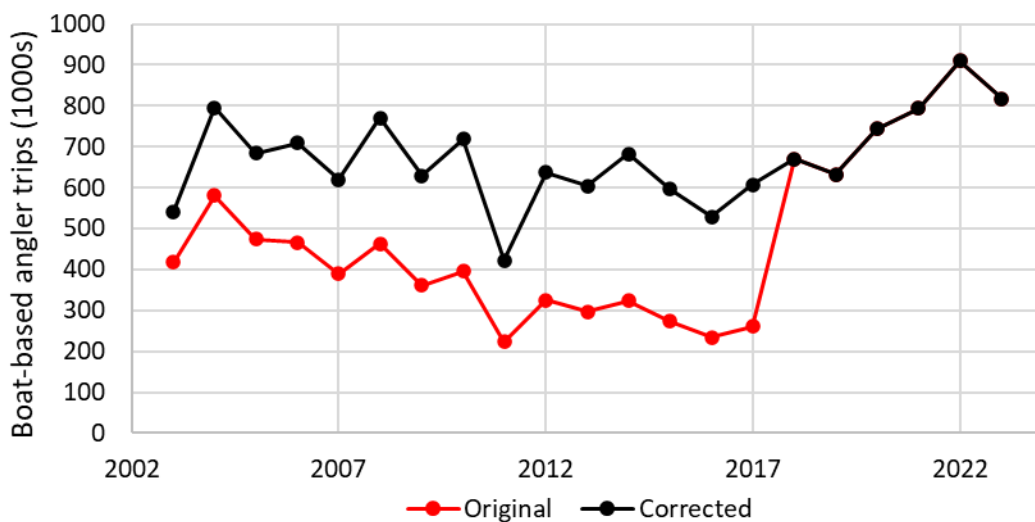


Figure 9. Comparison of HMRFS boat-based annual fishing effort between the original and corrected time series.

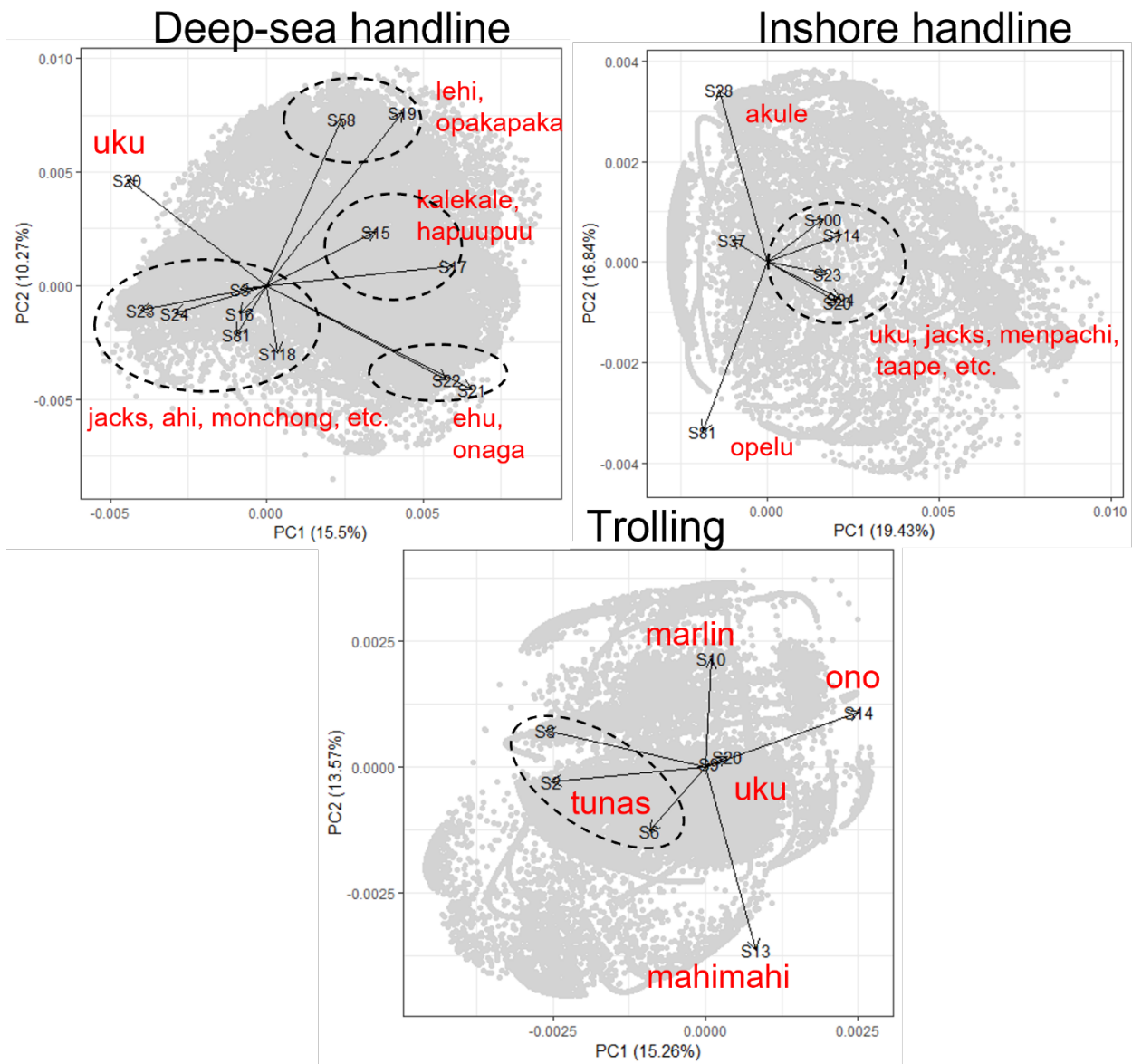


Figure 10. Principal Component Analysis (PCA) plots showing the principal axes included in the Direct Principal Component (DPC) analysis for the 3 gear types used for CPUE indices (note: *uku* are excluded from the actual DPC analyses but are presented here for reference).

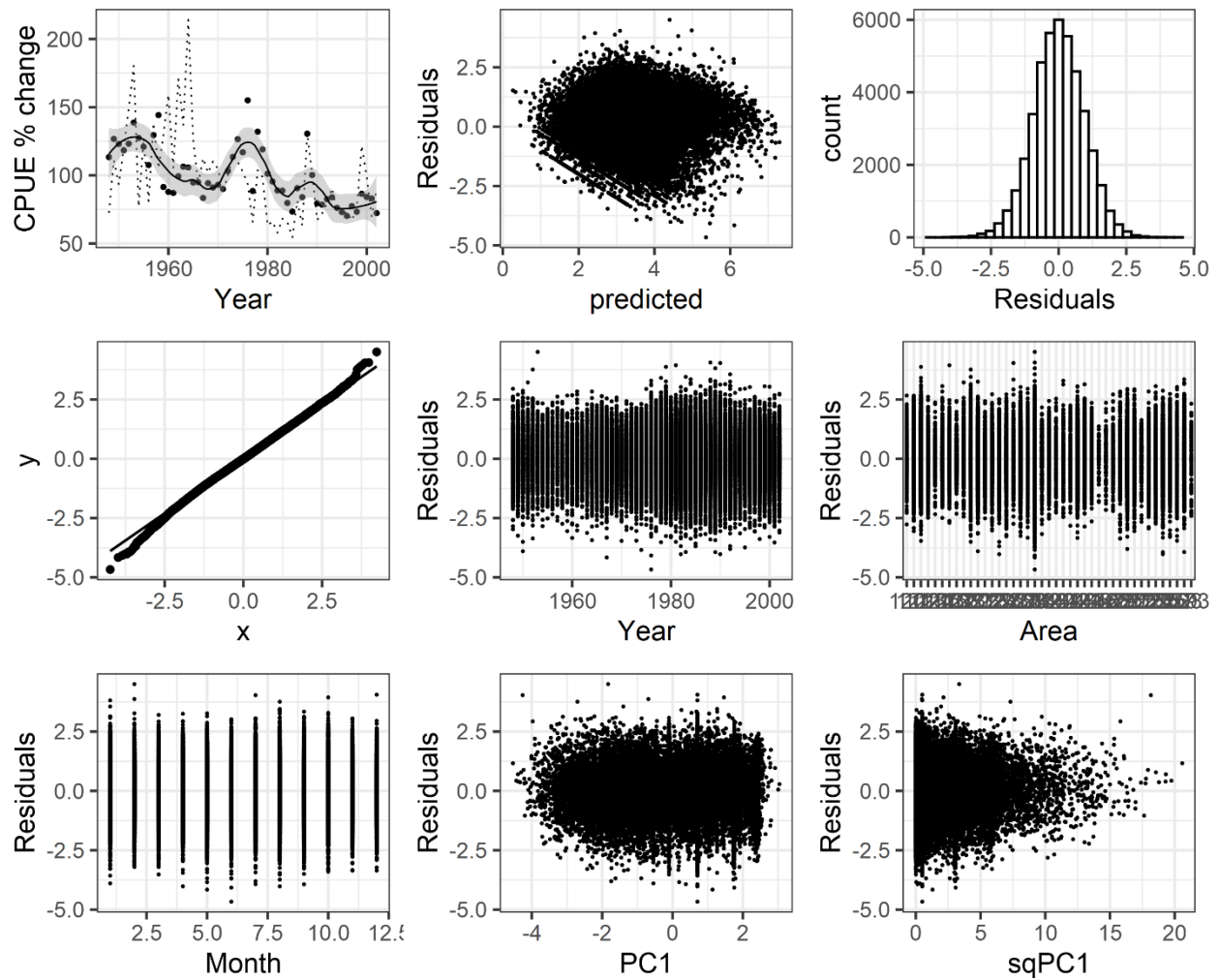


Figure 11. Diagnostics of the lognormal CPUE standardization model for the 1948–2002 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.

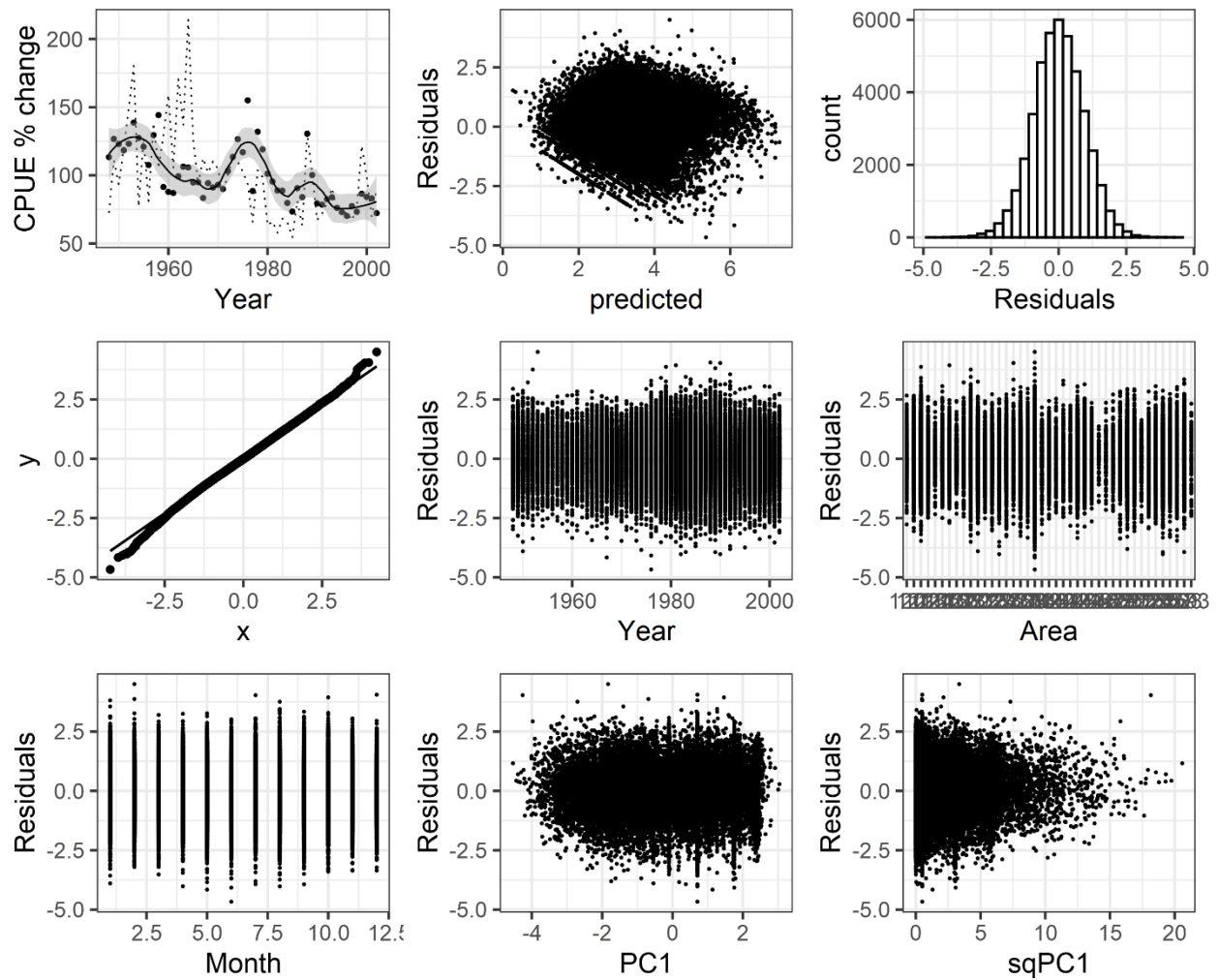


Figure 12. Diagnostics of the logistic CPUE standardization model for the 1948–2002 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.

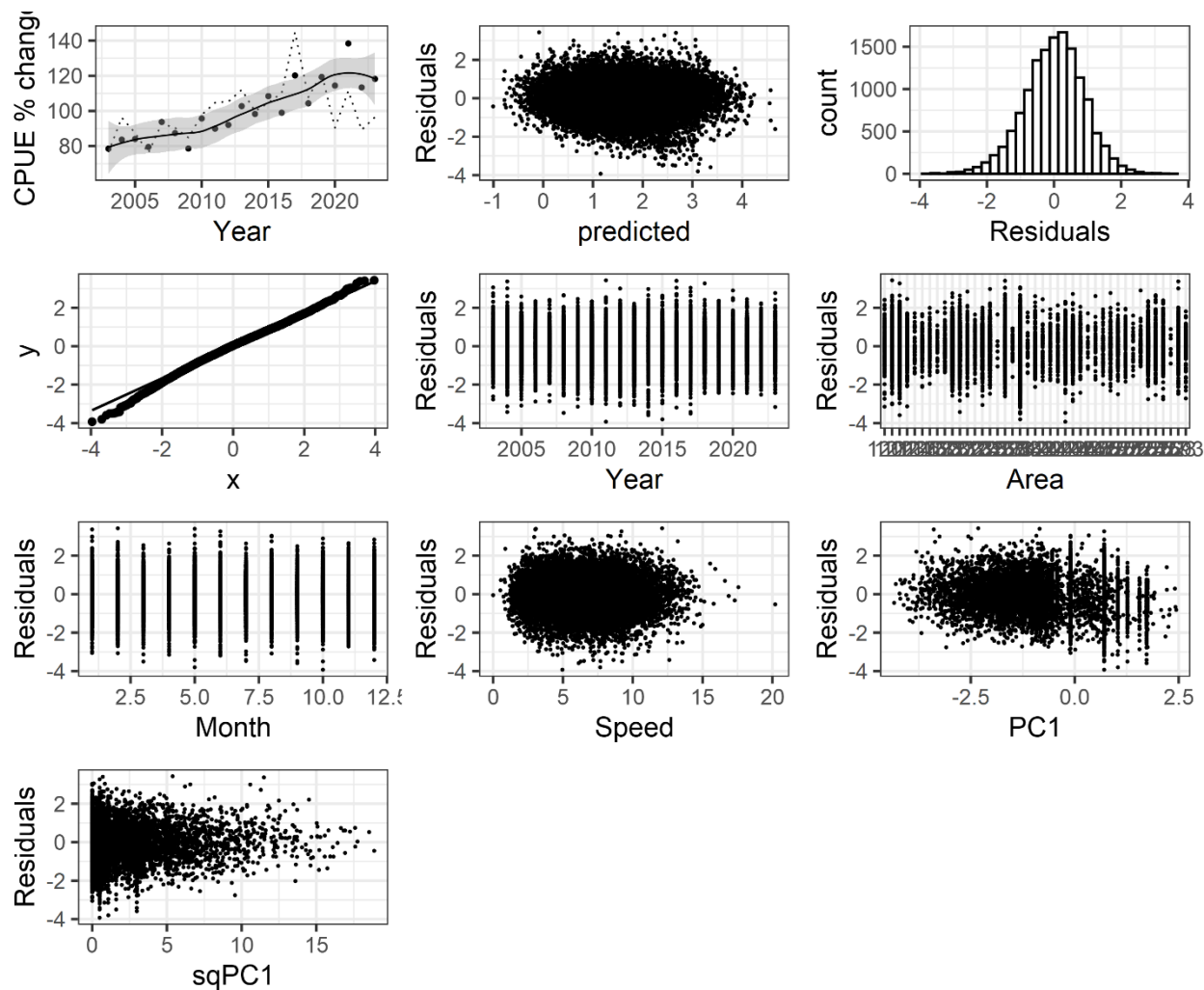


Figure 13. Diagnostics of the lognormal CPUE standardization model for the 2003–2023 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.

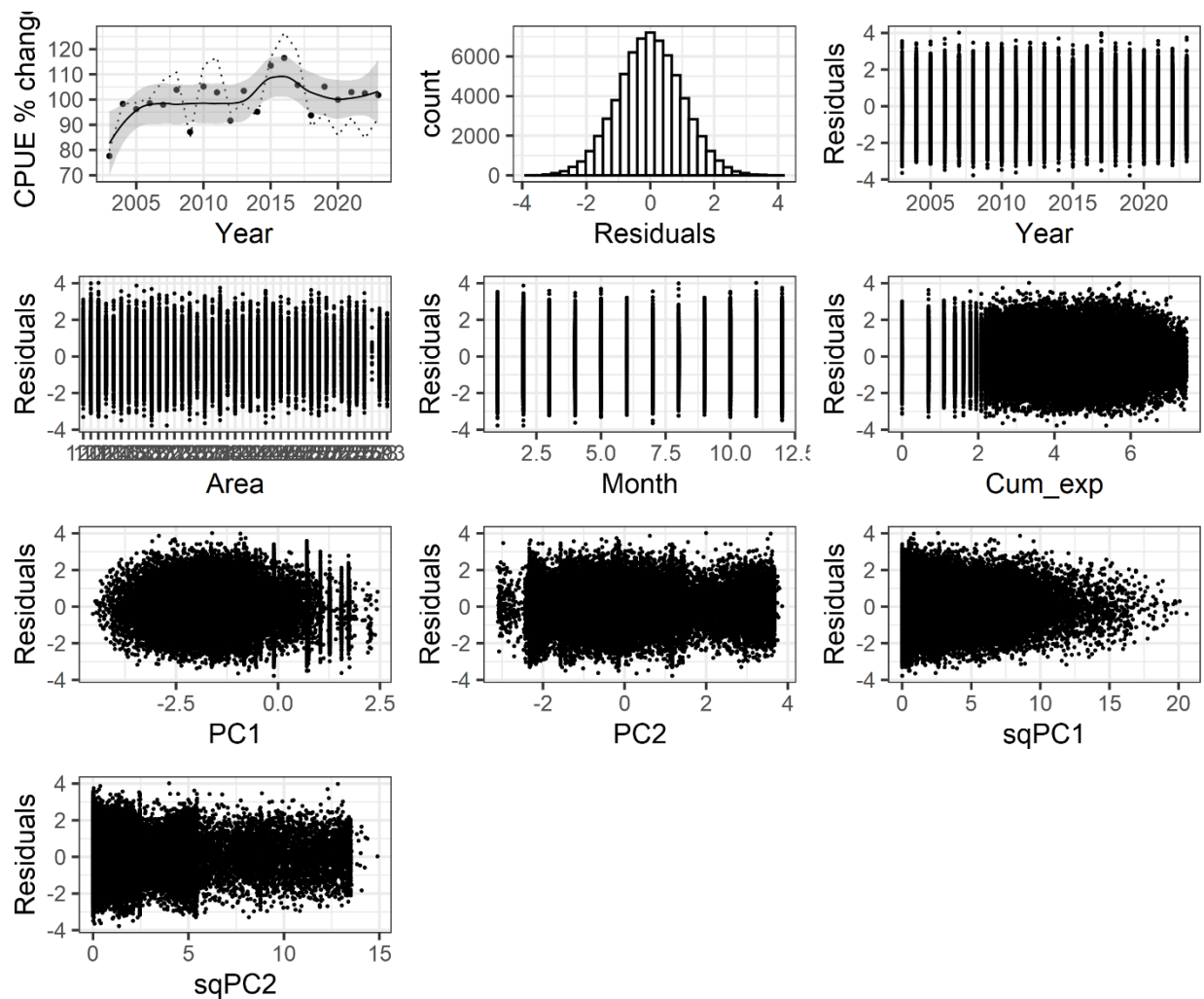


Figure 14. Diagnostics of the logistic CPUE standardization model for the 2003–2023 period (deep-sea handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.

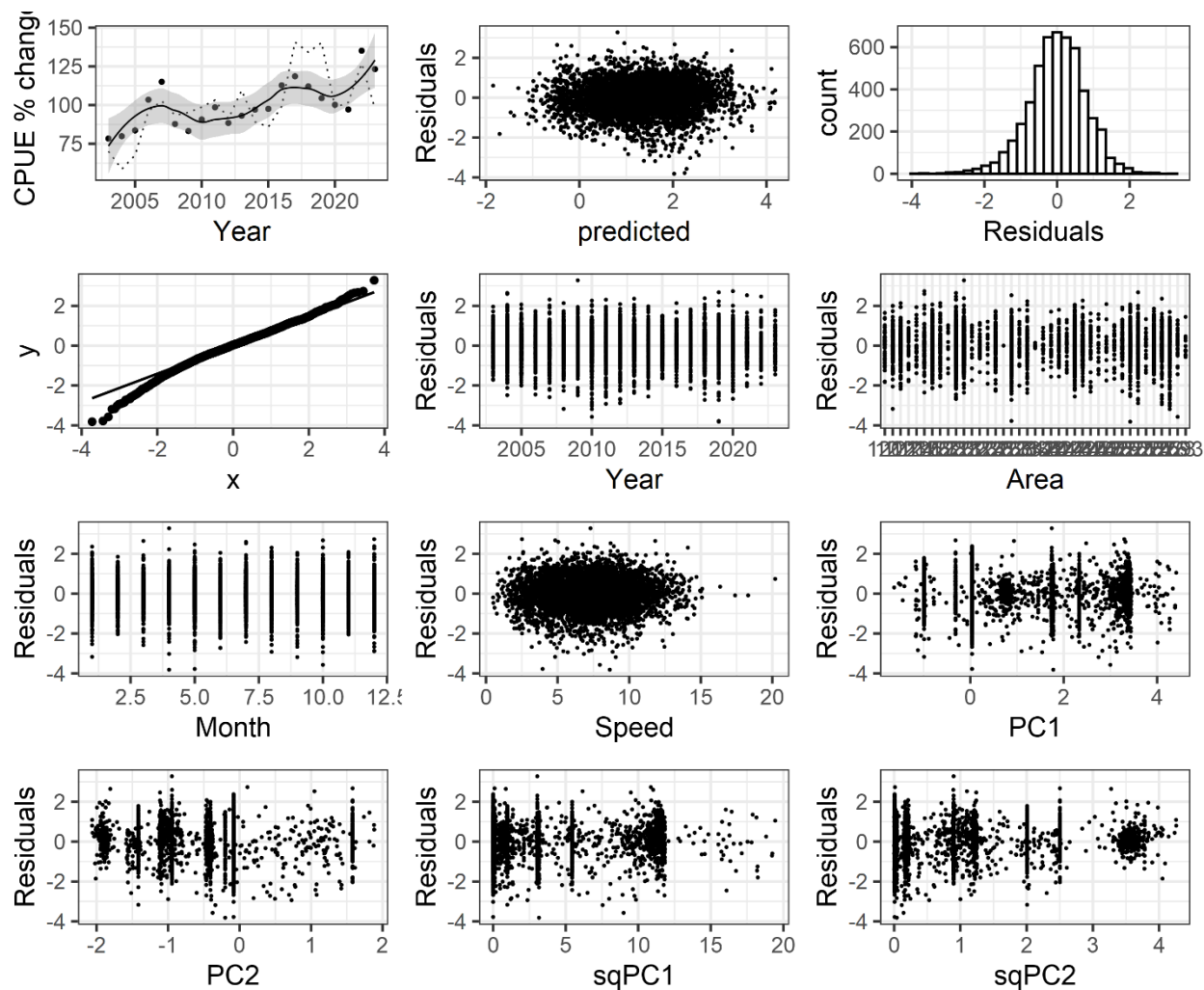


Figure 15. Diagnostics of the lognormal CPUE standardization model for the 2003–2023 period (inshore handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.

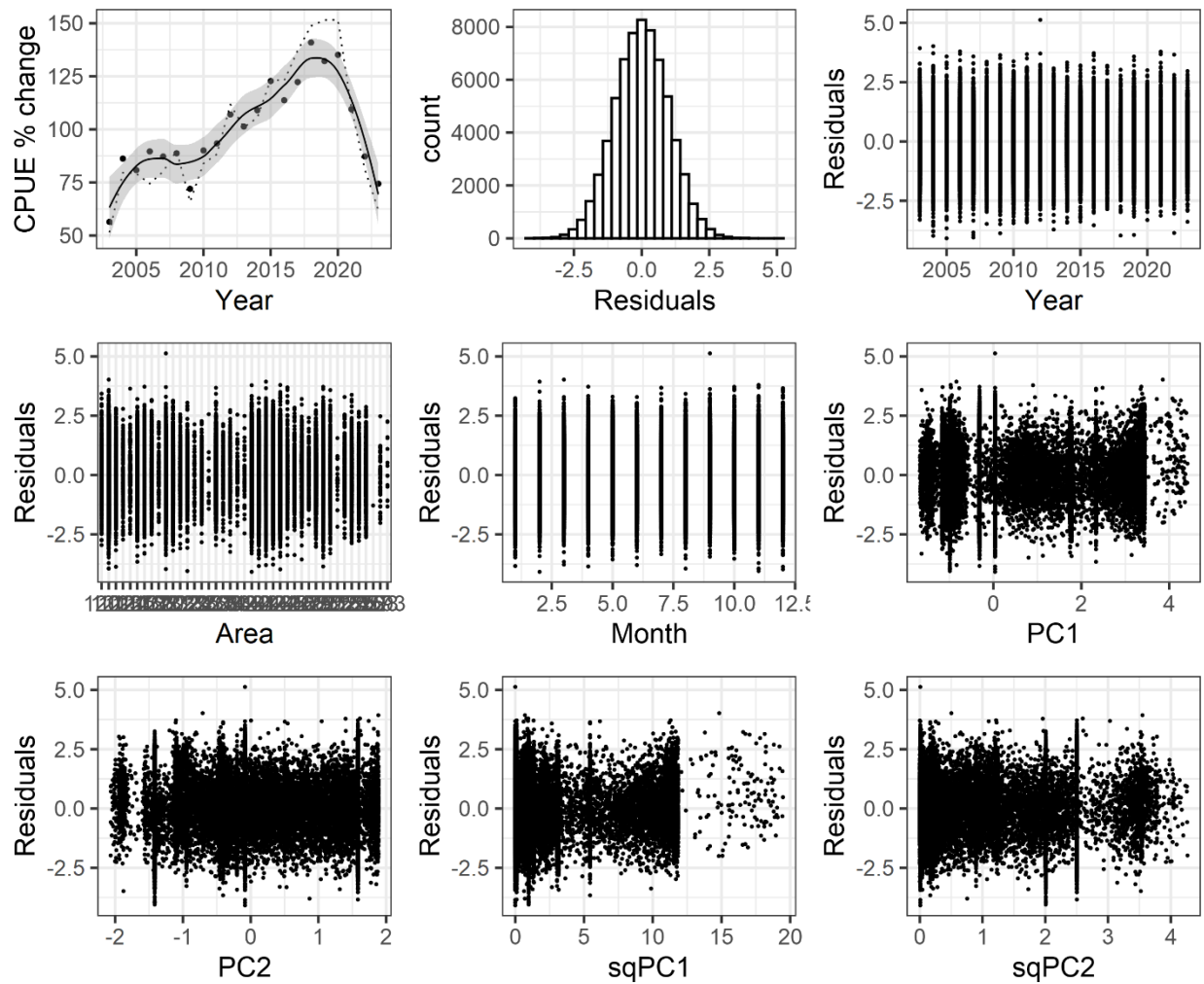


Figure 16. Diagnostics of the logistic CPUE standardization model for the 2003–2023 period (inshore handline gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.

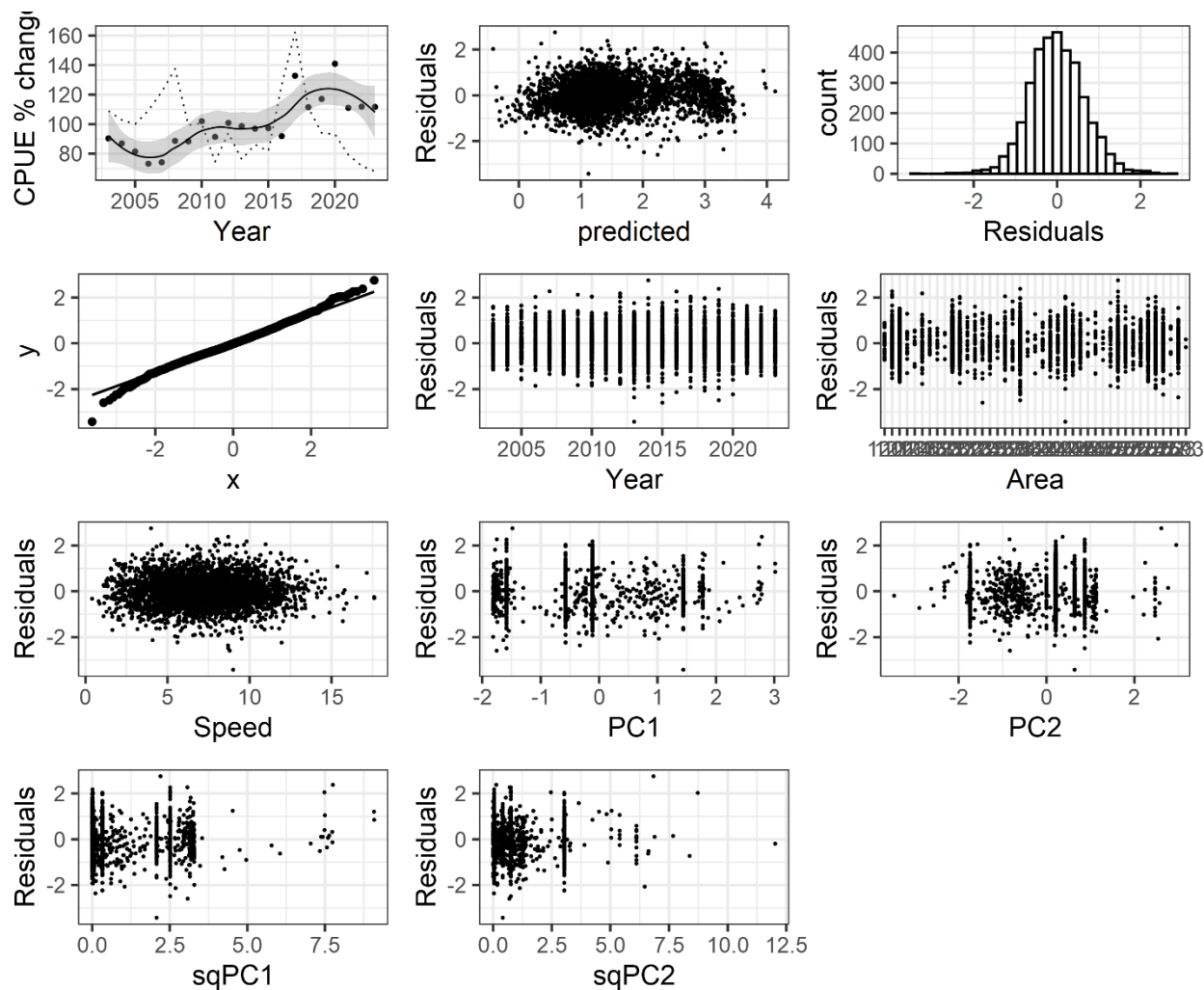


Figure 17. Diagnostics of the lognormal CPUE standardization model for the 2003–2023 period (trolling gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns including a quantile-quantile plot.

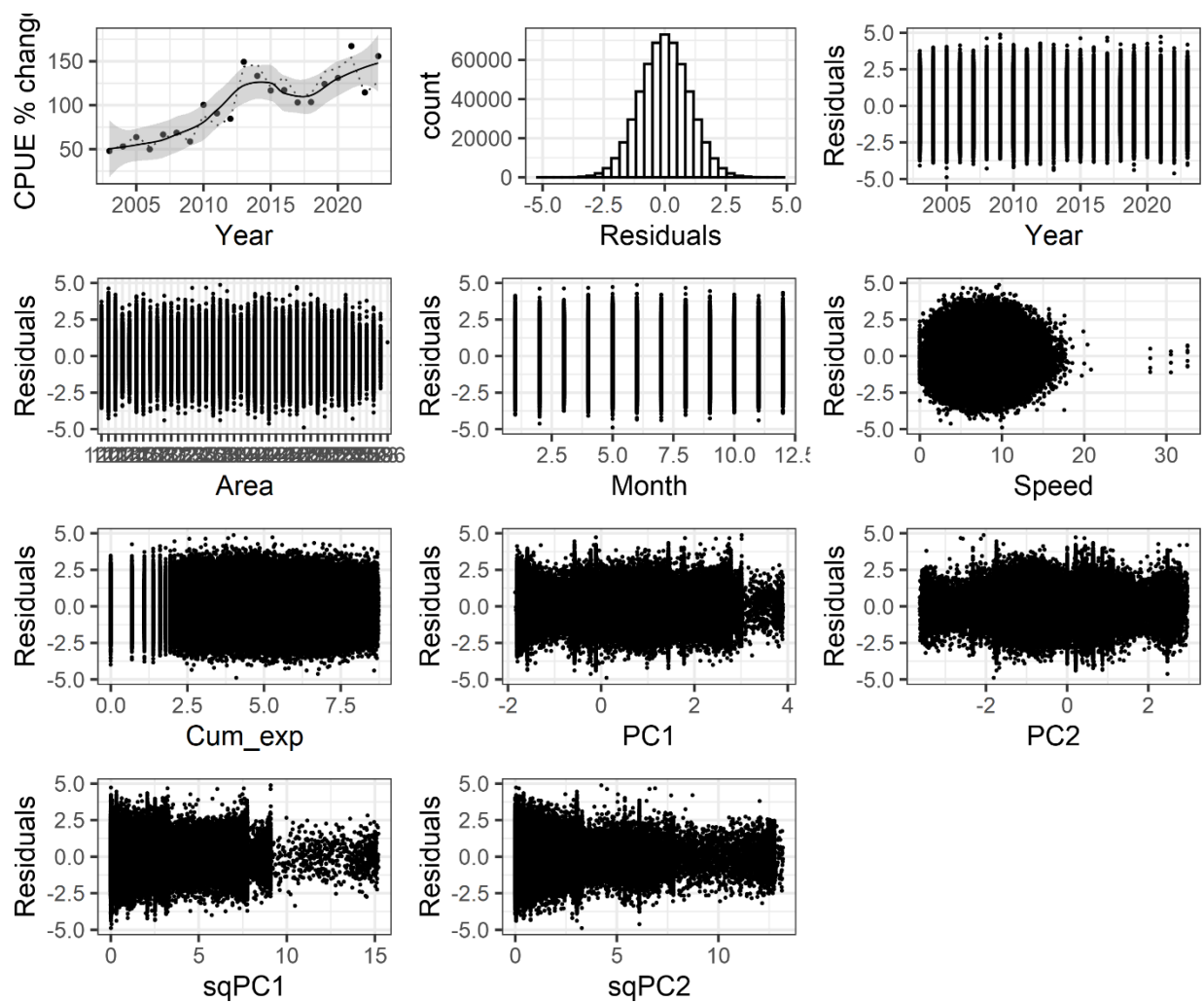


Figure 18. Diagnostics of the logistic CPUE standardization model for the 2003–2023 period (trolling gear). Top left graph shows nominal vs. standardized CPUE trends as a percent of their respective overall mean (dashed line=nominal, dots with loess curve=standardized). All other graphs present overall or variable-specific residual patterns.

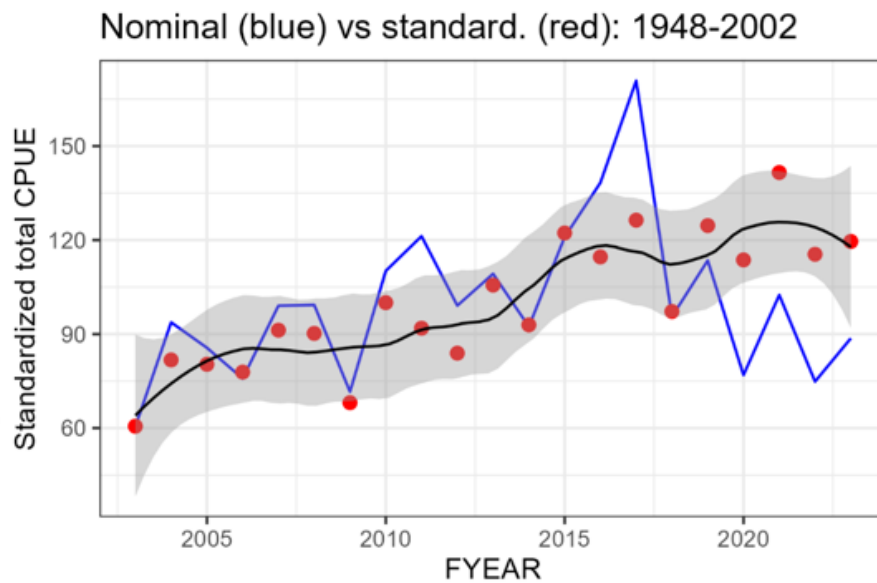
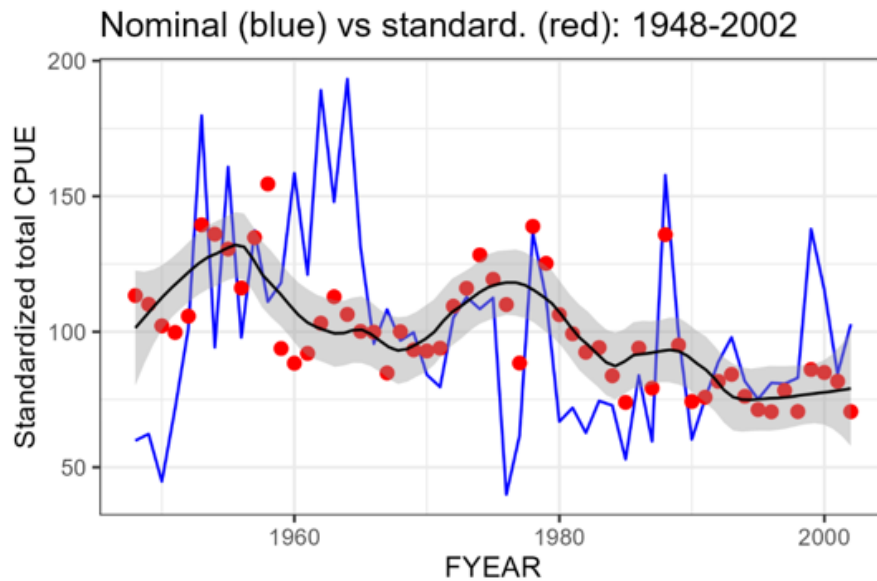


Figure 19. Overall nominal vs. standardized CPUE indices for deep-sea handline, generated by combining results from the lognormal and logistic standardization models.

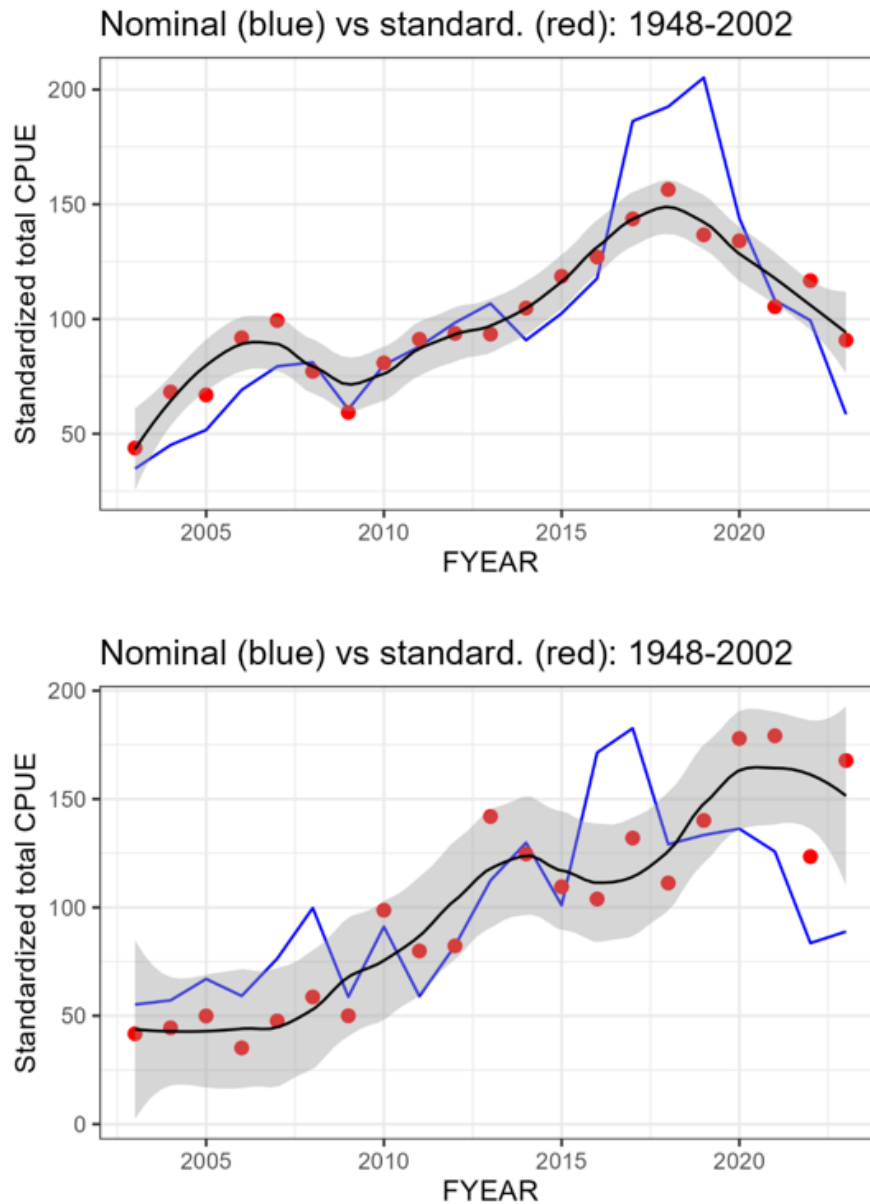


Figure 20. Overall nominal vs. standardized CPUE indices for inshore handline (top) and trolling (bottom) for the 2003–2023 period, generated by combining results from the lognormal and logistic standardization models.

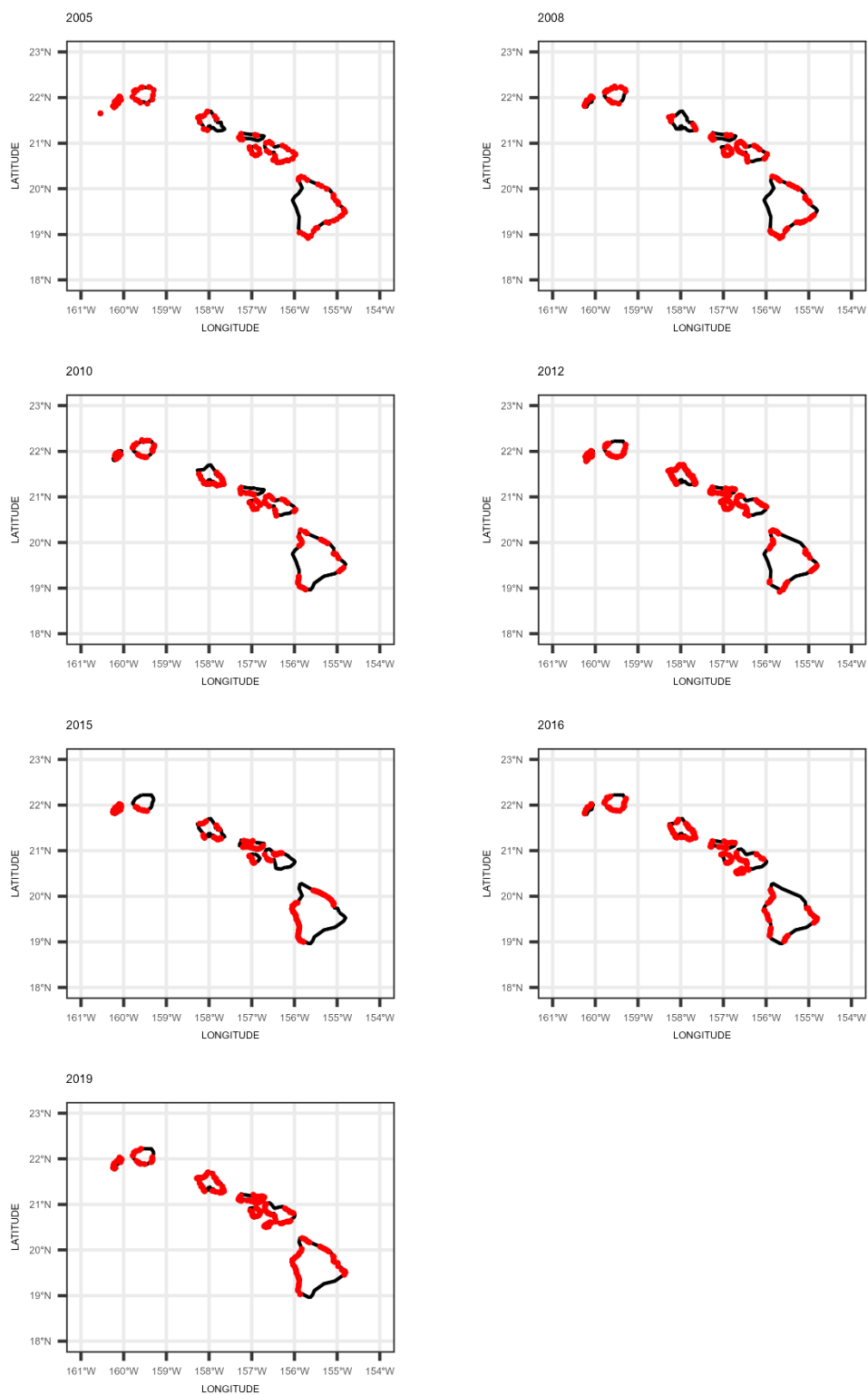


Figure 21. Distribution of survey sites (red dots) around the MHI for the NOAA diver surveys in each year.

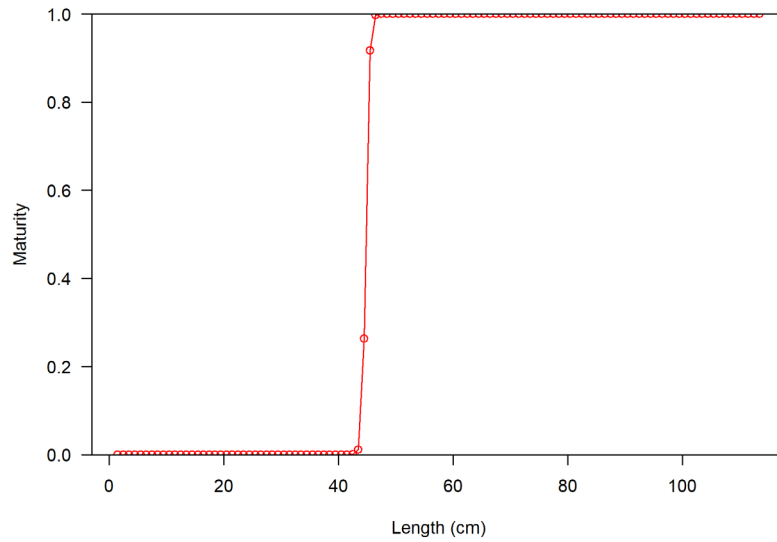


Figure 22. Maturity at length (FL) curve used in the stock assessment model, obtained from Everson and Williams (1989).

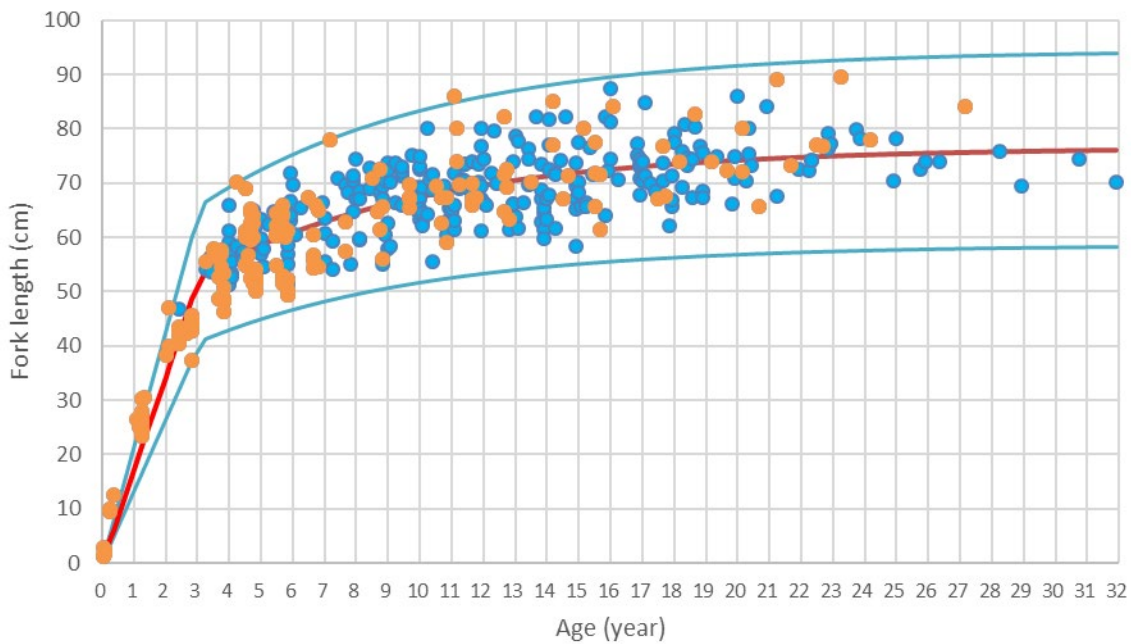


Figure 23. Age and length data collected in the MHI (orange points) and NWHI (blue points) with a two-stage von Bertalanffy curve, including the 95% confidence interval associated with the CV parameters (0.12 for both old and young individuals). Data provided by J. O'Malley, PIFSC.

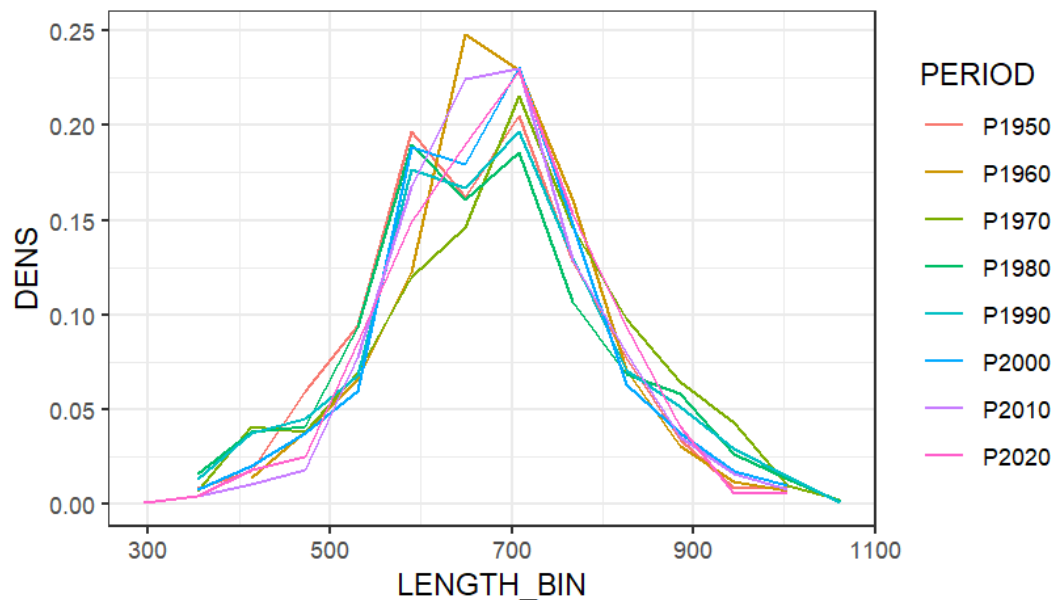


Figure 24. Size structures of uku from the FRS dataset and deep-sea handline gear summarized by decade.

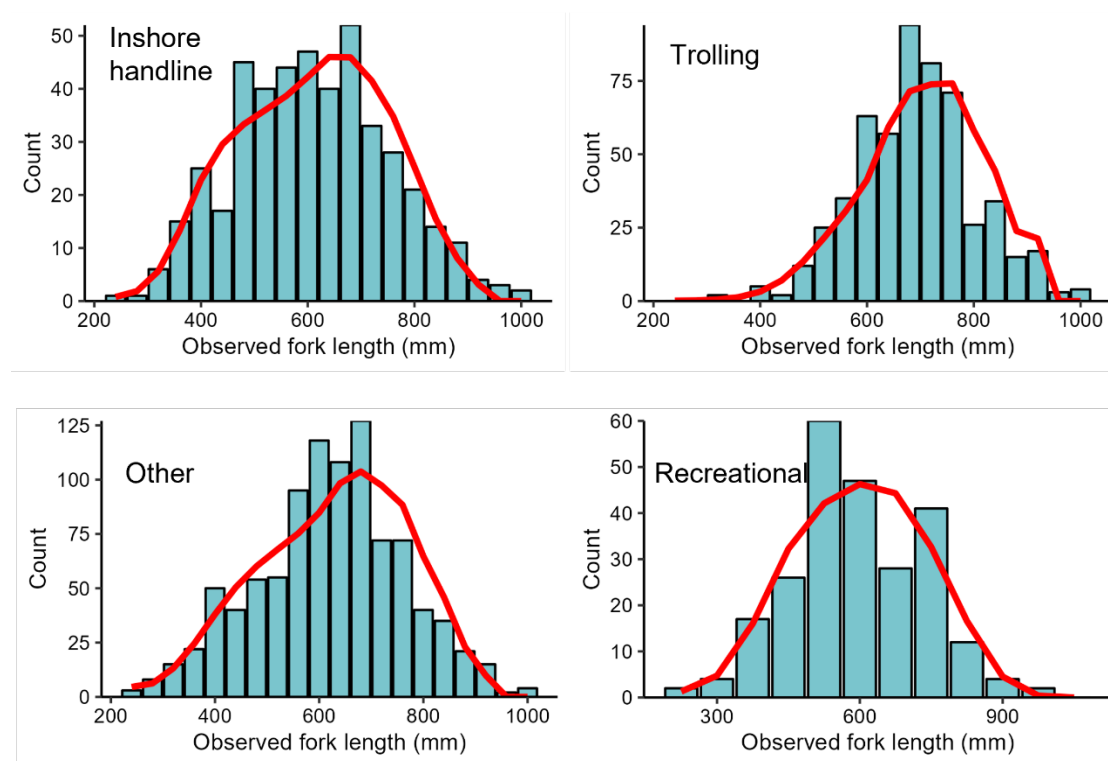


Figure 25. LBSPR model fit to size structure data for inshore handline, trolling, and “other” gear types from the FRS data set (2008–2023), as well as for recreational landings reported in the HMRFS dataset (2003–2023).

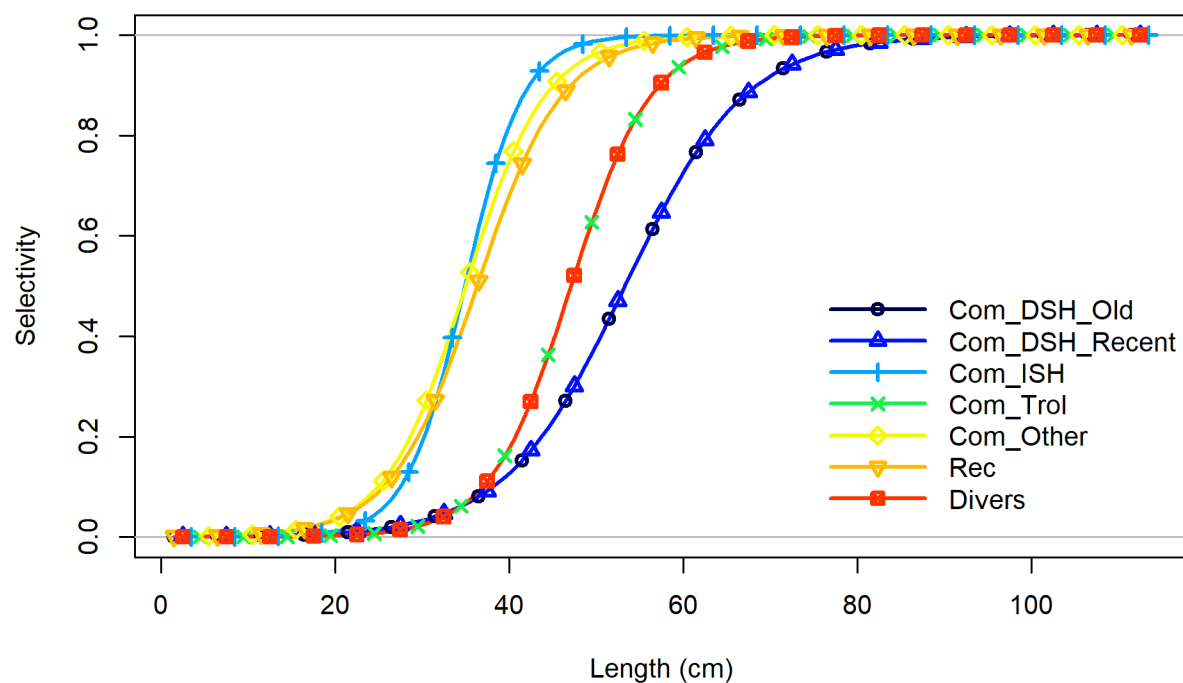


Figure 26. Length-based selectivity used in the Stock Synthesis model for all commercial fishing gear types and the recreational sector. Only the deep-sea handline selectivity was estimated in SS.

MFMT	MSST	B _{FLAG}
$F(B) = \frac{F_{MSY} B}{c B_{MSY}} \text{ for } B \leq c B_{MSY}$ $F(B) = F_{MSY} \text{ for } B > c B_{MSY}$	$c B_{MSY}$	B_{MSY}
where $c = \max(1-M, 0.5)$		

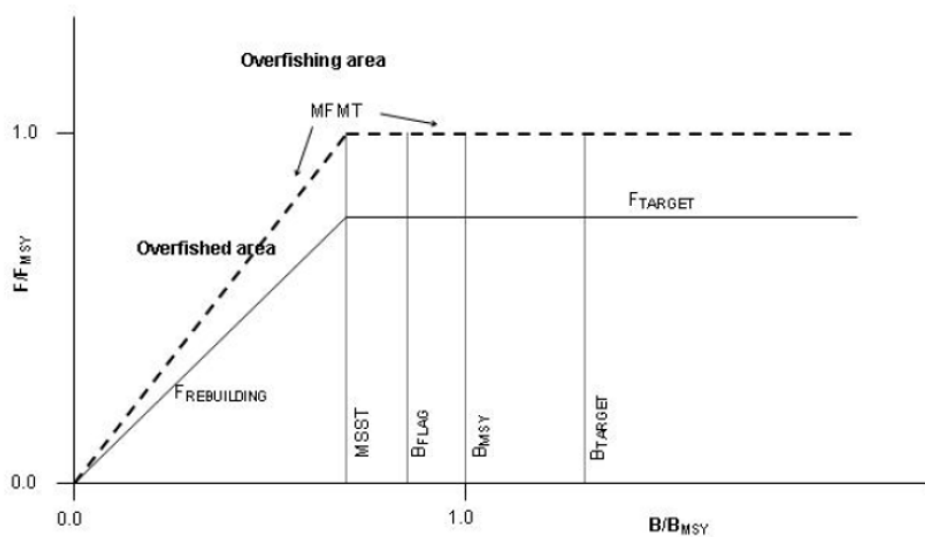


Figure 27. Fishery management control rules used for bottomfish in Hawai'i.

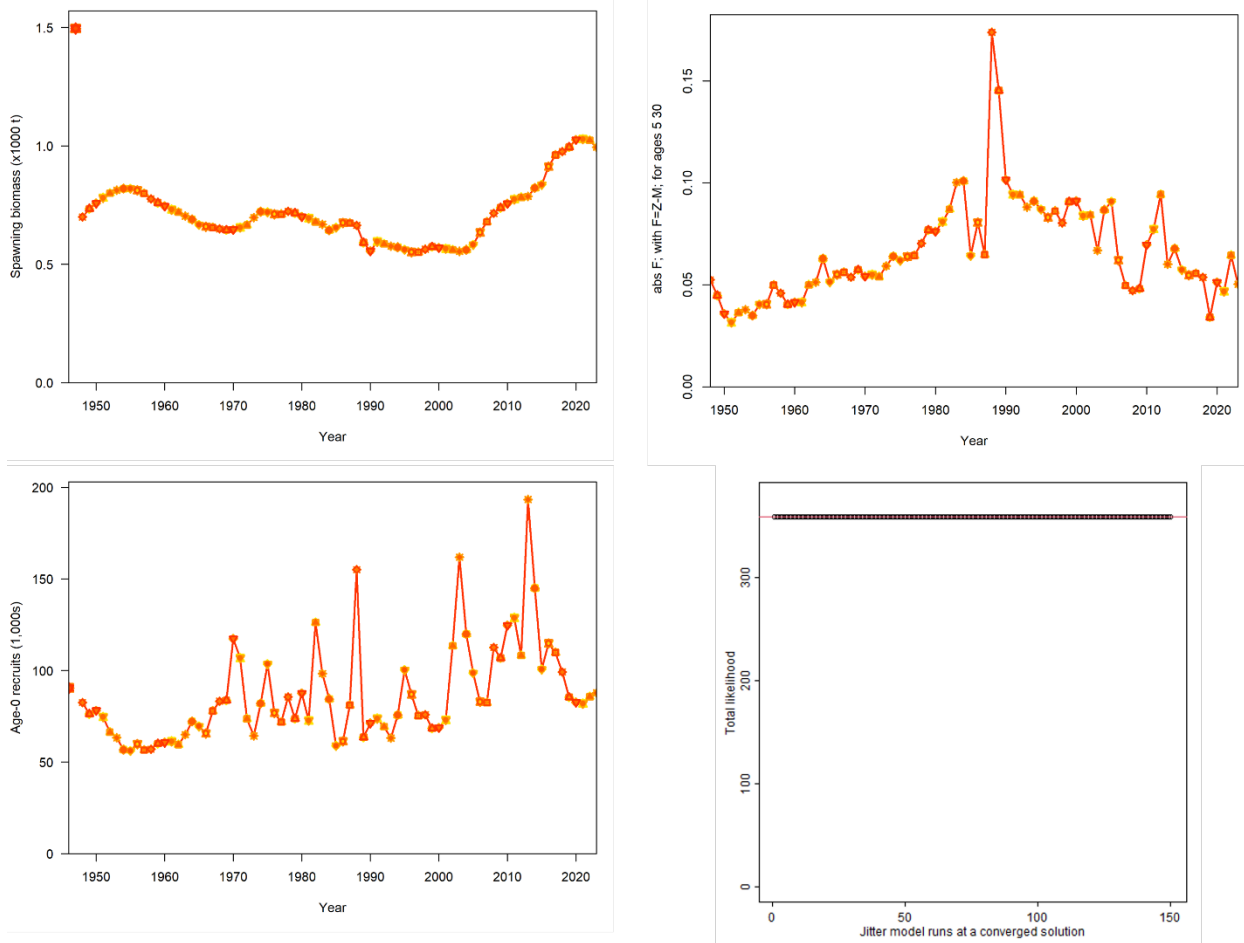


Figure 28. Jitter results for the base-case model, showing spawning biomass (top left), F (top right), recruitment (bottom left), and total likelihood (bottom right) for 150 model runs with different starting parameter values.

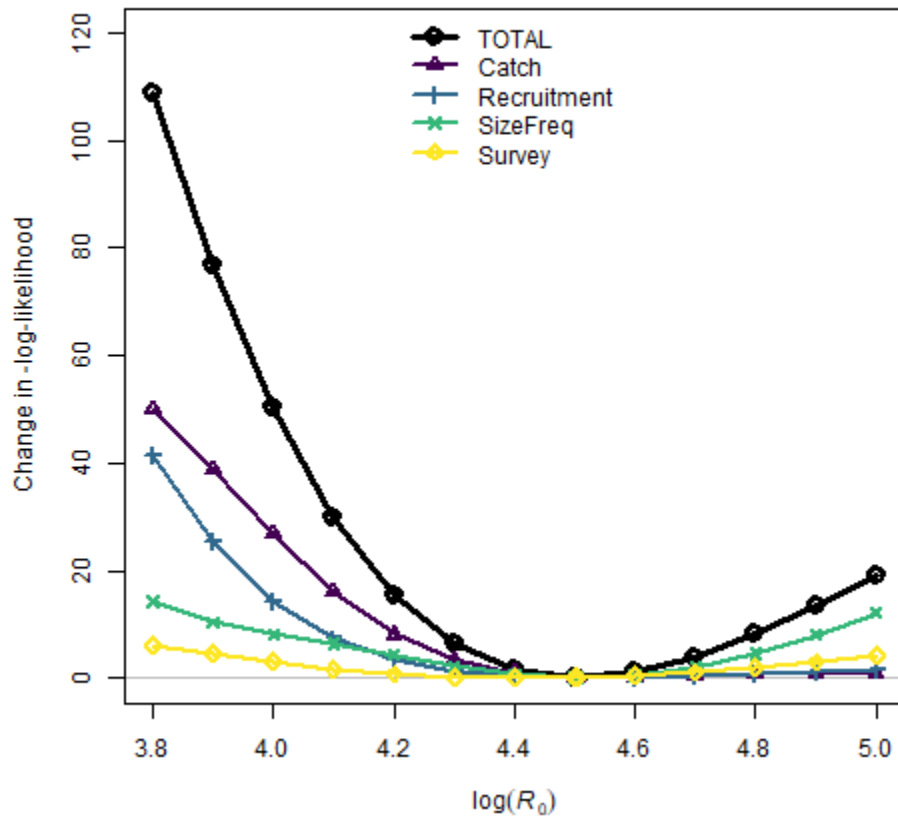


Figure 29. Profiles of the negative log-likelihoods relative to the minimum value of each component for the different likelihood components affecting the unfished recruitment parameter R_0 in log-scale for the base-case model. Recruitment represents the likelihood component based on the deviations from the stock-recruitment curve, catch is the joint likelihood component based upon the estimated catch for each fleet, survey is the joint likelihood component based upon fitting to the CPUE and OPUE indices, and SizeFreq represents the likelihood component for the deep-sea handline fleet based on the fish size composition data.

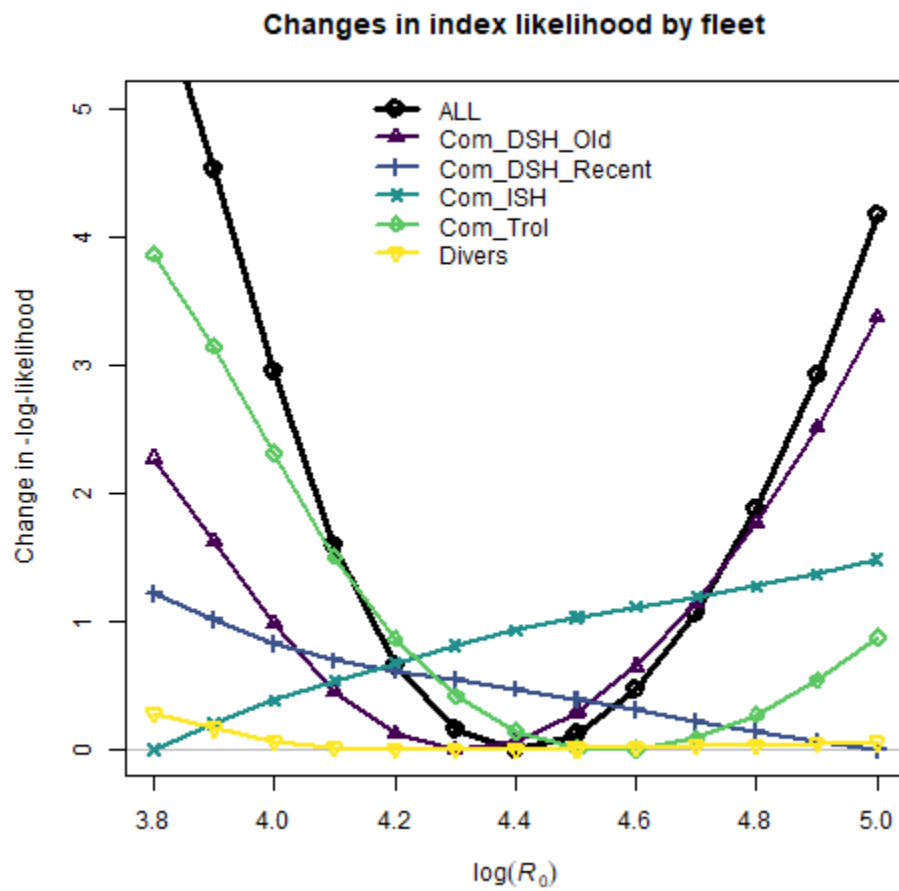


Figure 30. Profiles of the relative negative log-likelihoods by fleet-specific index likelihood components for the virgin recruitment in log-scale ($\log(R_0)$) ranged from 3.6 to 5.0 of the base-case scenario. DSH—deep-sea handline, ISH—inshore handline, and Trol—trolling.

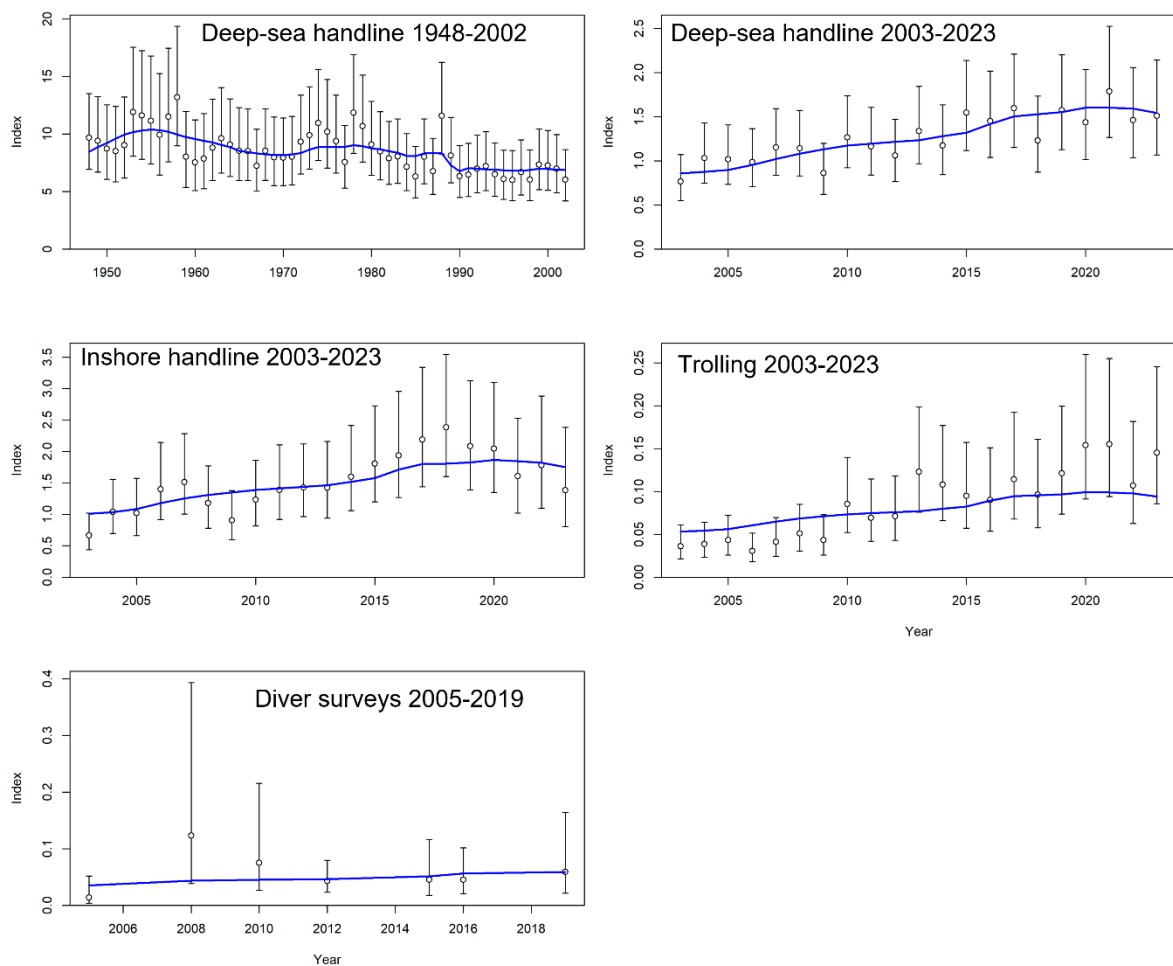


Figure 31. Plots of observed (open dots) and expected (blue lines) CPUE for the uku base-case model for all abundance indices used in the model. Bars around each observed data point represent the sum of the input SE and additional SE added within the model.

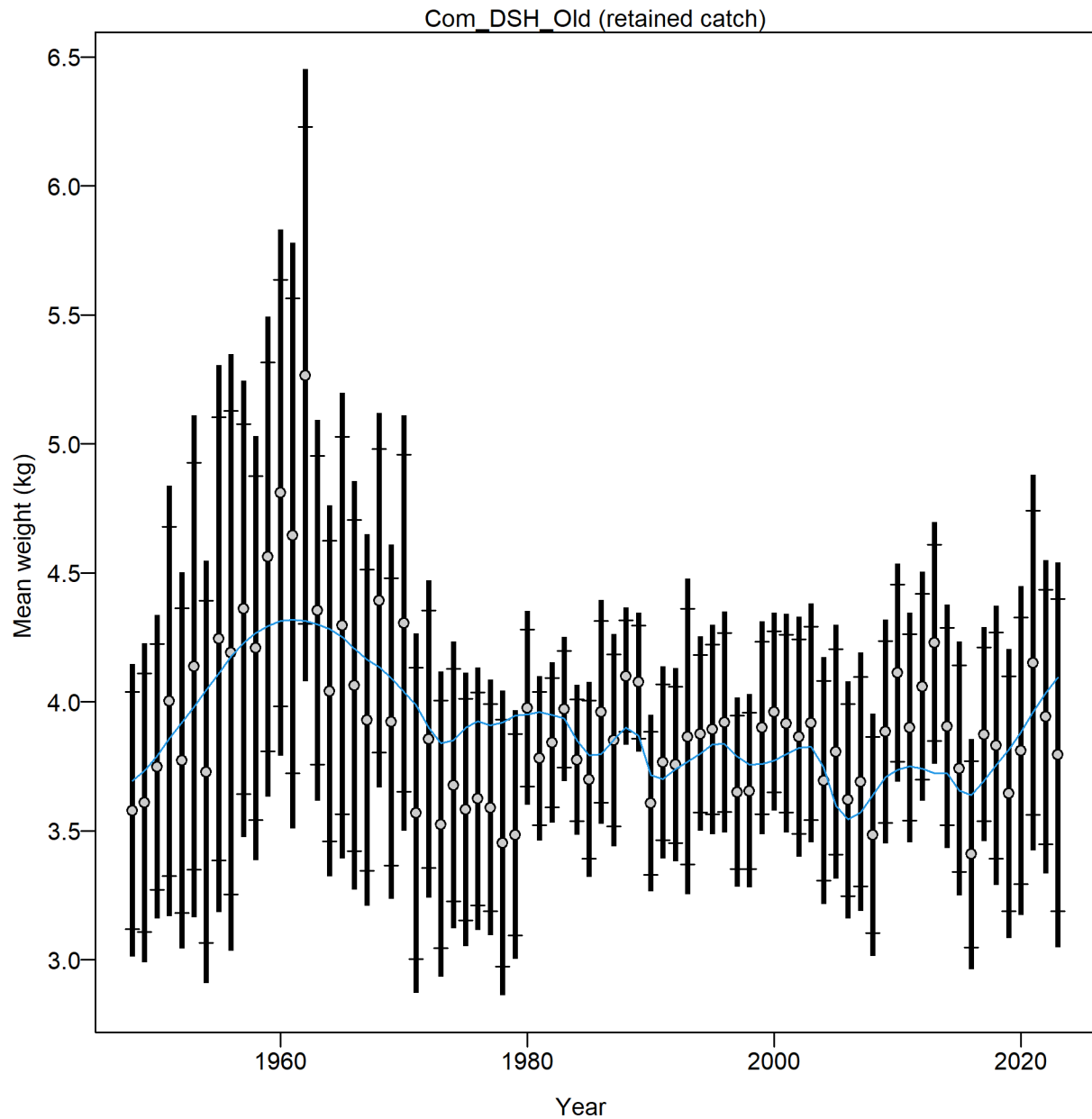


Figure 32. Model fit (lines) to mean weight (kg) of the composition data (points, showing the observed mean age and 95% credible limits around mean age (vertical lines)).

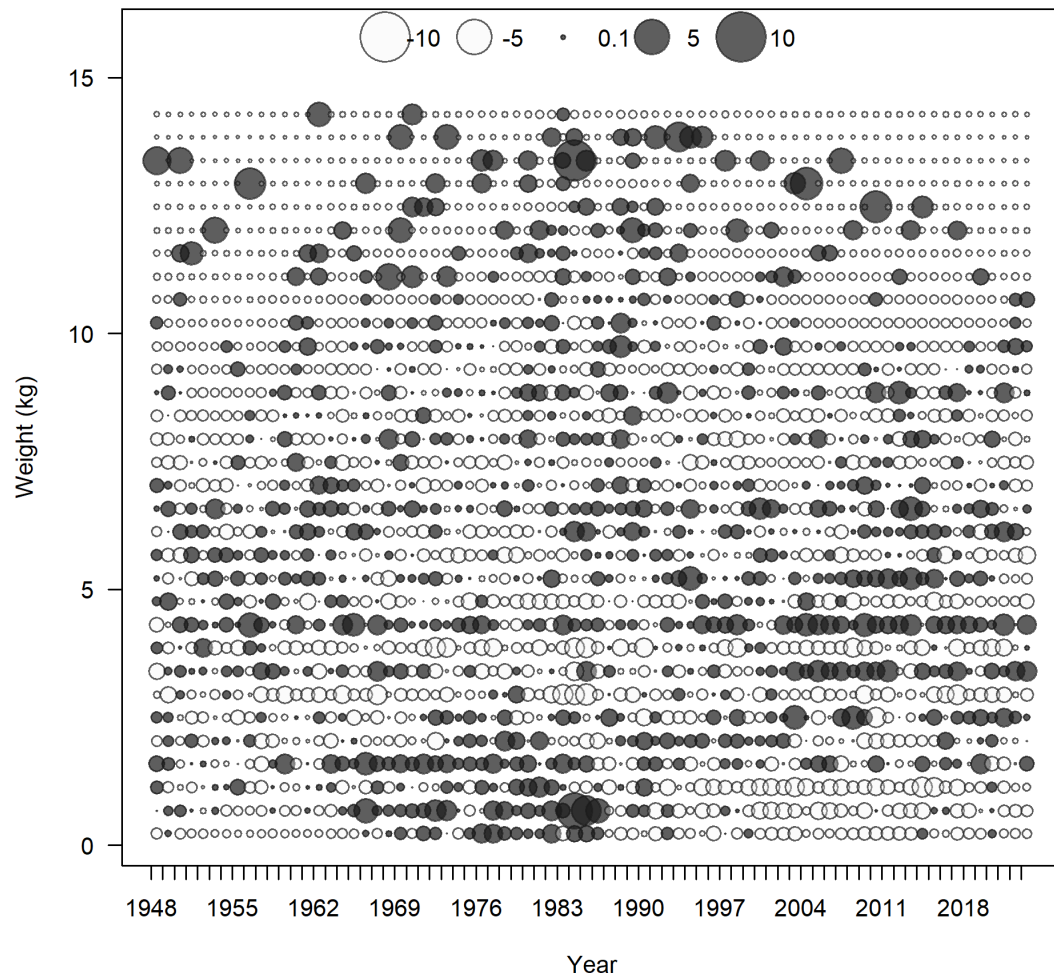


Figure 33. Pearson residual plots of model fits to the size composition data for deep-sea handline fishery targeting uku.

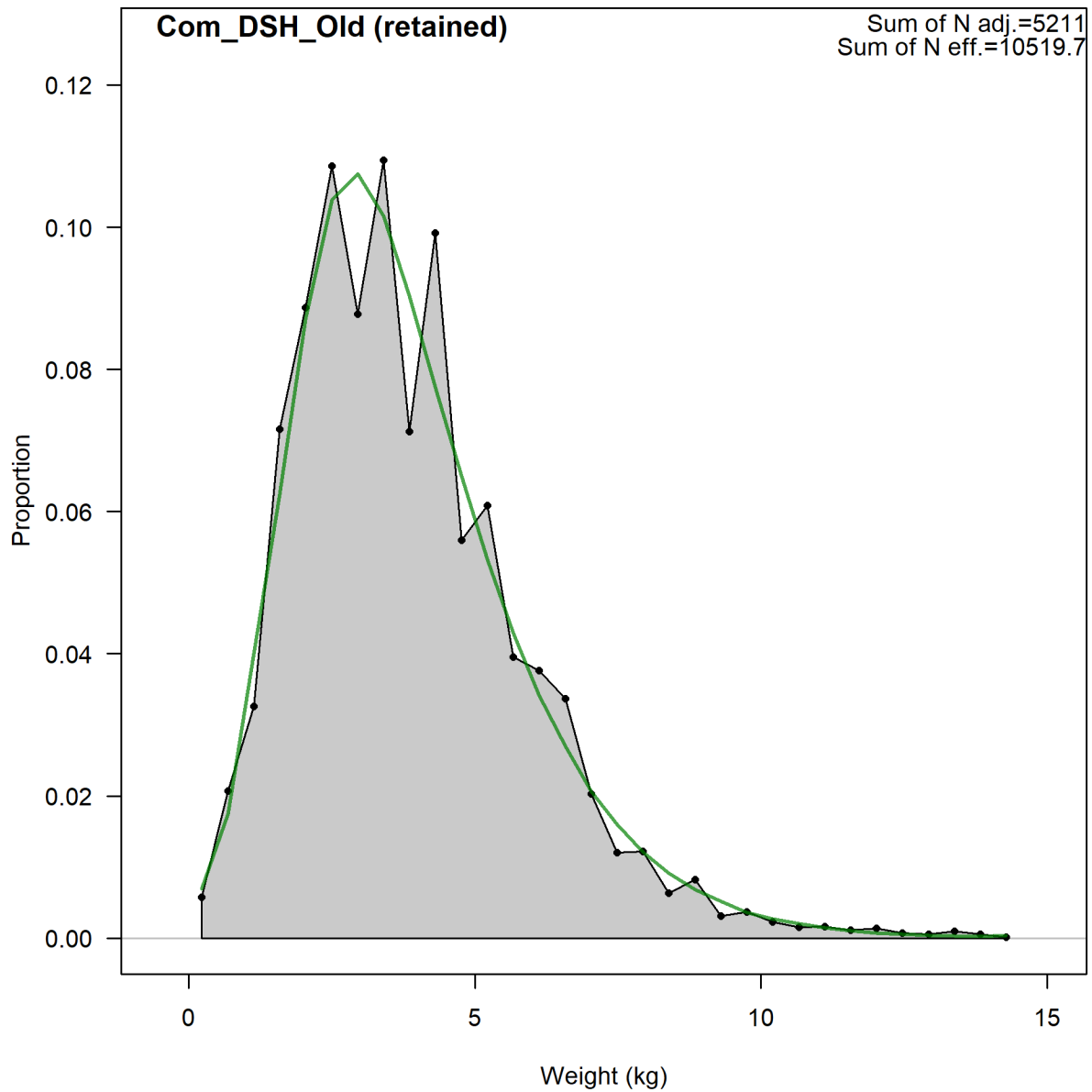


Figure 34. Comparison of observed (gray shaded area and black dots) and model predicted (green solid line) size composition data for the deep-sea handline fishery used in the stock assessment for uku.

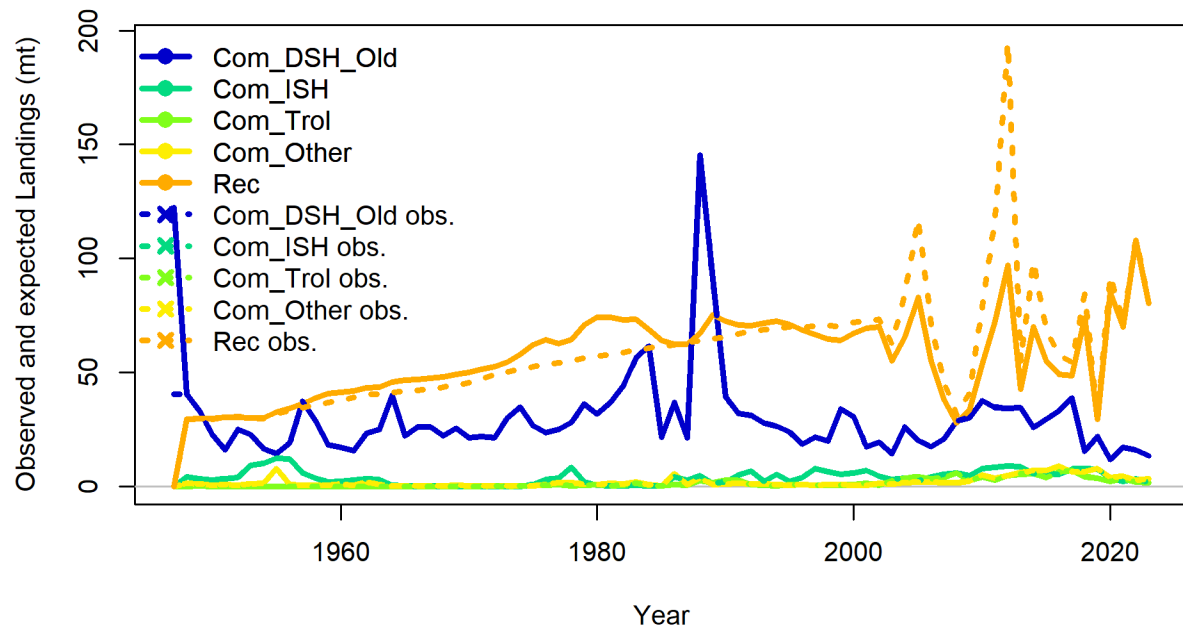


Figure 35. Observed (dashed line) vs. estimated (full line) catch in the Stock Synthesis model.

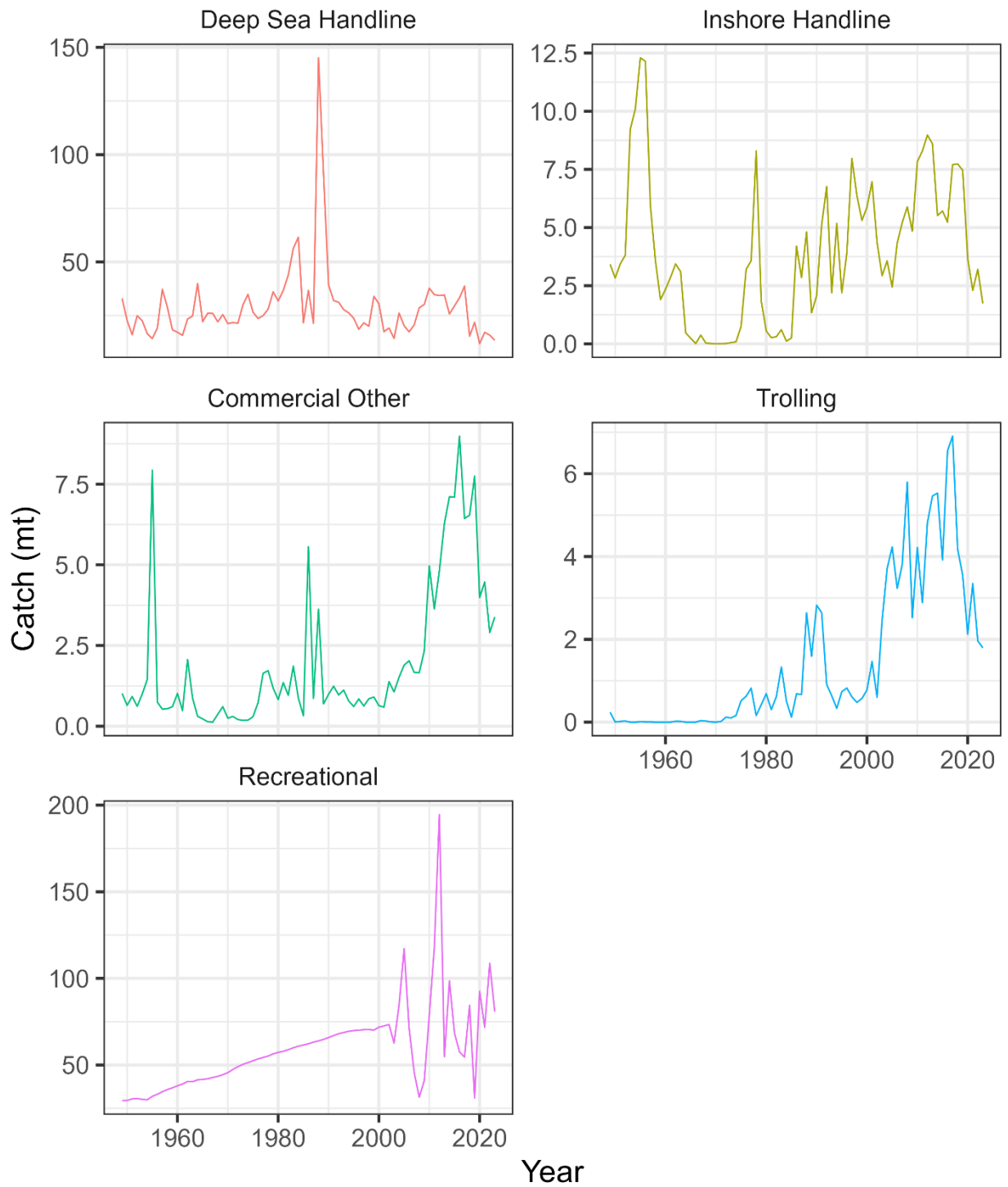


Figure 36. Observed catch by fishing gear.

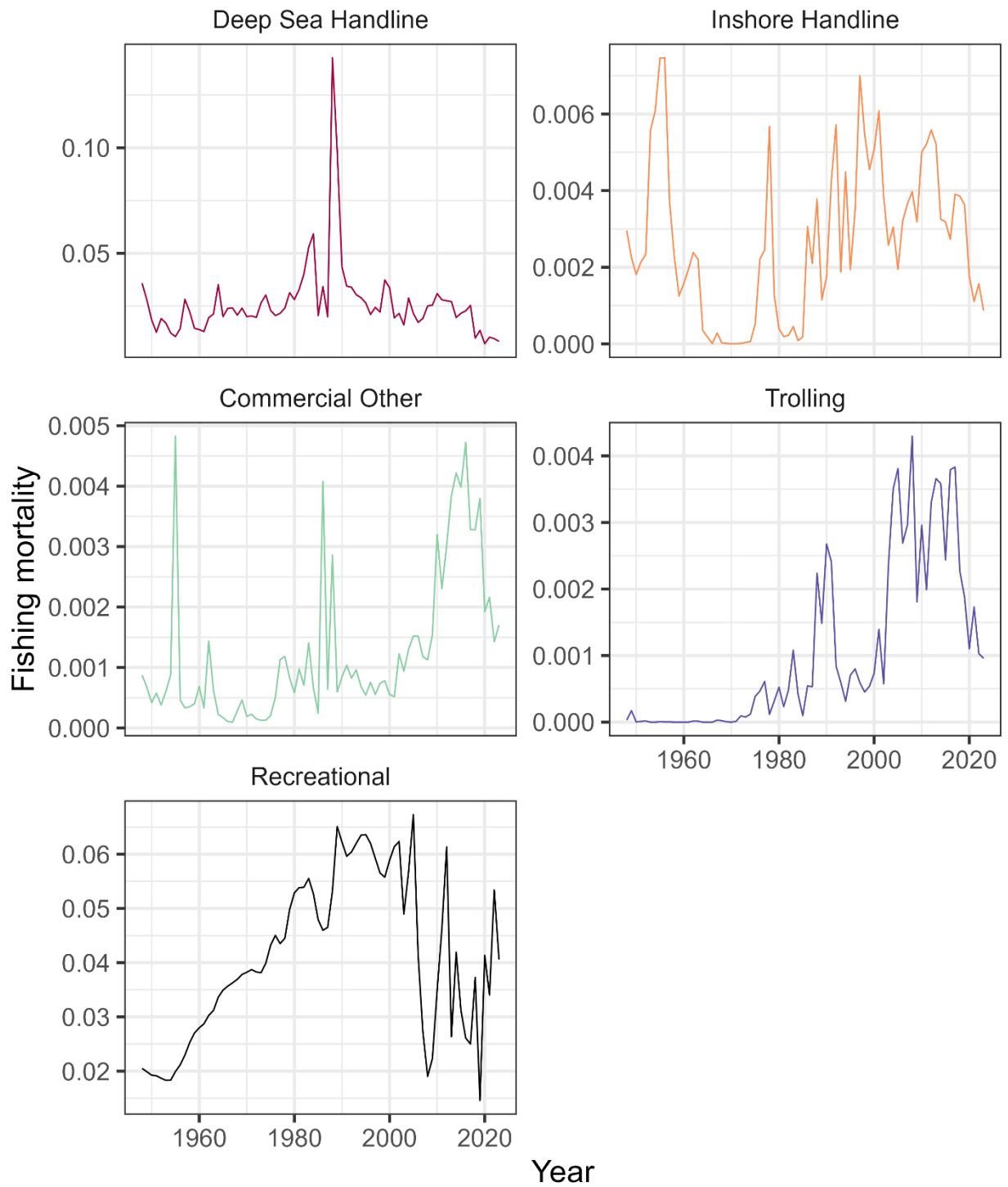


Figure 37. Annual fishing mortality estimated from the base-case model by fishing gear.

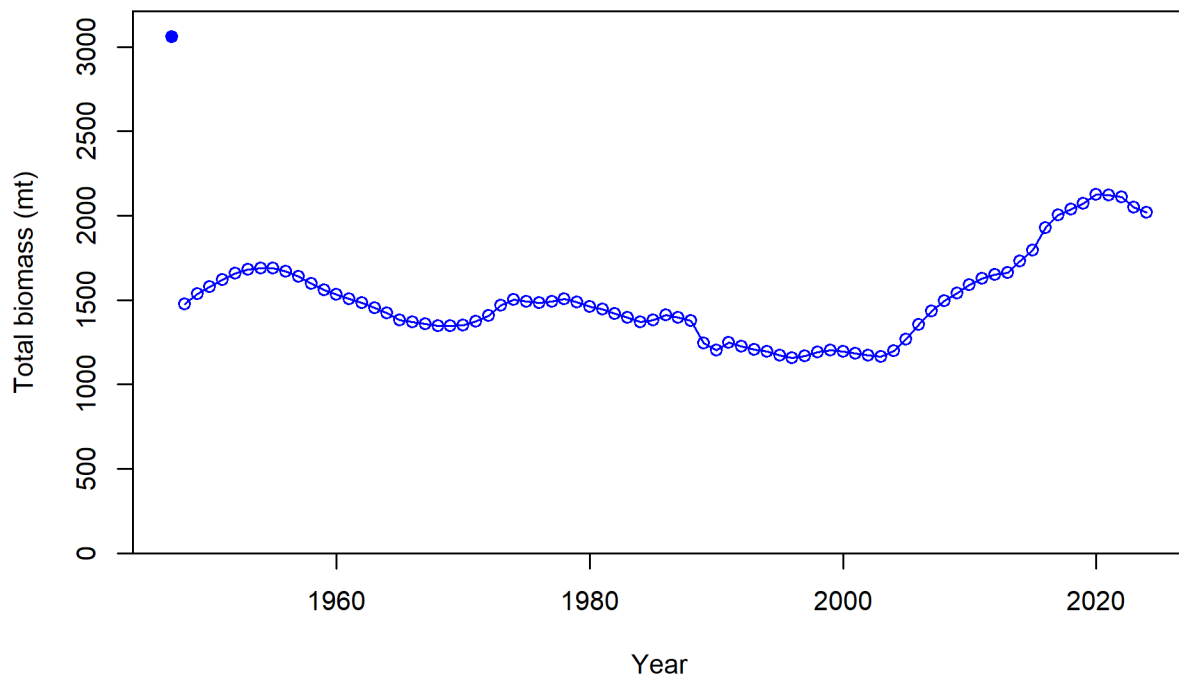


Figure 38. Time series of total biomass (age 1 and older, metric tons) for uku estimated in the base-case model. The first year indicates virgin biomass levels.

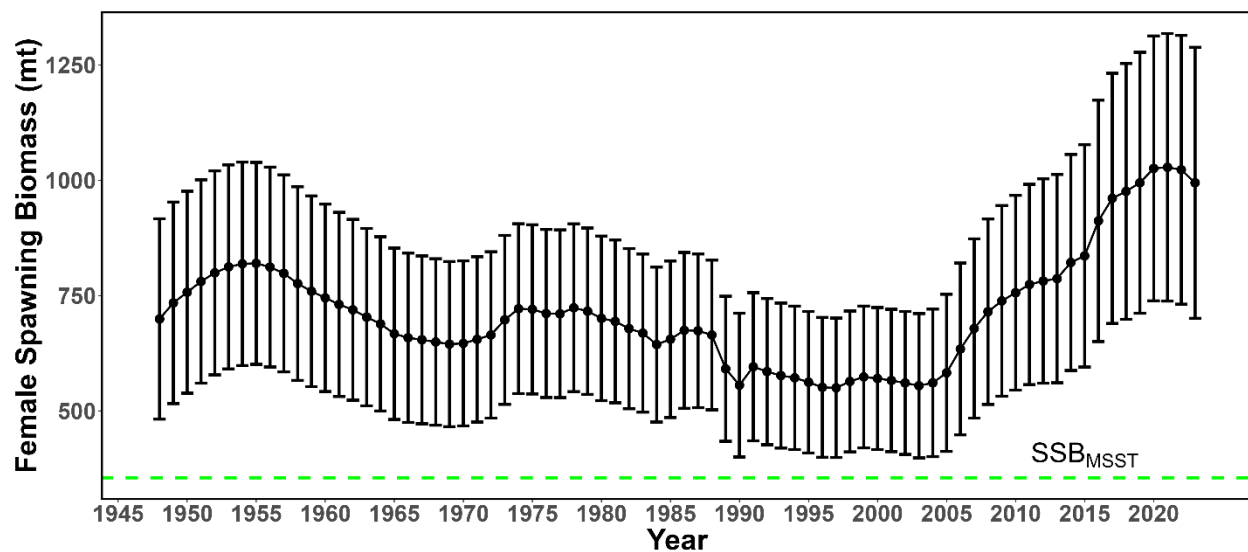


Figure 39. Time series of spawning biomass (metric tons) for uku estimated in the base-case model. The solid line with circles represents the maximum likelihood estimates and the error bars represent the uncertainty of the estimates (95% confidence intervals). The dashed horizontal line shows the spawning biomass to produce the MSST reference point (SSB_{MSST}).

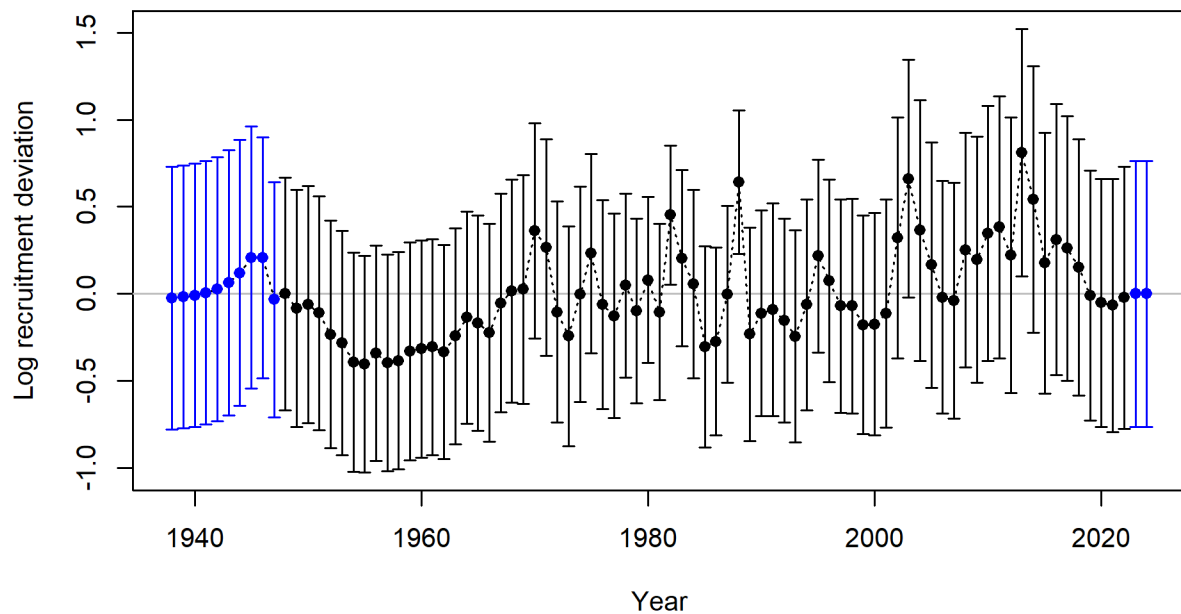


Figure 40. Estimated log recruitment deviations for the early (1938–1947, blue) and main (1948–2017, black) recruitment periods with associated 95% asymptotic confidence intervals.

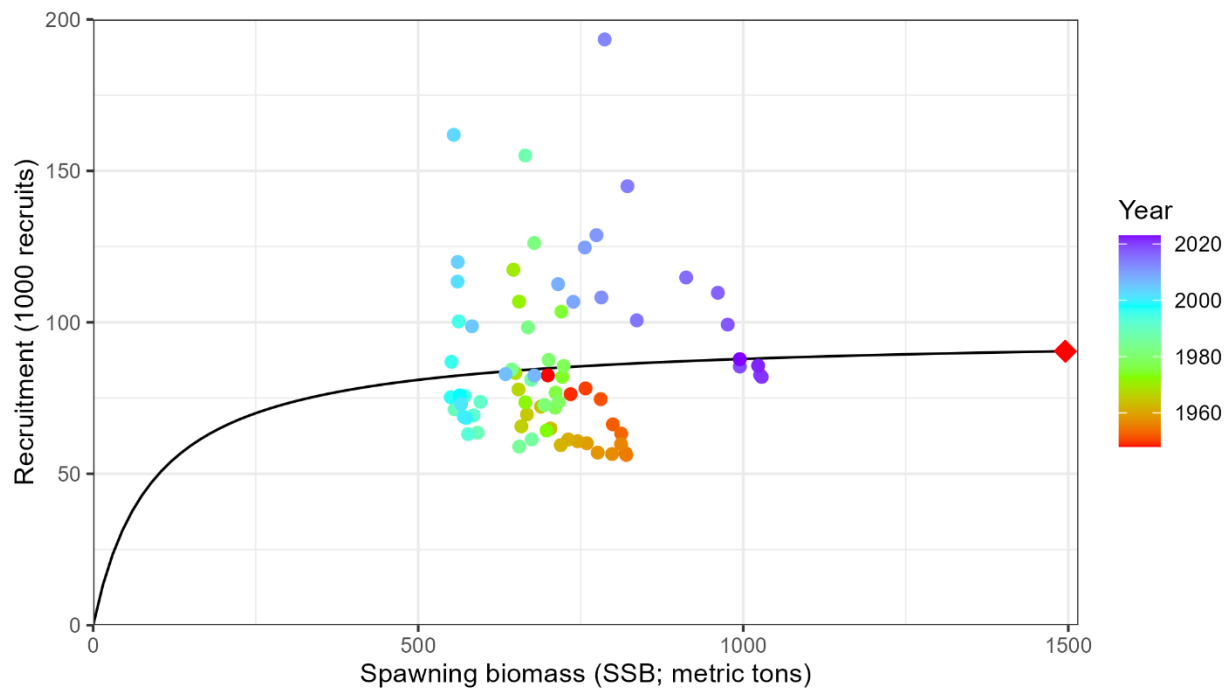


Figure 41. Expected recruitment from the stock-recruitment relationship (black line) and estimated annual recruitment (dots) from Stock Synthesis. Estimated virgin SSB and recruitment is indicated with a red diamond.

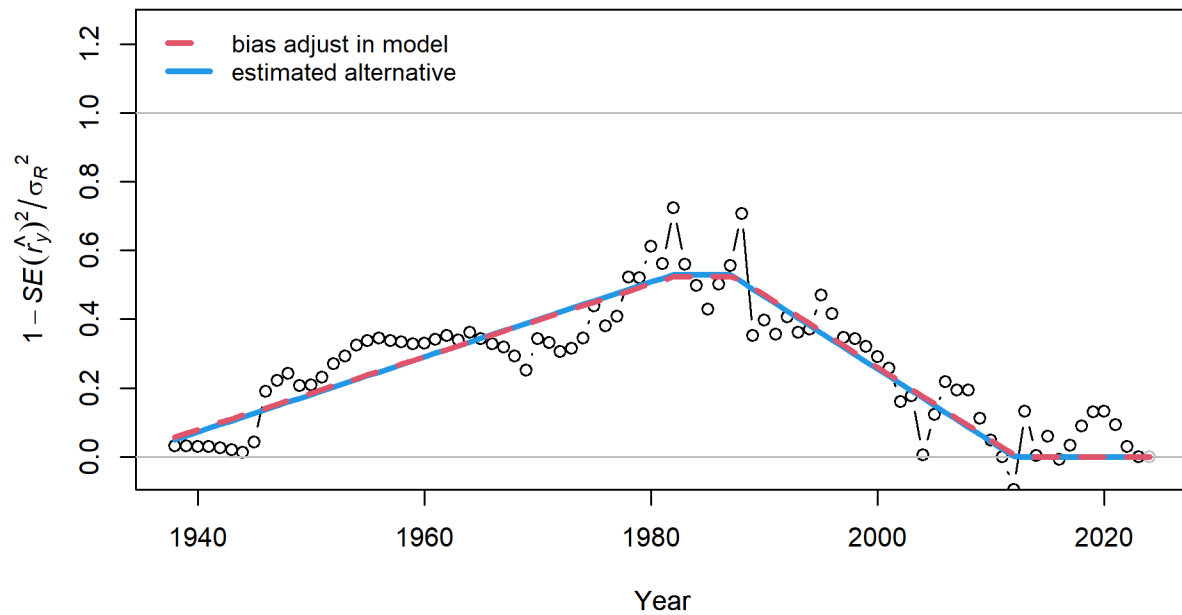


Figure 42. Bias adjustment applied to the stock-recruitment relationship (red stippled line) and the estimated alternative (blue line) obtained from the r4ss output.

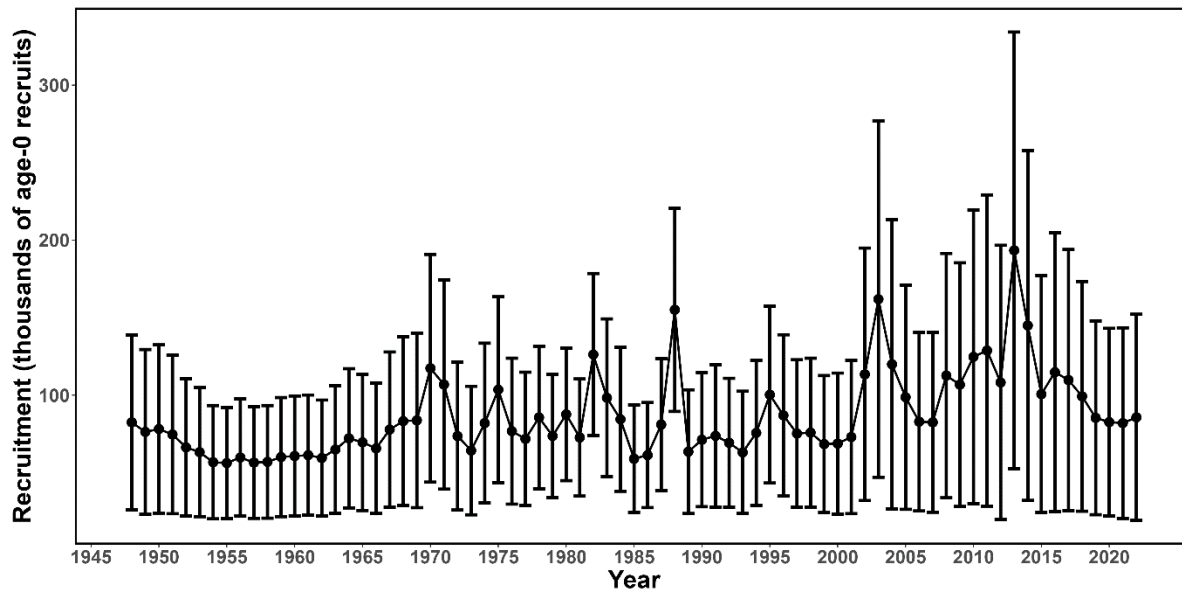


Figure 43. Time series of recruitment (thousands of age-0 fish) for uku estimated in the base-case model. The solid line with circles represents the maximum likelihood estimates and the error bars represent the uncertainty of the estimates (95% confidence intervals).

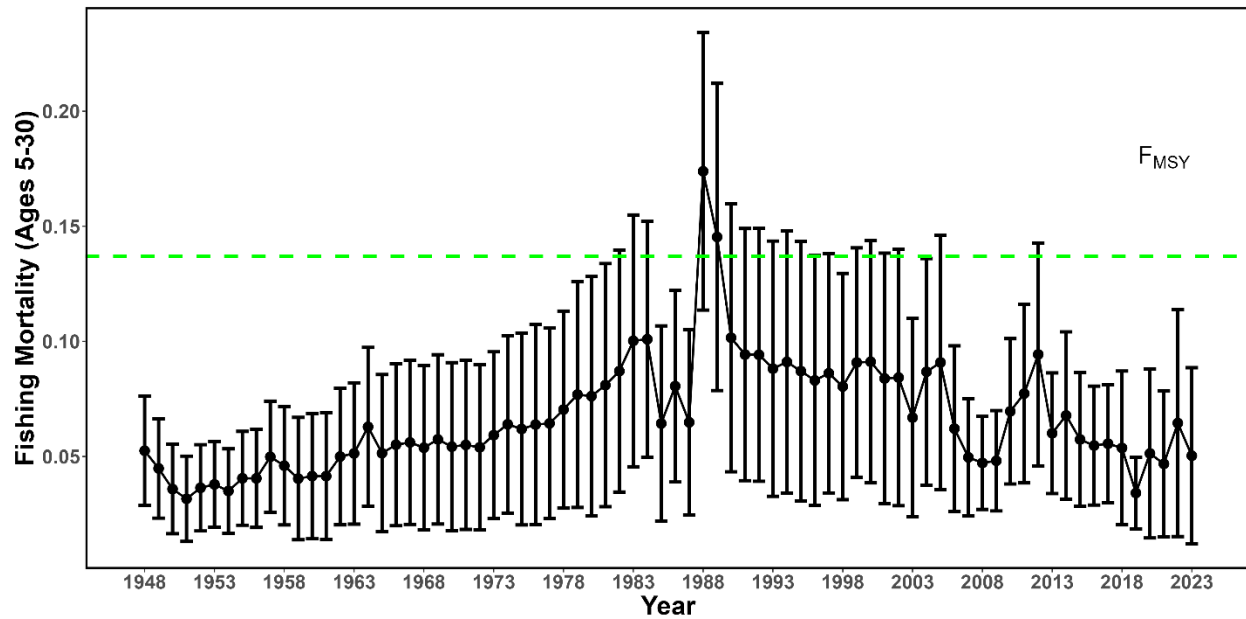


Figure 44. Time series of instantaneous fishing mortality (average for ages 5–30) for the uku estimated in the base-case model. The solid line with circles represents the maximum likelihood estimates, and the error bars represent the uncertainty of the estimates (95% confidence interval). The dashed horizontal line shows the fishing mortality to produce the MSY reference point (F_{MSY}).

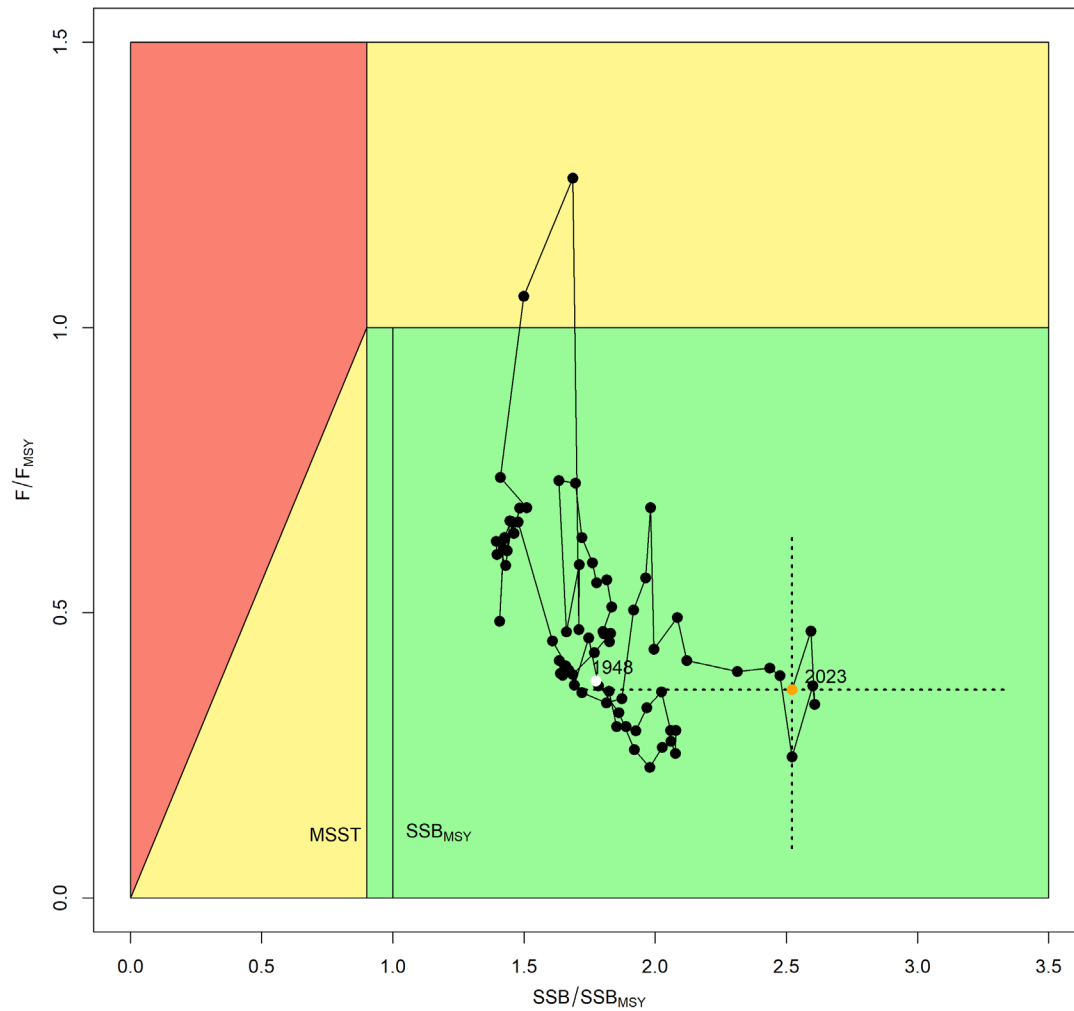


Figure 45. Kobe plot of the trends in estimates of relative fishing mortality (average of age 5–30) and spawning stock biomass of uku during 1948–2023. The white dot indicates 1948, the orange dot indicates 2023, and the dotted lines indicate the 95% confidence intervals around the final year values.

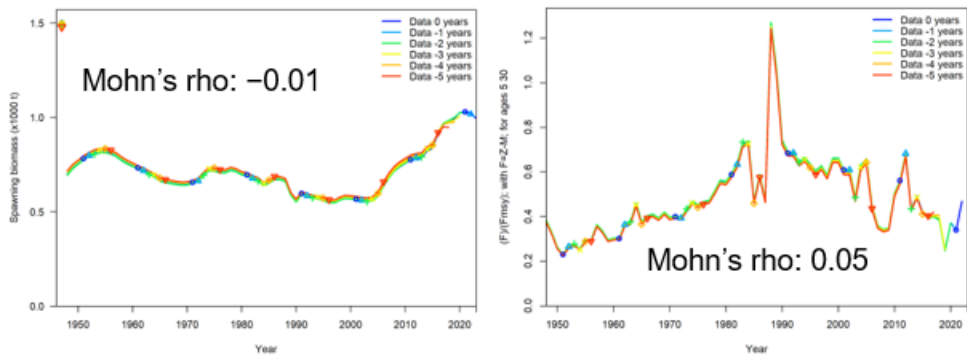


Figure 46. Retrospective analysis of spawning biomass (as SSB/SSB_{MST} , left) and fishing mortality (as F/F_{MSY} , right) consisting of 5 reruns of the base-case model each fitted with one additional year of data removed from the base-case model (black line, 1948-2023).

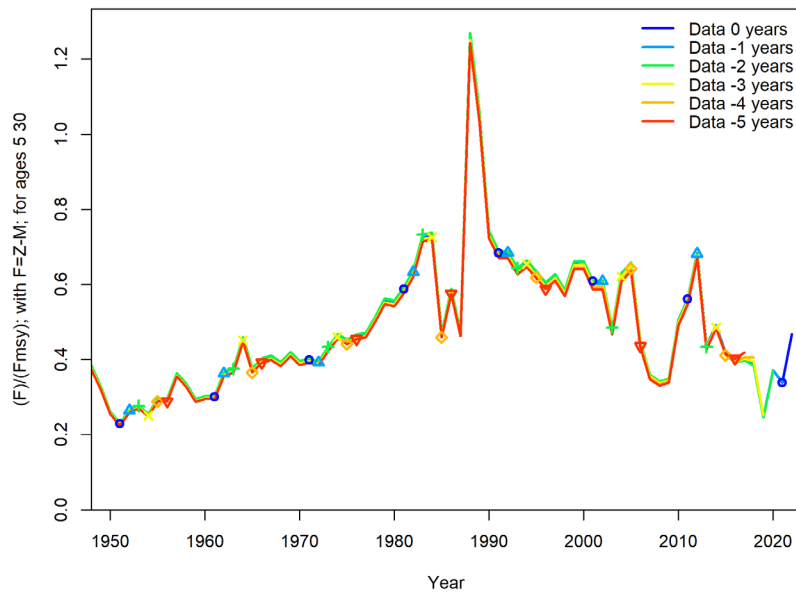


Figure 47. Plot of estimated Age-0 recruits (in 1000s of fish) for each run in the 5-year retrospective analysis. The dark blue line indicates the base-case model (1948-2023).

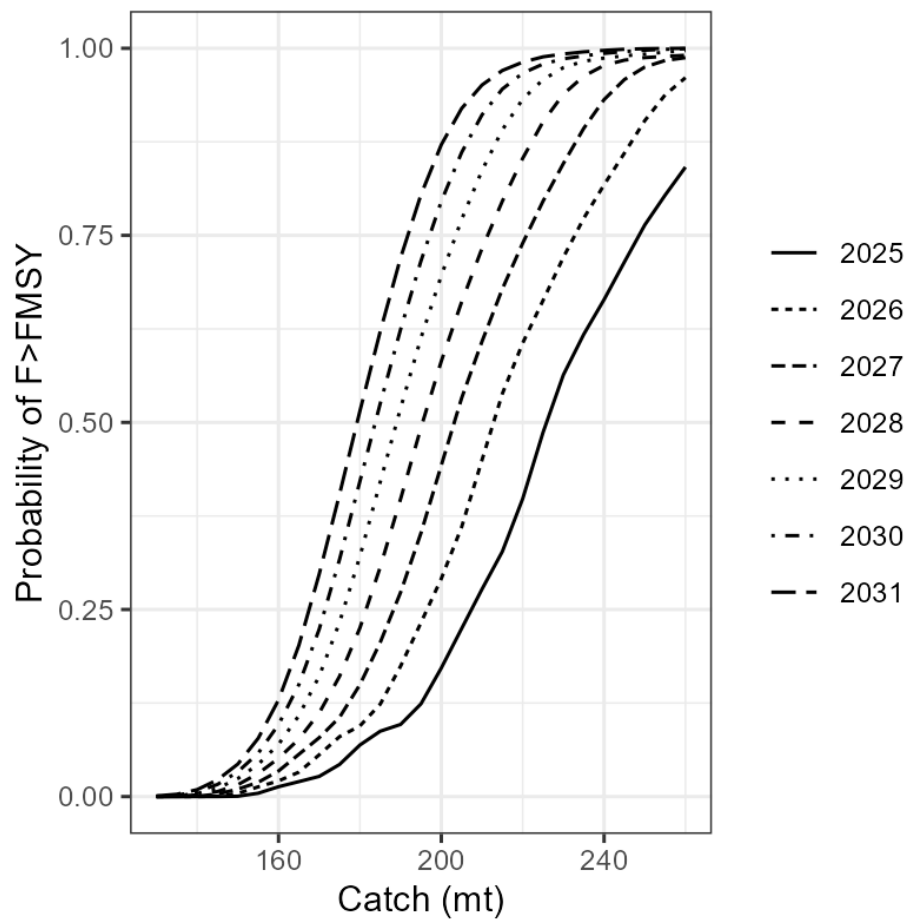


Figure 48. Probability of overfishing (i.e., $F/F_{MSY} > 1$) uku in the main Hawaiian Islands in fishing years 2025 through 2031, for a range of fixed catch scenarios.

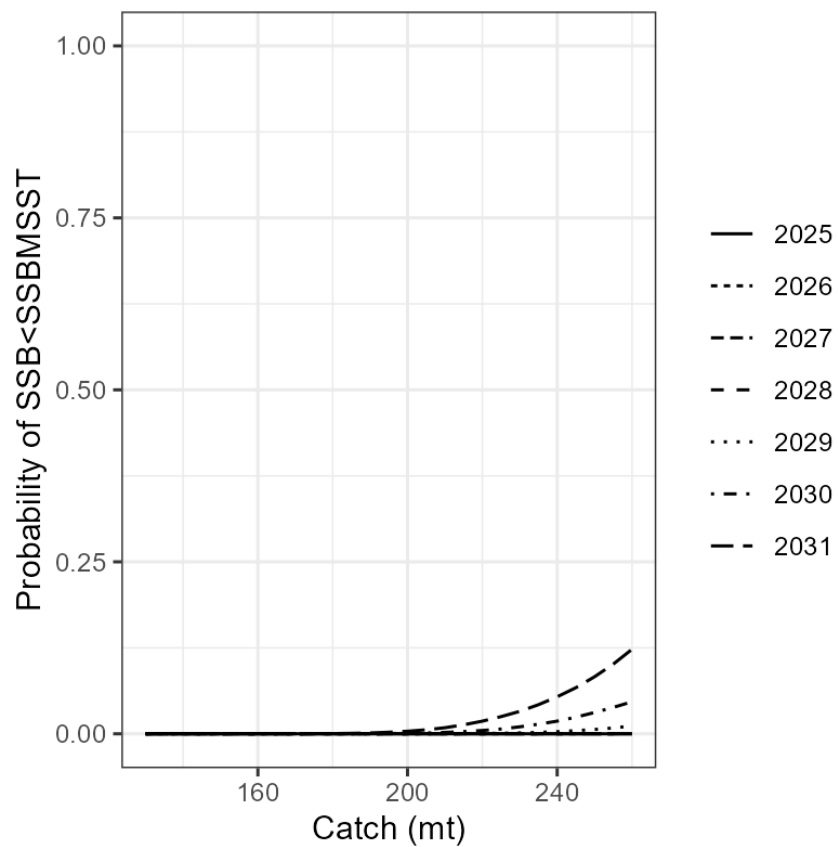


Figure 49. Probability of the stock being overfished (i.e., $SSB/SSB_{MSS} < 1$) for uku in the main Hawaiian Islands in fishing years 2025 through 2031 for a range of fixed catch scenarios.

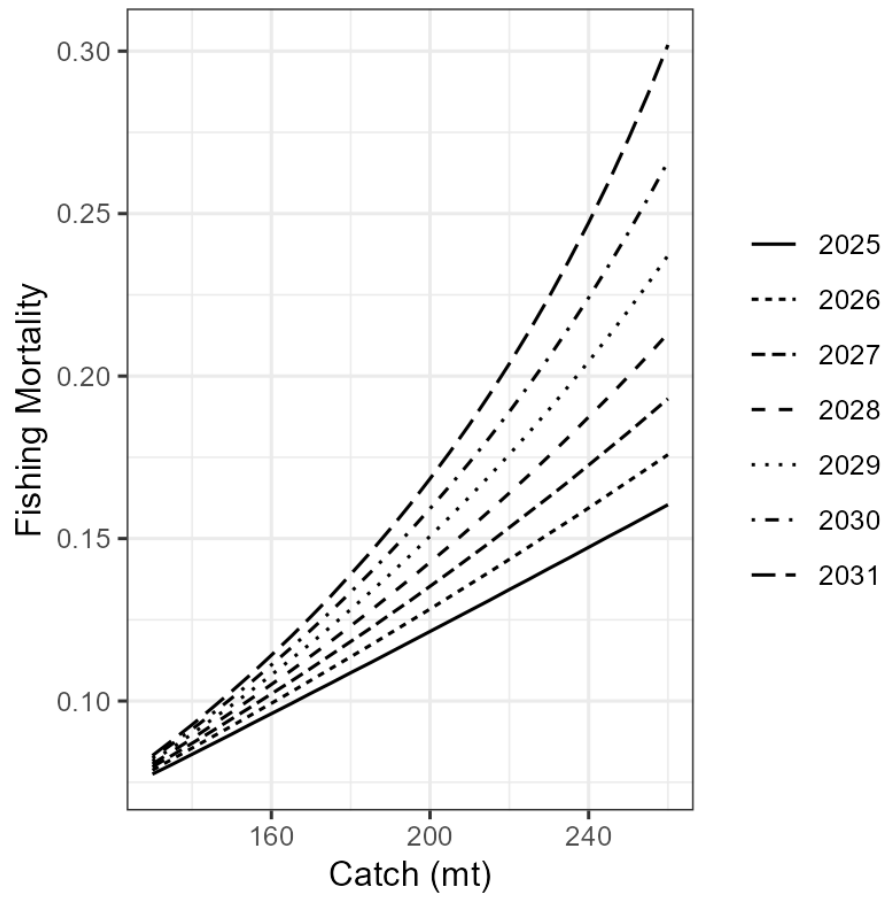


Figure 50. Mean fishing mortality for uku in the main Hawaiian Islands in fishing years 2025 through 2031 for a range of fixed catch scenarios.

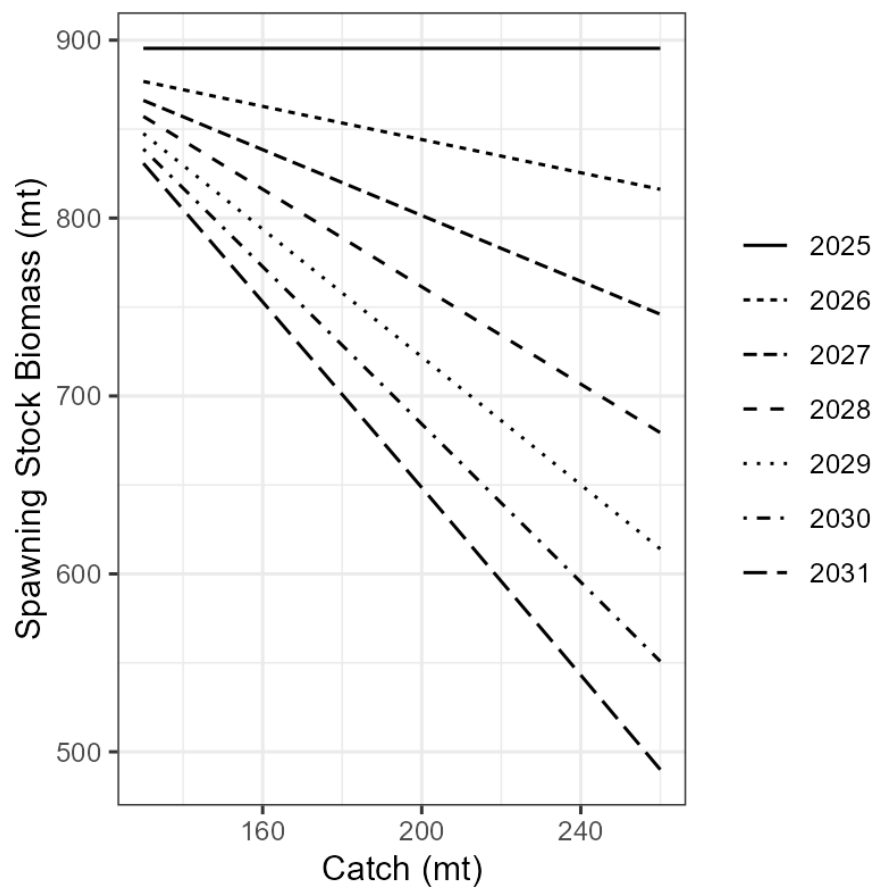


Figure 51. Mean spawning stock biomass for uku in the main Hawaiian Islands in fishing years 2025 through 2031 for a range of fixed catch scenarios.

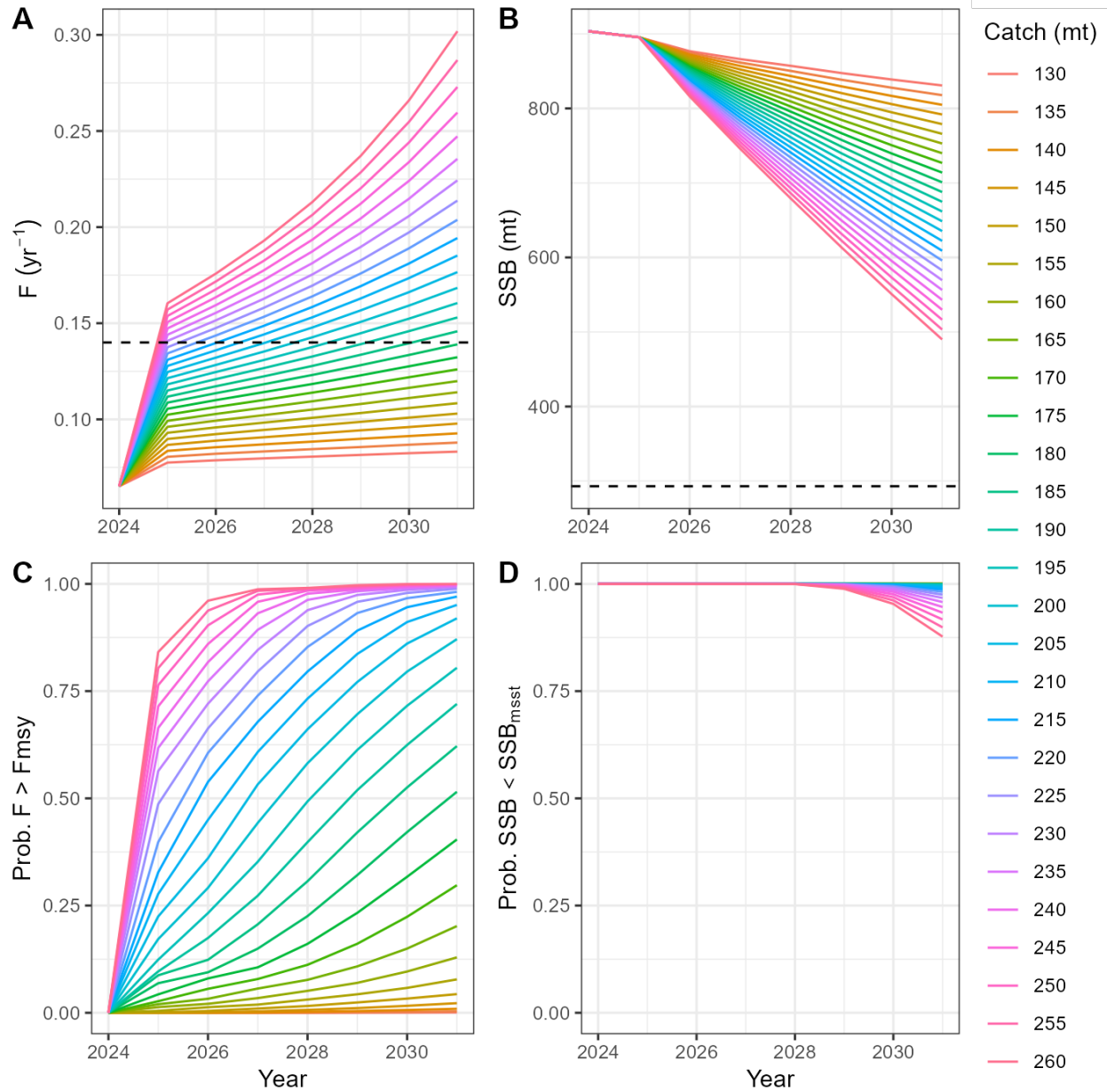


Figure 52. Fishing mortality with F_{MSY} reference line (a), Spawning Stock Biomass with SSB_{MSST} reference line (b), probability of overfishing (c), and probability of being overfished (d) for various fixed catch scenarios from 2024 to 2031.

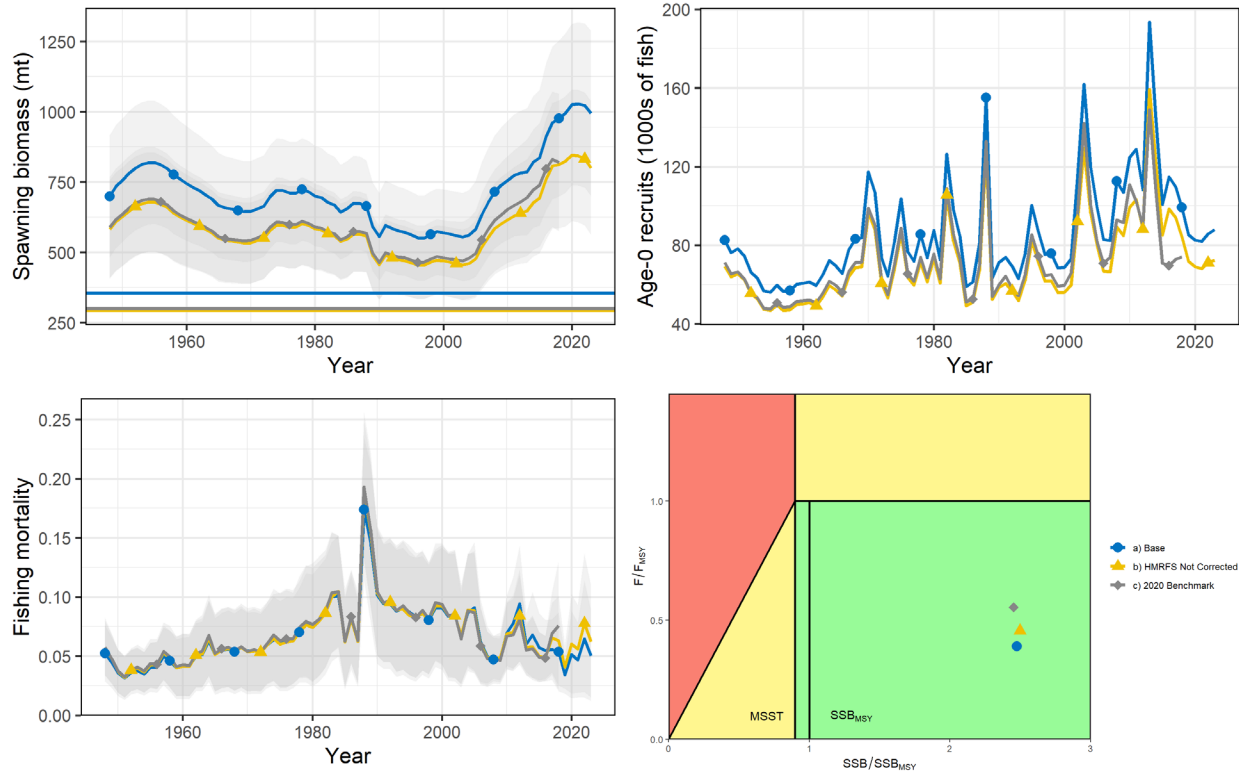


Figure 53. Comparisons of spawning biomass, fishing mortality, recruitment, and final year stock status (Kobe plot) for (a) the base model used in the current assessment update (blue), (b) the same model without the landline phone correction to the HMRFS catch (yellow), and (c) the 2020 benchmark model.